Chapter I

Stochastic I

1 Probability Distribution

Theorem 1.1 (Transformation Formula). *Given a random variable $X \in \mathbb{R}^n$ with joint density f, $g : \mathbb{R}^n \to \mathbb{R}^n$ a continuously differentiable and injective function, with non-vanishing Jacobian, then the density function of $Y = g(X)$ is:*

$$f^Y(y) = f^X(g^{-1}(y))|\det J_{g^{-1}}(y)|$$

Example.

2 Integration with Respect to probability measure

Theorem 2.1 (Markov Inequality). *Given a random variable X and **non-negative** monotone increasing function $f : [0,\infty) \to [0,\infty)$, then we have:*

$$\forall \varepsilon > 0 \textbf{ \textit{with} } f(\varepsilon) > 0 : \mathbb{P}(|X| \geq \varepsilon) \leq \frac{\mathbb{E}f(|X|)}{f(\varepsilon)}$$

Proof.

$$\begin{aligned}
\mathbb{E}[f(X)] &\geq \mathbb{E}[f(X)\mathbb{1}_{f(X)\geq f(\varepsilon)}] \\
&\geq \mathbb{E}[f(\varepsilon)\mathbb{1}_{f(X)\geq f(\varepsilon)}] \\
&= f(\varepsilon)\mathbb{P}(f(X) \geq f(\varepsilon)) \\
(1) &\leq f(\varepsilon)\mathbb{P}(|X| \geq \varepsilon)
\end{aligned}$$

where in (1) we use that f is an increasing function. □

Proposition 2.2 (Fubini-Tonelli). *Given X a non-negative random variable, then:*

$$\mathbb{E}X = \int_{\mathbb{R}} \mathbb{P}(X \geq t)dt$$

Proof. Observe:

$$\begin{aligned}\int_0^\infty \mathbb{P}(X(\omega) \geq x)dx &= \int_{\mathbb{R}^+}\int_\Omega \mathbb{1}_{X(\omega)\geq x} d\omega dx \\ &= \int_\Omega \int_0^{X(\omega)} dx d\omega \\ &= \int_\Omega X(\omega) d\omega \\ &= \mathbb{E}X\end{aligned}$$

□

3 Terminal σ-algebra and independence

This chapter we refer mainly to **Klenke**

3.1 Borel-Cantelli

Definition 3.1 (Limes superior and limes inferior). Given $(A_n)_{n\geq 1}$ a sequence of subsets in X. We define:

$$\limsup_n A_n := \{x \in X : x \in A_n \textbf{ for infinitely many } A_n\}$$

`(sequence is decreasing)`

$$\liminf_n A_n := \{x \in X : \exists n_0(x) \in \mathbb{N} \forall n \geq n_0(x) : x \in A_n(\textbf{ x is in all but finitely many } A_n)\}$$

`(sequence is increasing)`

equivalently we see:

$$\limsup_n A_n = \cap_{n=1}^\infty \cup_{k=n}^\infty A_n$$
$$\liminf_n A_n = \cup_{n=1}^\infty \cap_{k=n}^\infty A_n$$

We define the convergence of sets as :

$$\limsup_n A_n = \liminf_n A_n := \lim_n A_n$$

Theorem 3.1 (The first Borel-Cantelli Lemma). *If* $\sum_{n=1}^\infty \mathbb{P}(A_n) < \infty$*:*

$$\mathbb{P}(A_n \quad i.o.) = 0$$

Proof.

$$\begin{aligned}\sum \mathbb{P}(A_n) < \infty &\Rightarrow \lim_n \mathbb{P}(A_n) = 0 \\ &\Rightarrow \mathbb{P}(\lim_n A_n) = 0 \\ &\Rightarrow \mathbb{P}(\limsup A_n) = 0 \\ &\Rightarrow \mathbb{P}(A_n \quad i.o.) = 0\end{aligned}$$

□

Theorem 3.2 (The second Borel-Cantelli Lemma). *A_n independent and $\sum P(A_n) = \infty$, then*

$$\mathbb{P}(A_n \quad i.o.) = 1$$

Theorem 3.3. *$X_1, X_2, \ldots$ i.i.d. with $\mathbb{E}X_i = \infty$, then*

$$\mathbb{P}(|X_n| \geq n, i.o.) = 1$$

So if $S_n = X_1 + \cdots + X_n$ then:

$$\mathbb{P}(\lim \frac{S_n}{n} \in (-\infty, \infty) \quad i.o.) = 0$$

3.2 Kolmogorov 0-1 Law

Definition 3.2 (tail σ-algebra). Given a set of σ-field $(\mathfrak{A}_i)_{i \in I}$. The **tail σ-field** is defined as:

$$\mathfrak{T}\Big((\mathfrak{A}_i)_{i \in I}\Big) := \bigcap_{J < \infty, J \subset I} \sigma(\bigcup_{j \in I/J} (\mathcal{A}_j))$$

Remark. Related to the limes superior of the sets.

Theorem 3.4 (Kolomogorov 0-1 Law). *I countable index set, $(\mathcal{A}_i)_{i \in I}$ indepedant σ-field, than*

$$\forall A \in \mathfrak{T}\Big((\mathfrak{A}_i)_{i \in I}\Big) : \mathbb{P}(A) \in \{0, 1\}$$

Proof. □

Theorem 3.5 (Hewitt-Savage 0-1 Law). *Define $\mathcal{E}$ the σ-field of all the event stable under finite permutation(prove it's indeed a σ-field). Given $X_1, \ldots, X_n, \ldots$ i.i.d. random variable, if $A \in \mathcal{R}$, then $\mathbb{P}(A) \in \{0, 1\}$.*

3.3 Independence, Distribution and Expectation

Review 1 (central limits theory). The regularised mean of a sequence of random variable convergences in distribution to the standard normal distribution.(under certain condition)

$$Z := \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu) \xrightarrow{d} \mathcal{N}(0, 1) \tag{I.1}$$

Review 2 (Slutsky lemma). If a sequence of random variable convergences in probability to a constant while another convergences in distribution to a random variable, then the linearity holds (the linear combination convergences in distribution).

Review 3 (Portmanteau theory).

4 Convergence Concept of Random Variables

Definition 4.1 (convergence in probability).

Definition 4.2 (convergence in distribution).

Definition 4.3 (convergence in L^p).

5 Law of Large Numbers

Theorem 5.1 (strong law of large numbers). *Let $(X_n)_{n\geq 1}$ a sequence of i.i.d. random variables such that $X_j \in L^2(\Omega)$ with*

$$\mathbb{E}X_j = \mu, \quad \mathrm{Var}X_j = \sigma^2 \qquad \forall j = 1,\dots,n,\dots$$

Define the running sum $S_n := \sum_{j=1}^n X_j$, then it holds

$$\frac{S_n}{n} \xrightarrow{a.s} \mu, \quad \texttt{and} \quad \frac{S_n}{n} \xrightarrow{L^2} \mu.$$

Theorem 5.2 (L^2 weak law). *$X_1, X_2, \dots$ uncorrelated with $\mathbb{E}X_i = \mu$, and $X_i \in L^2$, then:*

$$\frac{S_n}{n} \xrightarrow{L^2} \mu, \qquad \frac{S_n}{n} \xrightarrow{P} \mu.$$

Lemma 5.3. *If $Y \geq 0$ and $p > 0$, then:*

$$\mathbb{E}Y^p = \int_0^\infty p y^{p-1} P(Y > y) dy$$

6 Central Limit Theory

6.1 Characteristic functions

Definition 6.1 (characteristic function). Given a random variable $X \in L^1(\mathbb{P})$. The characteristic function is the Fourier transform of the measurable function X:

$$\varphi(u) : \mathbb{R}^n \to C_b^0(\mathbb{R}^n), \quad u \rightarrowtail \mathbb{E}e^{iu\cdot x}$$

Example (The normal distribution).

Theorem 6.1 (The inversion formula). *Let $\varphi(t)$ be the characteristic function of the probability measure μ of the random variable X. If $a < b$, then:*

$$\lim_{T\to\infty} (2\pi)^{-1} \int_T^T \frac{e^{-ita} - e^{-itb}}{it} \varphi(t) dt = \mu(a,b) + \frac{1}{2}\mu(\{a,b\})$$

Theorem 6.2. *If $\int |\varphi(t)| dt < \infty$, then μ has the bounded continuous density:*

$$f(y) = \frac{1}{2\pi} \int e^{-ity} \varphi(t) dt$$

Theorem 6.3 (Continuity theorem of Levy). *Let μ_n, $1 \leq n \leq \infty$ be probability measures with ch.f. φ_n, then:*

1. $\mu_n \to \mu_\infty \quad \Rightarrow \quad \varphi_n(t) \to \varphi_\infty(t) \quad \forall t$
2. *$\varphi_n(t) \to \varphi(t)$ for some φ and φ is continuous at 0, then the associated sequence of distribution μ_n is tight and converges weakly to the measure μ with characteristic function φ.*

Definition 6.2. Let X be a $\mathbb{R}^n$ random variable, $kin\mathbb{R}^n$, then we define the characteristic function of X as:

$$\varphi^X(k) : k \to \int_{\mathbb{R}^n} e^{ik\cdot x} \mathbb{P}^X(dx)$$

Theorem 6.4 (uniqueness theorem).

Theorem 6.5 (central limit theorem I). *Given $X_1, \dots, X_n, \dots$ sequence of i.i.d. samples such that $X_i \in L^2(\Omega)$ with $\mathbb{E}X_i = \mu, \mathrm{Var}X_i = \sigma^2$. Define the running sum $S_n := \sum_{i=1}^n X_i$ and $Y_n := \frac{S_n - n\mu}{\sqrt{n}\sigma}$. Then it holds*

$$Y_n \xrightarrow{d} N(0,1).$$

Theorem 6.6 (central limit theorem II). *Given $X_1, \dots, X_n, \dots$ sequence of independent (but not necessarily identical) distributed samples such that $X_i \in L^2(\Omega)$ with $\mathbb{E}X_i = 0, \mathrm{Var}X_i = \sigma_j^2$. Define the running sum $S_n := \sum_{i=1}^n X_i$. If*

$$\sup_j \mathbb{E}|X_j|^{2+\varepsilon} < \infty, \quad \texttt{for some } \varepsilon > 0$$
$$\sum_{j=1}^{\infty} \sigma_j^2 = \infty$$

then it holds

$$\left(\sqrt{\sum_{i=j}^{n} \sigma_j^2}\right)^{-1} S_n \xrightarrow{d} N(0,1).$$

7 The Gaussian Random variable

8 Weak Convergence

Definition 8.1 (Separating Family). Given $F \subset M(E)$ a subset of Radon measure(inner regular Borel-measure). A family of measurable function C is the separating family for F, if $\forall \mu, \nu \in F$

$$\int f d\mu = \int f d\nu \quad \forall f \in C \cap \mathcal{L}^1(\mu) \cap \mathcal{L}^1(\nu) \quad \Rightarrow \quad \mu = \nu$$

Definition 8.2 (weak convergence of measure).

Definition 8.3. A sequence of random variables converge weakly or converges in distribution if its distribution function converges for all the continuity point of F.

Theorem 8.1. *If $F_n \to F_\infty$*

Theorem 8.2. *Every probability measure on a Polish space is tight.*

Theorem 8.3. *Every probability measure on a Polish space is regular*

Theorem 8.4 (Continuous Mapping theorem). *If $g : S \to T$ is continuous, X, T metric space then:*

$$X_n \xrightarrow{d} X \Rightarrow g(X_n) \xrightarrow{d} g(X)$$

Proof. Notice the fact that $\forall f \in C_b(X, \mathbb{R}$, $f \circ g$ is also bounded. □

Remark. If $X_n \overset{a.s.}{\Rightarrow} X$, then $g(X_n) \overset{a.s.}{\Rightarrow} g(X)$.

Theorem 8.5. *Convergence in distribution implies convergence of the cdf in any **continuous** point.*

Chapter II

Discrete Stochastic Process

1 The Conditional Expectation

The motivation of regular conditional expectation(probability) is that the classical definition of conditional probability $\frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A)}$ fails if $\mathbb{P}(A)=0$, which could be solved by defining the regular conditional probability.

First we observe the regular conditional probability from the functional analysis point of view.

1.1 Construction and Properties

Given a sub-σ-field $\mathcal{G}\subset\mathcal{F}$, we obtain the functional space $L^2(X,\mathcal{G},\mathbb{P})\subset L^2(X,\mathcal{F},\mathbb{P})$. It's easy to verify the subspace is closed and convex. Given a $X\in L^2(X,\mathcal{F})$, by the approximation theorem there exists a $Y\in L^2(X,\mathcal{G})$, such that the distance $Y=\min_{Z\in L^2(X,\mathcal{G})}\mathbb{E}|X-Z|^2$. Therefore we can define the conditional expectation in L^2 space.

Definition 1.1 (Conditional Expectation (L^2 version)). $X\in L^2(\Omega,\mathcal{F},\mathbb{P})$, $Y\in L^2(\Omega,\mathcal{F},\mathbb{P})$. The conditional expectation of X given Y: $\mathbb{E}[X|Y]$, is defined as the orthogonal projection of X on the functional space $L^2(X,\sigma(X),\mathbb{P})$.

The central result of this chapter is to extend the definition of conditional expectation in L^2 space to the L^1 space. The usual trick is to truncate the L^1 random variable and apply monotone convergence theorem.

Exercise 1.1.1. *Point out the error in this argument:*
$\sigma(Z_1=\mathbb{1}_A)=\sigma(Z_2=\mathbb{1}_{A^C})$
$\Rightarrow$ *the 2 random variables define the same L^2 subspace.*
$\Rightarrow$ *the conditional expectation of the 2 RVs are the same.*

Proof. The error lies in the first induction. notice the L^2 subspace is characterised by the measure of Z. □

Lemma 1.1 (L^2 properties of conditional expectation).

Theorem 1.2 (Existence and Uniqueness of the general conditional expectation). *For $Y\in\mathfrak{M}^+(\Omega,\mathcal{F})$ or $Y\in L^1(\Omega,\mathcal{F},\mathbb{P})$. There exists an unique $\mathbb{E}[Y|\mathcal{G}]$:*

$$\forall G\in\mathcal{G}:\quad \mathbb{E}[\mathbb{E}[Y|\mathcal{G}]\mathbb{1}_G]=\mathbb{E}[Y\mathbb{1}_G]$$

Proof. 1. For $Y\in\mathfrak{M}^+$: Define $Y_n:Y\wedge n$, $n\in\mathbb{N}$, notice $Y^n\uparrow$ monotone sequence of random variable. Then

$$\mathbb{E}[\mathbb{E}[Y_n|\mathcal{G}]\mathbb{1}_G]=\mathbb{E}[Y_n\mathbb{1}_G]$$

taking the limit:

$$\mathbb{E}[Y\mathbb{1}_G]=\mathbb{E}[\lim_{n\to\infty}Y_n\mathbb{1}_G]=\mathbb{E}[\lim_{n\to\infty}\mathbb{E}[Y_n|\mathcal{F}]\mathbb{1}_G]$$

By $\mathcal{G}$ measurability of Y we ensure the existence of $\mathbb{E}[Y\mathbb{1}_G]$ therefore the existence of $\lim_{n\to}\mathbb{E}[Y_n|\mathcal{G}]$.
2. For $Y\in L^1(X,\mathcal{F})$, decompose Y into Y^+ and Y^- □

At this stage we are able to define the L^1 version of the regular conditional expectation:

Definition 1.2 (Conditional expectation(L^1)). Given $(\Omega,\mathcal{F},\mathbb{P})$, X non-negative or $\in L^1$, $\mathcal{G}\subset\mathcal{F}$ a sub σ-field, we define Y a version of conditional expectation of X given $\mathcal{G}$ if:

- $\forall G\in\mathcal{G}:X\mathbb{E}\mathbb{1}_G=\mathbb{E}Y\mathbb{1}_G$

- Y is $\mathcal{G}$-measurable.

There are many properties of conditional expectation, the most frequently used technique is to apply the definition of conditional expectation and choose the suitable test sets.

Theorem 1.3 (Properties of the conditional expectation). *Let $Y \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G}$ a sub-σ-Algebra. We have:*

1. ***Tower Properties:*** $\mathcal{H} \subset \mathcal{G} \Rightarrow \mathbb{E}[\mathbb{E}[Y|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[Y|\mathcal{H}]$

2. ***Constantization of RVs.*** *Z is $\mathcal{G}$-measurable, $ZY \in L^1$ a.s. $\Rightarrow \mathbb{E}[ZY|G] = Z\mathbb{E}[Y|G]$*

1. $\mathbb{E}[\mathbb{E}[Y|\mathcal{G}]] = \mathbb{E}Y$

2. *Y is $\mathcal{G}$-measurable, $\mathbb{E}[Y|\mathcal{G}] = Y$ a.s.*

3. *If $Y \perp \mathcal{G}$, then $\mathbb{E}[Y|\mathcal{G}] = \mathbb{E}Y$ a.s.*

4. *(MCT) $Y_n \in M^+, Y_n \uparrow Y$ a.s $\Rightarrow \mathbb{E}[Y_n|\mathcal{G}] \uparrow \mathbb{E}[Y|\mathcal{G}]$ a.s.*

5. *(Fatou) $Y_n \in M^+ \Rightarrow \mathbb{E}[\liminf_n Y_n|\mathcal{G}] \leq \liminf_n \mathbb{E}[Y_n|\mathcal{G}]$ a.s.*

6. *(Jensen) φ convex, $\mathbb{E}[\varphi(Y)|\mathcal{G}] \geq \varphi(\mathbb{E}[Y|\mathcal{G}])$*

Let $Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G}$ a sub-σ-Algebra. We have:

1. $\mathbb{E}[\mathbb{E}[Y|\sigma(Z)]Z] = \mathbb{E}[YZ]$

Theorem 1.4. *Let X, Y independent random variable on $(\Omega, \mathfrak{F}, \mathbb{P})$(taking value in respective measurable space). Given a L^1 or non-negative function g of X, Y, we have:*

$$\mathbb{E}[g(X,Y)|\sigma(Y)] = \mathbb{E}[g(X,y)]|_{y=Y(\omega)}$$

In a word, if X and Y are independent, we may treat either one as constant when we compute the expectation.

Proof.

$$\begin{aligned}\mathbb{E}[g(X,Y)\mathbb{1}_{B(Y)}] &= \int_{E_X \times E_Y} g(x,y)\mathbb{1}_B(y)\mathbb{P}_{(X,Y)}(dx,dy) \\ &= \int_{E_X \times E_Y} g(x,y)\mathbb{1}_{B(Y)}\mathbb{P}_X(dx)\mathbb{P}_Y(dy)\end{aligned}$$

□

Example (Wald's identity). $(X_i)_{i\in\mathbb{N}}$ i.i.d.. N $\mathbb{N}$-valued random variable independent of X_i, we have:

$$\mathbb{E}S_N = \mathbb{E}N\mathbb{E}X_0$$

1.2 Regular Conditional Probability

Definition 1.3 (The stochastic kernal). Let $(\Omega_i, \mathfrak{F}_i)$, $i = 1,2$ be 2 measurable space. A mapping

$$\kappa : \Omega_1 \times \mathfrak{F}_2 \to [0,1]$$

is called a stochastic kernel from $(\Omega_1, \mathfrak{F}_1)$ to $(\Omega_2, \mathfrak{F}_2)$ if:

1. $\forall B_2 \in \mathfrak{F}_2 : \omega \to \kappa(\omega, B_2)$ is $\mathfrak{F}_1$-measurable.
2. $\forall \omega \in \Omega_1$: $B_2 \to \kappa(\omega, B_2)$ is a probability measure on $(\Omega_2, \mathfrak{F}_2)$.

Definition 1.4 (Regular Conditional Probability). Given a random variable $X : (\Omega, \mathfrak{F}) \to (E, \mathfrak{E})$, and a sub-$\sigma$-field $(\Omega, \mathfrak{G})$. A function $\mathbb{P}^X(\omega, B)$ is called a regular conditional probability if:

1. $\forall \omega \in \Omega$:

$$\mathbb{P}^X(\omega, B) : (E, \mathfrak{E}) \to ([0,1], \mathfrak{B}([0,1])$$

   is a probability measure.

2. $\mathbb{P}^X(;B)$ is a version of conditional expectation of $\mathbb{E}[\mathbb{1}_B|\mathfrak{G}]$ for any $B \in \mathfrak{E}$.

Theorem 1.5. *Let $\mathbb{P}^X(\omega, B)$ with $\omega \in \Omega, B \in B(\mathbb{R}))$ a regular conditional distribution, then for any measurable function f with $\mathbb{E}f < \infty$ we have:*

$$\mathbb{E}[f|\mathfrak{G}] = \int_x f(x)\mathbb{P}^X(\omega, dx)$$

Example (Markov chain). Given a canonical stochastic process $(X_n)_{n\in\mathbb{N}}$, $(E, \mathfrak{E})$ the state space, X is a Markov chain w.r.t. $\mathbb{P}$ and $\mathfrak{F}_n$ if:

$$\mathbb{P}(X_n \in B|\mathfrak{F}_m) = \mathbb{P}(X_n \in B|X_m) \quad \forall m \leq n, B \in \mathfrak{E}$$

Theorem 1.6. *If the random variable X is real-valued, i.e $(E, \mathfrak{E}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then there exists for every $\mathfrak{G} \subset \mathfrak{E}$ a regular conditional probability $\mathbb{P}^X(\omega, B)$.*

Proof. □

Example ((Durret 5.1.2) Bayes).

2 Martingale Theory

2.1 Introduction

Need to review: 2.8

2.2 Definition and Properties

Definition 2.1 (Martingale). Adapted not necessarily predictable.

1. $M_n \in L^1 \ \forall n \geq 0$
2. adapted
3. $\mathbb{E}[M_{n+1}|\mathcal{F}_n] = M_n$

Theorem 2.1 (Properties of martingale). *Given X a martingale, we have:*

1. $\mathbb{E}X_n = \mathbb{E}X_0 \ \forall n \in \mathbb{N}$
2. $\mathbb{E}[X_n|\mathcal{F}_m] = X_m \quad \forall m \leq n$
3. ***Martingale transformation:****Given H_n bounded* ***predictable*** *process, we have:*

$$Y_0 = 0, Y_n = \sum_{k=1}^{n} H_k(X_k - X_{k-1})$$

is a martingale

Remark. Notice **1.** does not hold for asymptotical behaviors. For $n \to \infty$, $\mathbb{E}X_\infty = \mathbb{E}X_0$ does not hold in general

Remark (the Martingale Transform). X_n a predictable process, M_n a martingale(with respect to the implied filtration). The martingale transform, or the **discrete stochastic integral**, could be defined as:

$$(X \bullet M)_0 = 0, \quad (X \bullet M)_n := \sum_{k=1}^{n} X_k(M_k - M_{k-1})$$

If X is bounded, then by the theorem above we know the martingale transform is again a martingale.

Proof. 1) and 2) are trivial by the tower property and expectation of the conditional expectation is the expectation.
3)i) $Y_n \in L^1$

$$Y_n = \sum_{i=1}^{n} H_i(X_i - X_{i-1}) \leq \|H_i\|_\infty \sum_{i=1}^{n}(X_i - X_{i-1}) = \|H_i\|_\infty (X_n - X_0)$$

ii) Y_n adapted.
iii) The Martingale equality:

$$\begin{aligned}
\mathbb{E}[Y_n|\mathcal{F}_{n-1}] &= \mathbb{E}[\sum_{i=1}^{n} H_i(X_i - X_{i-1})|\mathcal{F}_{n-1}] \\
&= \sum_{i=1}^{n} \mathbb{E}[H_i(X_i - X_{i-1})|\mathcal{F}_{n-1}] \\
&= \sum_{i=1}^{n-1} H_i(X_i - X_{i-1}) + \mathbb{E}[H_n(X_n - X_{n-1})|\mathcal{F}_{n-1}] \\
(1) &= \sum_{i=1}^{n-1} H_i(X_i - X_{i-1}) + H_n\mathbb{E}[X_n - X_{n-1}|\mathcal{F}_{n-1}] \\
&= Y_n
\end{aligned}$$

where in (1) the first part we used H is bounded and second H is predictable. □

Theorem 2.2 (Doob's decomposition). *Given $(X_n)_n$ a submartingale, there exists a a.s. unique decomposition of X_n:*

$$X_n = X_0 + M_n + A_n, \quad A_0 = M_0 = 0$$

*where A_n is a increasing **predictable** process and M_n is a martingale. And*

$$A_n := \sum_{k=1}^{n} \mathbb{E}[X_k - X_{k-1}|\mathcal{F}_{k-1}]$$

Remark. A predictable martingale is a.s. constant.

Proof. 1)Show A_n is a predictable process. (Also A_{n+1} a submartingale?)It's trivial to see that $\mathbb{E}[A_n|\mathcal{F}_{n-1}] = \mathbb{E}[X_n - X_{n-1}|\mathcal{F}_{n-1}] + A_{n-1}$. Notice the conditional expectation on the right-hand side is non-negative, therefore $A_{n-1} \leq \mathbb{E}[A_n|\mathcal{F}_{n-1}]$ a.s.

2)Show that $M_n := X_n - X_0 - A_n$ is a martingale:
i)adeptness and integrability is trivial.
ii) The martingale equality:

$$\begin{aligned}\mathbb{E}[M_n|\mathcal{F}_{n-1}] - M_{n-1} &= A_n - A_{n-1} + \mathbb{E}[X_n|\mathcal{F}_{n-1}] - X_{n-1} \\ &= \mathbb{E}[X_n - X_{n-1}|\mathcal{F}_{n-1}] - \mathbb{E}[X_n - X_{n-1}|\mathcal{F}_{n-1}] \\ &= 0\end{aligned}$$

□

Lemma 2.3. *M_n a martingale and $\varphi : \mathbb{R} \to \mathbb{R}$ a convex function with $\varphi \in L^1$, then $\varphi(M_n)$ is a submartingale. If φ is a increasing convex function and M_n a submartingale, then M_n is also a submaringale.*

Remark. In particular, $M_n^2 \in L^2$ is a submartingale.

Lemma 2.4. *$X_n \in L^p, M_n \in L^q$ with $\frac{1}{p} + \frac{1}{q} = 1$, $\Rightarrow (X \bullet M)_n$ is again a martingale.*

2.3 Stopping Times

Definition 2.2 (Stopping time). Stopping time is a random variable w.r.t. the filtration $(\mathcal{F}_t)_{t\geq 0}$ if

$$\{\tau = n\} \in \mathcal{F}_n \quad \forall n \geq 0$$

Lemma 2.5. *τ is a stopping time $\Leftrightarrow \forall n \in \mathbb{N} : \{\tau \leq n\} \subset \mathcal{F}_n$*

Proof.
$\Rightarrow$: $\{\tau \leq n\} = \cup_{i=1}^{n} \{\tau = i\}$
$\Leftarrow$: $\{\tau = n\} = \{\tau \leq n\}/\{\tau \leq n-1\}$ both of the latter is $\mathcal{F}_n$ measurable. □

Remark. We define the following notation:
$\mathbf{X}_\tau$: The random variable at time τ.
$\mathbf{X}^\tau$: The stopped process $X^\tau := X_{n\wedge\tau}$

Lemma 2.6. *Given X_τ and X^τ defined as above:*

- *X_τ is well-defined if τ is **finite***
- *X^τ is always well-defined*

Proof.
1) For $B \in \mathcal{B}(\mathbb{R})$,

$$\{X_\tau \in B\} = \bigcup_{n=1}^{\infty} \Big\{ \{X_n \in B\} \cap \{\tau = n\} \Big\}$$

since τ is finite, we only have finite intersection, therefore the set is still in the σ-field $\mathcal{F}$.
2) is trivial since we know by $\tau \wedge n$ that the process is bounded. □

Theorem 2.7 (Doob's Optional stopping). *(M_n) (sub)martingale, τ stopping time. Then the stopped process $(M_\tau^n) := (M_{\tau\wedge n})$ is again (sub)martingale.*

Proof. w.l.o.g show $M_{\tau\wedge n} - M_0$ is martingale. Define $C_k := \mathbb{1}_{\tau(\omega)\geq k}$ notice C_k is $\mathcal{F}_{k-1}$ measurable therefore predictable

$$\begin{aligned} M_{\tau\wedge n} - M_0 &= \sum_{k=0}^{\tau\wedge n} M_k - M_{k-1} \\ &= \sum_{k=0}^{n} C_k M_k - M_{k-1} \end{aligned}$$

By the property of martingale we know the stopped process is again a martingale. □

Remark. $\tau \wedge n$ comes to play to apply the martingale transformation. And

$$\mathbb{E}[X_{\tau\wedge n} | \mathfrak{F}_k] = X_{\tau\wedge k}$$

Example ($\mathbb{E}X_\tau \neq \mathbb{E}X_0$ even for finite τ, therefore not a martingale). We construct an example where $\mathbb{E}X_\tau \neq \mathbb{E}X_0$ for a martingale X_n in order to show that in general X_τ is not a martingale. (X_n) be i.i.d random variable with $\mathbb{E}X_n = 0$, $VarX_n < \infty$. Notice S_n is a martingale with its natural filtration. Observe the tail σ-field:

$$\mathcal{T} := \bigcap_{k\in\mathbb{N}} \mathcal{F}_{n\geq k}$$

Observe the event:

$$C := \{\limsup S_n = \infty\} \in \mathcal{T}$$

By Kolmogorov $\mathbb{P}(C) \in \{0,1\}$. Furthermore, by central limit theorem, S_n is $\log n$ normal:

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} \mathcal{N}(\mu, Var[X_1])$$

Select any $a \in \mathbb{R}$

$$\mathbb{P}(\lim_{n\to\infty}(S_n > \sqrt{n}\cdot a)) = 1 - \Phi(\frac{a}{\sqrt{VarX_1}})$$

(Convergence in distribution implies convergence of cdf in continuous point.) Notice it's strictly positive, by Kolmogrov:

$$\mathbb{P}(\limsup_{n\to\infty} S_n = \infty) = 1$$

Define $\tau = \inf\{n : S_n = 10\}$, by the argument above $\mathbb{P}(\tau < \infty) = 1$. Then:

$$\mathbb{E}S_\tau = 10 \neq \mathbb{E}S_0$$

Remark (Finiteness and Boundedness). Finiteness doesn't come up in the context of measurable function. We have shown that even for the bounded stopping time, the stopped process may not be a martingale. The fundamental reason is that **A finite stopped martingale may may not be bounded, or u.i.**

Corollary 2.7.1.

1. *)Let τ be a stopping time, if X is a martingale, and τ is **bounded**, then $\mathbb{E}X_\tau = \mathbb{E}X_0$.*
2. *If X is a L^1 bounded martingale($X_n < \infty$), then $\mathbb{E}X_\tau = \mathbb{E}X_0$.*
3. *If X is a non-negative supermartingale, then $\mathbb{E}X_\tau \leq \mathbb{E}X_0$*

Proof.
1)

$$\mathbb{E}X_\tau = \mathbb{E}X_{\tau\wedge c} = \mathbb{E}X_c^\tau = \mathbb{E}X_0$$

2)we have $\tau \wedge n \overset{a.s.}{\to} \tau$. Then we can apply DCT

$$\mathbb{E}X_0 = \mathbb{E}X_{\tau\wedge n} \overset{a.s}{\to} \mathbb{E}X_\tau$$

3)Notice the random variable is non-negative. We can apply Fatou

$$\mathbb{E}X_\tau = \mathbb{E}\liminf_{n\to\infty} X_{\tau\wedge n} \leq \liminf_{n\to\infty} \mathbb{E}X_{\tau\wedge n} \leq \mathbb{E}X_0$$

where the last inequality follows from the property of supermartingale □

Remark. For the supermartingale to preserve the inequality the stopping time does not to be bounded, since we apply Fatou here.

Theorem 2.8. *Let X be $(\mathcal{F}_n)$ adapted process, integrable. If:*

$$\mathbb{E}X_\tau = \mathbb{E}X_0 \quad \forall\tau \quad \textbf{\textit{bounded}}$$

then X is a martingale.

Remark. It's not enough to take all deterministic times $\mathbb{E}X_n = \mathbb{E}X_0$

Proof. The basic idea is, for any $A \in \mathfrak{F}_n$ we define a stopping time and show the martingale property:

$$\tau = \begin{cases} n, & \omega \in A^C \\ n+1 & \omega \in A \\ 0 & else \end{cases}$$

τ is bounded and notice $\{\tau \leq n\} = A^C \in \mathfrak{F}_n$. We have

$$\mathbb{E}[X_n \mathbb{1}_A + X_n \mathbb{1}_{A^C}] = \mathbb{E}X_n \overset{deter.st.}{=} \mathbb{E}X_0 \overset{\tau<\infty}{=} \mathbb{E}X_\tau = \mathbb{E}[X_n \mathbb{1}_{A^C} + X_{n+1}\mathbb{1}_A]$$

We have

$$\mathbb{E}\mathbb{1}_A X_n = \mathbb{E}\mathbb{1}_A X_{n+1}$$

□

Definition 2.3 (Stopped Filtration). Let τ be a $\mathcal{F}_n$ stopping time. We define

$$\mathcal{F}_\tau := \left\{ A \cap \{\tau \leq n\} | \quad A \in \mathcal{F} \quad \forall n \in \mathbb{N}_0 \right\}$$

Lemma 2.9. *Let τ be a stopping time*

- *$\mathcal{F}_\tau$ is a σ-field*
- *If $\tau \equiv n$, then $\mathcal{F}_\tau = \mathcal{F}_n$*
- *If τ is **finite** and X adatpted, then X_τ is $\mathcal{F}_\tau$-measurable.*

Theorem 2.10 (Doob's Optional Sampling). *Let (M_n) be a (sub)martingale and σ, τ bounded stopping times with $\sigma \leq \tau$. Then*

$$\mathbb{E}[M_\tau | \mathcal{F}_\sigma] = M_\sigma \quad (\mathbb{E}[M_\tau | \mathcal{F}_\sigma] \geq M_\sigma)$$

Proof. Define $\rho = \mathbf{1}_A \sigma + \mathbf{1}_{A^C} \tau$. By optional stopping we have:

$$\mathbb{E}M_\rho = \mathbb{E}M_0 = \mathbb{E}M_\tau \quad (*)$$

We observe:

$$\{\rho = n\} = \Big(\{\sigma = n\} \cap A \Big) \bigcap \Big(\{\tau = n) \cap A^C \Big) \in \mathcal{F}_n$$

Therefore ρ defines a stopping time. By $(*)$

$$\mathbb{E}[\mathbb{1}_A M_\sigma + \mathbb{1}_{A^C} M_\tau)] = \mathbb{E}M_\rho = \mathbb{E}M_\tau = \mathbb{E}[\mathbb{1}_A M_\tau + \mathbb{1}_{A^C} M_\tau]$$

Consequently: $\forall A \in \mathcal{F}_\sigma$:

$$\mathbb{E}[\mathbb{1}_A M_\tau] = \mathbb{E}[\mathbb{1}_A M_\sigma]$$

□

Closed Martingale

Now we turn to the question: what if the stopping time is unbounded. We know:

$$X \in L^1 \Rightarrow \mathbb{E}[X | \mathcal{F}_n] \text{ is a martingale}$$

The question now is to find out a way to characterise this kind of stochastic process.

Definition 2.4 (closed martingale). A martingale $(X)_n$ is closed if: $\exists Y \in L^1 \forall n \geq 0 : \mathbb{E}[Y|\mathcal{F}_n] = X_n$. If the martingale is closed by Y, we may define

$$\mathcal{F}_\infty := \sigma(\bigcup_n \mathcal{F}_n)$$

The smallest filtration contains all $\mathcal{F}_n$. And

$$X_\infty := \mathbb{E}[Y|F_\infty]$$

X_∞ is a $\mathcal{F}_\infty$ measurable random variable. Notice that the X_∞ satisfies all the martingale properties, therefore

$$(X_n)_{n \in \mathbb{N} \cup \{\infty\}}$$

is a martingale as well.

With the introduction of closable martingale, we may generalise the result of optional sampling:

Theorem 2.11 (The second optional sampling). *Let (X_n) be a martingale **closed** by a random variable Y, and $\sigma \leq \tau$ **any** stopping time, then:*

$$\mathbb{E}[X_\tau|F_\sigma] = X_\sigma$$

Proof. We would like to prove:

$$\forall A \in \mathcal{F}_\sigma : \mathbb{E}[X_\tau \mathbb{1}_A] = \mathbb{E}[X_\sigma \mathbb{A}]$$

Observe:

$$\begin{aligned}
\mathbb{E}[X_\tau \mathbb{1}_A] &= \mathbb{E}[\sum_{n=1}^{\infty} X_\tau \mathbb{1}_{A \cap \{\tau = n\}}] \\
&= \mathbb{E}[\sum_{n=1}^{\infty} X_n \mathbb{1}_{A \cap \{\tau = n\}}] \\
&= \mathbb{E}[\sum_{n=1}^{\infty} X_\infty \mathbb{1}_{A \cap \{\tau = n\}}] \\
(Fubini) &= \mathbb{E}[X_\infty \sum_{n=1}^{\infty} \mathbb{1}_{A \cap \{\tau = n\}}] \\
&= \mathbb{E}[X_\infty \mathbb{1}_A]
\end{aligned}$$

Analog we can show

$$\mathbb{E}[\mathbb{1}_A X_\tau] = \mathbb{E}[\mathbb{1}_A X_\infty]$$

□

Example (Ruin time of the insurance company). We define: V_n: capital of time n, $c > 0$ premium collected at time, Y_n payment in year n, which are i.i.d. We would require $\mathbb{E}Y_n < c$, and we have the capital dynamic:

$$V_n = V_{n-1} + c - Y_n$$

We define the stopping time: $T := \inf\{n \in \mathbb{N}_0 | V_n < 0\}$, the time of bankruptcy. We are interested in the upper bound of $\mathbb{P}(T < \infty)$. Assume we have λ_0 as the solution of:

$$\mathbb{E}e^{\lambda Y_1} = e^{\lambda c}$$

2.4 Martingale Inequalities

Question: When can we close a martingale? X_∞ some limit of X_n. Somehow we would like to bound the running maximum to get closeness and convergence of the martingale. We define the **Running Maximum till time** n of a martingale $(X_n)_{n\in\mathbb{N}}$ as

$$X_n^* := \sup_{i\in\{1,\cdots,n\}} |X_i|$$

It's worth noticing that $X_n^* \in L^1$ since it's less than the finite sum of the integrable random variable. Using the Markov inequality we would get

$$\mathbb{P}(X_n^* \geq \alpha) \leq \frac{\mathbb{E}X_n^*}{\alpha}$$

We prove we can further improve this upper bound.

Theorem 2.12 (Doob's martingale Inequality). *X is a martingale, or a positive submartingale. then we have:*

$$\mathbb{P}(X_n^* \geq \alpha) \leq \frac{\mathbb{E}[|X_n|\mathbb{1}_{X_n^*\geq\alpha}]}{\alpha} \leq \frac{\mathbb{E}|X_n|}{\alpha}$$

In particular,

$$\mathbb{P}(X_\infty^* \geq \alpha) \leq \sup_{n\in\mathbb{N}_0} \frac{\mathbb{E}|X_n|}{\alpha}, \qquad X_\infty^* := \sup_n |X_n|$$

Proof. We define the stopping time:

$$\tau := \inf\{k : X_k \geq \alpha\}$$

Then the tail probability may be written as:

$$\begin{aligned}
\mathbb{P}(X_n^* \geq \alpha) &= \mathbb{E}\mathbb{1}_{\tau\leq n} \\
(1) &\leq \mathbb{E}[\frac{|X_{\tau\wedge n}|}{\alpha}\mathbb{1}_{\tau\leq n}] \\
&\leq \frac{(\mathbb{E}|X_n|\mathbb{1}_{X_n^*\geq\alpha})}{\alpha} \\
(2) &\leq \frac{\mathbb{E}|X_n|}{\alpha}
\end{aligned}$$

In (1) by $\tau \leq n$ we know $\max_{k=1}^n X_k \geq a$, therefore $\frac{X_{\tau\wedge n}}{\alpha} \geq 1$. (2) follows from $|X_n|$ is a submartingale and optional stopping. □

Example (Random walk). For $S_n = \sum_{k=0} X_k$, $\mathbb{E}X_k = 0$, $Var(X_k) = \sigma^1$. We have

$$\mathbb{P}(S_n^* \geq a) \leq a^{-2}\sigma^2$$

The results follows from S_n^2 is a positive sub-martingale, we then apply the theorem above to S_n^2.

Theorem 2.13 (Doob's L^p martingale inequality). *Given $X \in L^p$, $p \in (1,\infty)$.*

$$\mathbb{E}|X_n^*|^p \leq \frac{p}{1-p}\mathbb{E}|X_n|^p$$

Proof.

$$\begin{aligned}
\mathbb{E}|X_n^*|^p &= \mathbb{E}(\int_0^{X_n^*} px^{p-1}dx) \\
&= \int_0^\infty px^{p-1}\mathbb{P}(X_n^* \geq x)dx \\
(1) &\leq \int_0^\infty px^{p-2}\mathbb{E}\left[|X_n|\mathbb{1}_{(X_n^* \geq x)}\right] \\
&= \mathbb{E}\left[|X_n|\int_0^{X_n^*} px^{p-2}dx\right] \\
&= \frac{p}{p-1}\mathbb{E}[|X_n^*|^{p-1}|X_n|]
\end{aligned}$$

Apply Holder,

$$\begin{aligned}
\mathbb{E}|X_n^*|^p &\leq \frac{p}{p-1}(\mathbb{E}|X_n^*|^{pq-q})^{\frac{1}{q}}\|X_n\|_p \\
&= \frac{p}{p-1}\|X_n^*\|_p^{\frac{p}{q}}\|X_n\|_p \\
\|X_n^*\|_p^{p-\frac{p}{q}} &\leq \frac{p}{p-1}|X_n\|_p \\
\|X_n^*\|_p &\leq \frac{p}{p-1}\|X_n\|_p
\end{aligned}$$

where (1) follows from the martingale inequality □

Example (The L^p inequality fails when $p = 1$). Define $X_n := S_{\tau\wedge n}$ where S_n is a simple random walk and $\tau := \inf\{n \in \mathbb{N} : S_n = 0\}$, $S_0 = 1$. By the Doob's inequality:

$$\mathbb{P}(X_n^* \geq M) \leq \frac{\mathbb{E}X_n}{M} = \frac{1}{M}$$

Therefore

$$\mathbb{E}X_n^* = \sum_{M=1}^\infty \mathbb{P}(X_n^* \geq M) = \sum_{M=1}^\infty \frac{1}{M} = \infty$$

However

$$\mathbb{E}|X_n| = 1$$

Theorem 2.14. *If X_n is a martingale with $\sup\mathbb{E}|X_n|^p < \infty$, $p > 1$, then $X_n \to X$ a.s. and in L^p.*

Proof. □

Doob's upcrossing lemma

motivation: Convergence of a random variable X_n is equivalent to the $\liminf X_n = \limsup X_n$, which is in turn equivalent to, for any interval, there are only finite X_n lying outside the interval($\liminf$ cannot exceeds the $\limsup$), which is equivalent to there are only finite many **upcrossings**. That's the reason for the following **Doob's upcrossing theorem**. We denote for a random process X, $a < b$:

$T_0 = 0$,

$B_{j+1} := \min\{k \geq T_j : X_k \leq a\}$

$S_{j+1} := \min\{k \geq S_j : X_k \geq b\}$

$U_n := \max\{j \geq 0 : T_j \leq n\}$ number of upcrossings by time n.

Theorem 2.15 (Doob's upcrossing theorem). *X a submartingale, and $a < b$, then*

$$\mathbb{E}U_n(a,b) \leq \frac{1}{b-a}\mathbb{E}(X_n - a)^+$$

Proof. 1. By induction we know S_j and T_j define stopping times.

$$\{S_j = n\} = \{B_j < n\} \bigcap \left\{ \bigcap_{T_j \leq k \leq n} \{X_k < b\} \right\} \bigcap \{X_n \geq b\}$$

Inductively if $\{B_j \leq n\}$ is $\mathcal{F}_n$-measurable, we know $\{S_j = n\}$ is $\mathcal{F}_n$-measurable.
2. Notice $U_n(a,b)(b-a) \leq$ (process of buying at $X \leq a$ then selling at $X \geq b$). With this intuition we define the predictable process:

$$H_m = \begin{cases} 1 & B_j < m \leq S_{j+1} \\ 0 & else \end{cases}$$

Notice it's a predictable process.
3. Define $Y_n := (X_n - a)^+ + a$, the max function preserves the submartingale property:

$$\mathbb{E}[(X_n - a)^+ + a | \mathcal{F}_{n-1}] \geq (\mathbb{E}[X_n | \mathcal{F}_{n-1}] - a)^+ + a \geq (X_{n-1} - a)^+ + a$$

By the argument above we have

$$\mathbb{E}(H \cdot X)_n \geq (b-a)U_n(a,b)$$

On the other hand, define $K := 1 - H$, by construction we have:

$$X_n - X_0 = (H \cdot X)_n + (K \cdot Y)_n$$

since Y_n is a submartingale, we have $\mathbb{E}[(X \cdot K)_n] \geq \mathbb{E}[(X \cdot K)_0] = 0$. Hence by the independence of X_n:

$$\mathbb{E}[X_n - Y_0] = \mathbb{E}[(X_n - a)^+] \geq \mathbb{E}(H \cdot X)_n \geq U_n(a,b)(b-a)$$

□

2.5 Martingale Convergence theorem

In certain sense, the (sub)martingale could be seen as a monotone increasing sequence. Purpose of this chapter is to show that, under certain boundedness conditions, the monotone sequence converges.(in different type).

Theorem 2.16 (The First Martingale convergence theorem). *X be a submartingale, s.t.* $\sup_n \mathbb{E}|X_n^+| < \infty$

$$\lim_{n\infty} X_n = X_\infty \quad \textit{exists a.s.} \quad X_\infty \in L^1$$

Proof. By monotone convergence theorem

$$U_n(a,b) \uparrow U(a,b)$$

Observe $(X-a)^+ \leq X^+ + |a|$, we have

$$U_n(a,b) \leq \frac{1}{b-a}\mathbb{E}(X_n - a)^+ \leq \frac{1}{b-a}(\mathbb{E}X_n^+ + |a|)$$

If $\sup_n \mathbb{E}X_n^+ < \infty$, we have

$$\mathbb{E}U(a,b) = \mathbb{E}\liminf_n U_n(a,b) \leq \liminf_n \mathbb{E}U_n(a,b) < \infty$$

For all $a,b \in \mathbb{Q}$, we only have finite number of upcrossing, therefore

$$\mathbb{P}(\bigcup_{a,b\in\mathbb{Q}} \{\liminf X_n \leq a \leq b \leq \limsup X_n\}) = 0$$

Therefore $\liminf X_n = \limsup X_n$ a.s. The limit exists. To see $X_\infty \in L^1$,

$$\mathbb{E}X^+ \leq \liminf \mathbb{E}X_n^+ < \infty$$

and

$$\mathbb{E}X_n^- = \mathbb{E}X_n^+ - \mathbb{E}X_n \leq \mathbb{E}X_n^+ - \mathbb{E}X_0$$

hence

$$\mathbb{E}X^- \leq \liminf_n \mathbb{E}X_n^- \leq \liminf_n \mathbb{E}X_n^+ - \mathbb{E}X_0 < \infty$$

□

Remark. The reason we only have the positive part in the assumption is: for a submartingale we have increasing expectation, therefore only the positive parts is of the interest.

Example (The L^1 convergence doesn't holds in general. And the existence of the L^1 limit doesn't apply the martingale is closed by it.). Given i.i.d. X_n with $\mathbb{P}(X_n = 0) = \mathbb{P}(X_n = 2) = \frac{1}{2}$. Observe

$$M_n = \prod_{k=1}^n X_k$$

we have

$$M_n \overset{a.s.}{\to} 0$$

However

$$\lim_{n\to\infty} \mathbb{E}|M_n - 0| = \lim_n \prod_{k=1}^n \mathbb{E}X_k = 1$$

Also the martingale is **not closed** by the limit, indeed

$$\mathbb{E}[0|\mathfrak{F}_n] \neq M_n$$

Corollary 2.16.1. *Give X a supermartingale or a martingale bounded from above or below,then*

$$X_\infty = \lim X_n$$

exists a.s and $X_\infty \in L^1$

The Second martingale convergence and the Uniform Integrability

Additional assumptions are required for the L^1-convergence. We at least need this for the closeness because we must have constant expectation.

Definition 2.5 (Uniformly integrable). A family of random variables $(X_i)_{i\in I}$ is uniformly integrable if

$$\lim_{R\to\infty}\sup_{i\in I}\mathbb{E}\left[|X_i|\mathbf{1}_{\{|X_i|>R\}}\right]=0$$

i.e. the value of the RV vanish on the edge of the measurable space.

Remark. It's a lot stronger than the L^1 boundedness for a family of random variable. After proving L^1 boundedness we still need $L^p(p>1)$ boundedness or vanishing value on shrinking measure to have u.i.

Example ($\mathbb{E}|X_n|<\infty \not\Rightarrow$ u.i). Observe:

$$X_i=2^i\mathbb{1}_{z\in[0,2^{-i}]}$$

For any $c\in\mathbb{R}$, we can find $2^i>c$, such that

$$\mathbb{E}[|X_i|\mathbb{1}_{X_i>c}]=\mathbb{E}|X_i|=1$$

therefore

$$\lim_{c\to\infty}\sup_{i\in\mathbb{N}}\mathbb{E}[|X_i|\mathbb{1}_{X_i\geq c}]\neq 0$$

Example (Closed martingale is u.i.). For $Z\in L^1(\Omega,\mathfrak{F})$, observe

$$\left\{X_i=\mathbb{E}[Z|\mathfrak{F}_i]:\ i\in I,\mathfrak{F}_i\subset\mathfrak{F}\right\}$$

Since $Z\in L^1(\Omega)$,

$$\forall\varepsilon>0\exists\delta>0\forall A\in\mathfrak{F},\mathbb{P}(A)\leq\delta:\mathbb{E}Z\mathbb{1}_A\leq\varepsilon$$

Apply Jensen

$$\begin{aligned}\mathbb{E}\left[|\mathbb{E}[Z|\mathfrak{F}_i]|\mathbb{1}_{X_i\geq C}\right]&\leq\mathbb{E}[\mathbb{E}[|Z||\mathfrak{F}]\mathbb{1}_{X_i\geq C}]\\&=\mathbb{E}[|Z|\mathbb{1}_{X_i\geq C}]\end{aligned}$$

Apply Markov, choose C big enough such that

$$\mathbb{P}(X_i\geq C)\leq\frac{\mathbb{E}|X_i|}{C}\leq\frac{\mathbb{E}|Z|}{C}\leq\delta$$

Then we have

$$\forall i\in I:\lim_{c\to\infty}\mathbb{E}[|X_i|\mathbb{1}_{X_i\geq C}]\leq\mathbb{E}[|Z|\mathbb{1}_{X_i\geq C}]\to 0$$

Example (superlinear operator). If there exists a superlinear operator G satisfying

$$\forall X\in\mathfrak{X}:\mathbb{E}G(X)<\infty$$

then $\mathfrak{X}$ is u.i. Normally we choose

$$G(x)=x^p$$
$$G(x)=x\log^+x$$

It follows that **the L^p boundedness implies u.i.**.

Theorem 2.17 (characterisation of uniform integrability). *$\mathfrak{X} \subset L^1$, then the following are equivalent:*

1. *$\mathfrak{X}$ is u.i.*
2. *$\mathfrak{X}$ is L^1 bounded ($\sup_{x\in\mathfrak{X}} \mathbb{E}|X| < \infty$), $\forall \varepsilon > 0 \exists \delta > 0 \forall A, \mathbb{P}(A) \leq \delta : \sup_{x\in\mathfrak{X}} : \mathbb{E}|X|\mathbb{1}_A \leq \varepsilon$*
3. *There exist a G with superlinear growthrate ($\frac{G(X)}{x} \to \infty$ as $x \to \infty$) such that $\sup_{X\in\mathfrak{X}} \mathbb{E}G(X) < \infty$*

Proof.
1)⇒2) By uniform integrability, let $\varepsilon > 0$, then there exists $c(\varepsilon)$:

$$\sup_{x\in X} \mathbb{E}\mathbb{1}_{|x|\geq c} \leq \frac{\varepsilon}{2}$$

Then we choose $\delta = \frac{\varepsilon}{c}$, then we know $\forall A : \mathbb{P}(A) \leq \delta$:

$$\begin{aligned}\mathbb{E}X\mathbb{1}_A &\leq \mathbb{E}X\mathbb{1}_{A\cap\{x\leq c\}} + \mathbb{E}X\mathbb{1}_{x\geq c} \\ &\leq c\cdot\frac{\varepsilon}{c} + \frac{\varepsilon}{2}\end{aligned}$$

2)⇒1) Since $\mathfrak{X} \in L^1$, $\forall \varepsilon > 0$, we may have:

$$\mathbb{E}[X_i\mathbb{1}_{X_i\geq C_i}] \leq \varepsilon$$

By 2) we know $\mathbb{P}(X_i \geq C_i)$ is controlled by δ uniformly:

$$\forall X_i \in \mathfrak{X} : \mathbb{P}(X_i \geq C_i) \leq \delta$$

Choose $C := \sup C_i$, C exist since then we have:

$$\sup_{x\in\mathfrak{X}} \mathbb{E}[X\mathbb{1}_{X\geq C}] \leq \varepsilon$$

3)⇒1), with the superlinear growth of G:

$$\forall \varepsilon > 0 \exists C_0 \forall x \geq C_0 : \frac{G(x)}{x} \geq \frac{1}{\varepsilon}$$

i.e

$$\forall \varepsilon > 0 \exists C_0 > 0 \forall x > C : X \leq \varepsilon G(X)$$

Therefore:

$$\sup_{x\in\mathfrak{X}} \mathbb{E}[X\mathbb{1}_{X\geq c}] \leq \varepsilon \sup_{x\in\mathfrak{X}} \mathbb{E}[G(X)\mathbb{1}_{X\geq C}] \overset{(1)}{\to} 0$$

In (1) we used the boundedness □

Corollary 2.17.1. *If $\sup_{x\in X} \mathbb{E}|X|^p < \infty$ for $p > 1$, X is u.i.*

Proof. L^p boundedness implies L^1 boundedness as well as L^1 bounded superlinear function. □

Theorem 2.18 (Vitalli). *Let $X_n, X_\infty \in L^1(\Omega)$ then the followings are equivalent:*

1. *$X_n \overset{L^1}{\to} X_\infty$*

2. $X_n \xrightarrow{P} X_\infty$ *and* X_i *u.i.*

Proof. 1)⇒2) L^1 converges implies converges in probability. And for the uniform integrability observe

$$\forall C > 0 : \mathbb{E}[|X_n|\mathbb{1}_{X_n \geq C}] - \mathbb{E}[|X_\infty|\mathbb{1}_{X_n \geq C}] \leq \mathbb{E}[|X_\infty - X_n|\mathbb{1}_{X_n > C}]$$
$$\mathbb{E}|X_n|\mathbb{1}_{X_n \geq C} \leq \mathbb{E}|X_\infty|\mathbb{1}_{X_n \geq C} + \mathbb{E}|X_\infty - X_n|\mathbb{1}_{X_n \geq C}$$

By integrability of X_∞, for all $\frac{\varepsilon}{2} > 0$, we can choose δ_1 such that for all A with $\mathbb{P}(A) \leq \delta_1$, $\mathbb{E}X_\infty \mathbb{1}_A \leq \frac{\varepsilon}{2}$. We choose C_1:

$$\mathbb{P}(X_\infty \geq C_1) \leq \frac{\mathbb{E}|X_\infty|}{C_1} \leq \delta_1$$

By the L^1 convergence we can find n_0 such that $\forall n \geq n_0$:

$$\mathbb{E}|X_n|\mathbb{1}_{X_n \geq C} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \leq \varepsilon$$

As for the finite n_0, By the integrability of X_n, for all $\varepsilon > 0$, we can find $\delta_2 > 0$, such that for all A with $\mathbb{P}(A) \leq \delta_2$, $\mathbb{E}X_k \mathbb{1}_A \leq \varepsilon$ for all $k \in \{1, \ldots, n_0\}$. Analog apply Markov we find C_2 such that $\mathbb{P}(X_k \geq C) \leq \delta_2$ for all $k \in \{1, \ldots n_0\}$. Now choose $\hat{C} = \max\{C_1, C_2\}$, we have

$$\mathbb{E}X_n \mathbb{1}_{X_n \geq \hat{C}} \leq \varepsilon \quad \forall n$$

2)⇒1) Define the truncated functions:

$$\varphi_M = \begin{cases} K & x > K \\ x & -K \leq x \leq K \\ -K & x < -K \end{cases}$$

Observe

$$\mathbb{E}|X_\infty - X_n| \leq \mathbb{E}|X_\infty - \varphi_M(X_\infty)| + \mathbb{E}|\varphi_M(X_\infty) - \varphi_M(X_n)| + \mathbb{E}|\varphi_M(X_n) - X_n|$$

The second term converges by the boundedness and converges in probability, the first and third part converges by the uniform integrability. □

Theorem 2.19 (The second martingale convergence theorem). *Let X_n be an u.i. (sub)martingale, then*

$$X_n \xrightarrow{L^1} X_\infty, \quad X_\infty \in L^1$$

and it's closed:

$$\mathbb{E}[X_\infty|\mathcal{F}_n] = (\geq) X_n$$

Proof. □

Corollary 2.19.1. *A martingale is closed iff it's u.i.*

Proof. "⇒":Shown by the example above.
"⇐" By the L^1 convergence we know $\mathbb{E}[X_\infty|\mathfrak{F}^n]$ forms a martingale. □

Corollary 2.19.2 (Kolmogorov 0-1 law). *Given (X_i) i.i.d, events in the tail-field happens either a.s. or a.s not.*

Proof. Notice

$$\mathfrak{T} \subset \mathfrak{F}_\infty$$

Let $A \in \mathfrak{T}$ and

$$X_n := \mathbb{E}[\mathbb{1}_A | \mathfrak{F}_n]$$

Observe it's a bounded, therefore u.i martingale. Then

$$\mathbb{1}_A = \mathbb{E}[1_A | \mathfrak{F}_\infty] = \lim_n \mathbb{E}[1_A | \mathfrak{F}_n] = \mathbb{E}1_A$$

Therefore $\mathbb{P}(A) \in \{0,1\}$. □

Example (Polya's Urn model). Define:

- X_n: # of black balls at time n.
- Y_n: # of red balls at time n.

At each time we draw a ball and put it back with c balls of the same color, define

$$V_n = \frac{X_n}{X_n + Y_n}$$

1) show V_n is a martingale.
2) show $V_n \overset{a.s.}{\to} V_\infty$, and compute the distribution of X_∞.

Proof.
1) measurability is clear, also $\mathbb{E}V_n \leq 1$, $V_n \in L^1$. The martingale inequality comes from:

$$\begin{aligned}
\mathbb{E}V_{n+1}\mathbb{1}_{V_n = \frac{b}{b+r}} &= \frac{b}{b+r} \cdot \frac{b+c}{b+r+c} + \frac{r}{b+r} \cdot \frac{b}{b+r+c} \\
&= \frac{b}{b+r} \\
&= \mathbb{E}V_n \mathbb{1}_{V_n = \frac{r}{b+r}}
\end{aligned}$$

2)First we show the convergence, since $\sup_n \mathbb{E}V_n < \infty$, we know the a.s. limit exists.
Second, we compute the distribution. Assume we get black at first m times and then red on the next $l = n - m$ times.

$$\frac{b}{b+r} \frac{b+c}{b+r+c} \cdots \frac{b+(m-1)c}{b+r+(m-1)c} \frac{r}{b+r+(l-1)c} \cdots \frac{r+(l-1)c}{b+r+(n-1)c}$$

Notice any n drawn with m black and l red has the same probability since the denominator remains the same while the nominator permuted. Therefore we have for $b = 1, c = 1, r = 1$:

$$\mathbb{P}(X_n = m) = \frac{1}{n+1}$$

Therefore V_∞ uniformly distributed in $(0,1)$ □

2.6 Backwards martingales

Recall if we have an u.i. martingale, then we proved it implies L^1 boundedness, a.s. convergence(hence convergence in probability), L^1 convergence and the martingale is closed by a L^1 random variable.

But the key property: uniformly integrable is not easy to show. But for a backwards martingale, it's always u.i.

Definition 2.6 (Backwards martingale). Let $(\mathcal{F}_{-n})_{n\in\mathbb{N}_0}$ be a filtration that satisfies $\forall n : \mathfrak{F}_{-n-1} \subset \mathfrak{F}_{-n}$, we call $(X_{-n})_{n\in\mathbb{N}_0}$ if it satisfies:

- $X_{-n} \in L^1$
- adapted
- $\mathbb{E}[X_{-n}|\mathcal{F}_{-n-1}] = X_{-n-1}$

Intuitively, $\mathcal{F}_0 \approx \mathcal{F}_\infty$ is the largest filtration(smallest σ-algebra contains all $\mathfrak{F}_n$) in the previous section

Theorem 2.20. *Let (X_{-n}) be a backward martingale, set*

$$\mathcal{F}_{-\infty} := \bigcap_{n\in\mathbb{N}_0} \mathcal{F}_{-n}$$

then

$$X_{-n} \to X_{-\infty} \quad a.s.$$

Moreover, the martingale is closable by X_0 hence the convergence is also in L^1.

Proof. we would like:

$$\mathbb{E}U_{-n}(a,b) \leq \frac{1}{b-a}\mathbb{E}[(X_0-a)^+] < \infty$$

We would transform it to a forward martingale to apply Doob, we cannot just change the sign of index but we can control the problem in finite dimension. Define:

$$Y_k := X_{k-n} \quad 0 \leq k \leq n$$

Y_n is then a forward martingale, apply Doob on Y_k:

$$\mathbb{E}U_{-n} \leq \frac{1}{b-a}\mathbb{E}[(Y_n-a)^+] = \frac{1}{b-a}\mathbb{E}[(X_0-a)^+]$$

it only works in finite dimension but it doesn't matter. To show the limits is in L^1, observe:

$$\mathbb{E}X_\infty^+ \leq \liminf_n \mathbb{E}X_\infty^+ = \liminf_n \mathbb{E}[X_\infty]^+$$

□

Example (Strong Law of large numbers). X_n i.i.d. $\mathbb{E}X_n < \infty$, then

$$\lim_{n\to\infty} \frac{S_n}{n} = \mathbb{E}X_1 \quad a.s.$$

Proof. Let $\mathcal{F}_{-n} = \sigma(S_n, \dots, S_{n+m}, \dots)$, Hence

$$M_{-n} := \mathbb{E}[X_1 | \mathcal{F}_{-n}]$$

By the result in the exercise(1.4):

$$M_{-n} = \mathbb{E}[X_1 | \mathcal{F}_n] = \cdots = \mathbb{E}[X_n | \mathcal{F}_n]$$

therefore:

$$M_{-n} = \frac{1}{n}\mathbb{E}[S_n | \mathcal{F}_{-n}] = \frac{1}{n} s_s n$$

By the theorem above we have a a.s. convergent limit:

$$M_{-\infty} = \lim_n M_{-n}, \quad \mathbb{E}M_{-\infty} = \mathbb{E}X_1$$

since $M_{-\infty}$ is a random variable in the tail-σ field, we know it's a.s. constant. □

Example (The Ballot theorem).

Lemma 2.21. *Given a martingale with bounded increment, then*

$$\{\lim_n X_n \textit{ exists and finite } \} = \{\sup_k X_k < \infty\}$$

Proof.
"$\subset$": trivial
"$\supset$": Define $\tau_a := \inf\{n \in \mathbb{N} : X_n \geq a\}$.

$$\{\lim X_n \textbf{ exists and finite}\} \supset \bigcup_{a \in \mathbb{R}} \{\tau_a = \infty\} \supset \{\sup X_n < \infty\}$$

□

Theorem 2.22. *Let (X_n) be a martingale with **bounded increments**, then either of the following event **must occur***

1. $\lim_n X_n$ *exists and is finite*
2. $\liminf = -\infty$ *and* $\limsup = +\infty$.

Proof. We apply the lemma to X and $-X$,
$\{\lim X_n \text{ exists and finite } \} = \{\sup X_n < \infty\} = \{\limsup X_n < \infty\}$

□

2.7 The Radon-Nikodym Theorem

Motivation: $(\Omega, \mathcal{F}, \mathbb{P})$ the probability space. Given:

$$X \in L^1, X \geq 0, \mathbb{E}_P X = 1$$

then

$$A \rightarrowtail \mathbb{E}_P X \mathbb{1}_A := Q(A)$$

defines a probability measure admitting the same null set as $\mathbb{P}$.

Definition 2.7 (Absolutely continuous).

The next theorem provides the characterisation of absolute continuity.

Theorem 2.23. *Given a probability space, The followings are equivalent:*

- $Q \ll \mathbb{P}$
- $\forall \varepsilon > 0 \exists \delta > 0 : \mathbb{P}(A) \leq \delta \Rightarrow Q(A) \leq \varepsilon$

Proof.
2)⇒1): By 2) $Q(A)$ cannot be big if $\mathbb{P}(A)$ is small. So if $\mathbb{P}(A) = 0$, then $\mathbb{P}(A) \leq \delta \quad \forall \delta > 0$, therefore $Q(A) = 0$.
1)⇒2): Assuming by contradiction there exists $\varepsilon > 0 \forall \delta > 0 : \mathbb{P}(A) < \delta \Rightarrow Q(A) > \varepsilon$, then choose $\delta_n = \frac{1}{2^n}$, then $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, we have:

$$\mathbb{P}(\limsup_n A_n) \to 0$$

On the other hand,

$$\begin{aligned} Q(\limsup_n A_n) &= \mathbb{E}^Q(\limsup_n \mathbb{1}_{A_n}) \\ &\geq \limsup_n \mathbb{E}^Q(\mathbb{1}_{A_n}) \\ &= \limsup_n Q(A_n) \\ &\geq \varepsilon \end{aligned}$$

Contradiction. □

Definition 2.8 (Atom). Given a measurable space, for $\omega \in \Omega$, we define

$$a_\omega := \bigcap_{E \in \mathcal{F}, \omega \in E} E$$

as the atom.

Remark.

1. The atom is not necessarily in the σ-field.
2. The atom forms a disjoint partition of Ω
3. In general $A(\mathcal{E})$ does not belong to $\mathcal{F}$, if the σ-field is countably generated. If the σ-field is finite generated, w.l.o.g. we can assume it's generated by the atoms.

Theorem 2.24 (Radon-Nikodym I). *Given $(\Omega, \mathcal{F}, \mathbb{P})$, $\mathcal{F}$ countably generated, $\mathbb{Q} \ll \mathbb{P}$ for some probability measure $\mathbb{Q}$. Then there exists an $\mathbb{P} - a.s.$ uniquely defined random variable $\frac{dQ}{d\mathbb{P}}$, such that:*

$$\forall A \in \mathcal{F} : Q(A) = \mathbb{E}^{\mathbb{P}}[\frac{dQ}{d\mathbb{P}} \mathbb{1}_A]$$

Proof. Define:

$$\mathcal{F}_n := \sigma(A_1, \ldots, A_n)$$

w.o.l.g. we $A_1, \ldots, A_n$ is a partition of Ω?(why partition) .
1)We construct a problem in finite dimension, □

Corollary 2.24.1.

Corollary 2.24.2. *(M_n) is a (super)martingale, then $\exists M_\infty \in L^1 : M_n \to M_\infty$ $\mathbb{P}a.s.$ and*

$$\mathbb{E}M_\infty \leq \liminf_n \mathbb{E}M_n = \inf_n \mathbb{E}M_n$$

Lemma 2.25.

1. *$(X_i)_{i\in I}$ u.i. $\Leftrightarrow X_i$ is L^1 bounded and $\forall \varepsilon > 0 \exists \delta > 0 : \mathbb{P}(A) < \delta \Rightarrow \sup_{i\in I} \mathbb{E}\{|X_i| \mathbf{1}_A\} < \varepsilon$*

2. *$\sup_{i\in I} \mathbb{E}|X_i|^p < \infty, p > 1 \Rightarrow (X_i)_{i\in I}$ u.i.*

3. *All X_i are bounded by the same $Y \in L^1$.*

3 Markov chains

3.1 Introduction

The Markov chain indicates that the probability conditioned on the past is entirely dependent on the last state.

Definition 3.1 (Markov chain). Given a canonical stochastic process $(X_n)_{n\in\mathbb{N}}$, $(E,\mathfrak{E})$ the state space, X is a Markov chain w.r.t. $\mathbb{P}$ and $\mathfrak{F}_n$ if:

$$\mathbb{P}(X_n \in B|\mathfrak{F}_m) = \mathbb{P}(X_n \in B|X_m) \quad \forall m \leq n, B \in \mathfrak{E}$$

Lemma 3.1. *If $X = (X)_{n\geq 0}$ is a (μ, P) Markov chain, then the past and future are independent:*

$$\mathbb{P}(X_0 = i_0, \dots, X_{m+n} = i_{m+n}, \dots | X_m = i) = \mathbb{P}(X_0 = i_0, \dots, X_m = i_i | X_m = i)\delta_{i,i_m} p_{i_n,i_{n+1}} \dots, p_{i_{m+n-1},i_{m+n}} \dots$$

Proof. For $A \in \mathcal{F}_m$, $B \in \mathfrak{G} = \sigma(X_m, X_{m+1}, \dots)$:

$$\begin{aligned}
\mathbb{P}(A \cap B|X_m) &= \mathbb{E}[\mathbb{1}_A \mathbb{1}_B|X_m] \\
&= \mathbb{E}\left[\mathbb{1}_A \mathbb{E}[\mathbb{1}_B|\mathfrak{G}]|X_m\right] \\
\left(\mathbb{E}[X|\mathfrak{G}] \in L^1(\Omega, \mathcal{F}^{X_m})\right) &= \mathbb{E}[\mathbb{1}_A|X_m]\mathbb{E}[\mathbb{1}_B|X_m] \\
&= \mathbb{P}(A|X_m)P(B|X_m)
\end{aligned}$$

□

Remark. By the Markov probability the regular conditional probability always exist, therefore there exists the transition kernel:

$$\mathbb{P}(X_{n+1} \in B|X_n(\omega)) = \bar{P}_{n+1}(X_n(\omega), B)$$

Definition 3.2 (The transition probability and initial distribution). A Markov chain is **homogeneous** if the transition law is state-independent, i.e., $\forall n \in \mathbb{N}_0$:

$$\bar{P}_{n+1} = \bar{P}_n$$

and the probability distribution:

$$\bar{\mu}(B) = \mathbb{P}(X_0 \in B)$$

is called the **initial distribution**

Remark (homogeneosity). We can always construct a homogeneous Markov chain from an inhomogeneous Markov chain by extending the state space to $E \times \mathbb{N}$. But it comes with a price. We may lost the property of E such as the compactness.

Remark (transition matrix). If we have a countable state space, then we call $u = (u_i)_{i\in E}$ a probability vector, if

$$\sum_i \mu_i = 1$$

and $P = (p_{ij})_{i,j\in E}$ the stochastic matrix, if:

$$p_{ij} \geq 0, \qquad \sum_{j\in E} p_{ij} = 1 \quad \forall i \in E$$

Observe the transition matrix of the Markov chain is a stochastic matrix with $p_{ij} = P(X_{n+1} = j|X_n = i)$.
We call a Markov chain on the countable state space a (μ, P) Markov chain if it admits the initial distribution of μ and the transition kernel of P.

Example (M/G/1 queue). The model describe the customers waiting for a service. The customers arrive according to a Poisson process with rate λ. We denote

- X_n: the number of customers in line when n-th customer enters.
- $F(t)$: the distribution of service time.
- a_k: $a_k = \int_0^\infty e^{-\lambda t}\frac{(\lambda t)^k}{k!}dF(t)$ probability that k customers arrive.
- ξ_i: net customer arriving at i-th service time.

Observe the MC defined as

$$X_{n+1} = (X_n + \xi_{n+1})^+$$

The transition matrix:

$$p(0,0) = a_0 + a_1$$
$$p(j, j+k-1) = a_k \quad j \geq 1, or, k > 1$$

3.2 Basic Properties

Lemma 3.2. *Let X be a (μ, P) Markov chain, then conditioned on $X_m = i$, the process*

$$(X_{n+m})_{n\geq 0} \text{ is a } (\delta_i, P)\text{-Markov chain}$$

and is independent of $\mathcal{F}_m^X$.

Proof. Let

$$A = \{X_0 = i_0, \ldots, X_m = i_m\} \in \mathcal{F}_m^X$$

Independence of past and future yields

$$\begin{aligned}\mathbb{P}(X_0 = i_1, \ldots, X_n = i_n|X_m = i) &= \mathbb{P}(A \cap \{X_{m+1} = i_{m+1}, \ldots, X_{m+n} = i_{m+n}\}|X_m = i)\\ &= \mathbb{P}(A|X_m = i)\mathbb{P}(X_{m+1} = i_{m+1}, \ldots, X_{m+n} = i_{m+n}|X_m = i)\\ &= \mathbb{P}(A|X_m = i)\delta_i(i_m)p_{i_m i_{m+1}} \cdots p_{i_{m+n-1} i_{m+n}}\end{aligned}$$

We see $\forall A \in \mathcal{F}^m$, A is independent from the event of future given $X_m = i$. □

Remark. independence means finitely many independence

Theorem 3.3 (Chapman-Kolmogorov). *The transition matrix satisfies the Chapman-Kolmogorov equation:*

$$P(t_1,t_3) = P(t_1,t_2)P(t_2,t_3)$$

Given a (μ,P)-Markov Chain, we have:

$$\mathbb{P}(X_0 = i_0,\dots,X_n = i_n) = \mu_{i_0} p_{i_1 i_2} \dots p_{i_{n-1} i_n}$$

Now we introduce some notation:

- $(\mu P)_j = \sum_{i\in E} \mu_i p_{ij}$: From the initial distribution to state j
- $(P^2)_{ik} = \sum_{j\in E} p_{ij} p_{jk}$: Take 2 steps from i to j.
- $(P^n)_{ik} = \sum_{j\in E} p_{ij}^{n-1} p_{jk}$: Take n steps from i to j.

Corollary 3.3.1. *Let X be a (μ,P)-MC, then*

1. $(\mu P^n)_j = \mathbb{P}(X_n = j)$
2. $p_{ij}^{(n)} = \mathbb{P}(X_n = j|X_0 = i) = \mathbb{P}(X_{n+m} = j|X_n = i)$ *or* $\left(p_{ij}^n = \mathbb{P}_i(X_n = j) = \mathbb{P}(X_{m+n} = j|X_m = i)\right)$

Proof. □

Example (M/G/1 queue).

Example (Polya's urn model).

Example (Ehrenfest Model). $S = \{0,1,\dots,r\}$. And

1. $p(k,k+1) = \frac{r-k}{r}$
2. $p(k,k-1) = \frac{k}{r}$
3. $p(i,j) = 0$ otherwise.

Definition 3.3 (communicating classes). 1. For, $i,j \in E$, we say that i leads to j if:

$$\mathbb{P}(X_n = j|X_0 = i) > 0 \quad \textbf{for some } n$$

We write $i \to j$.

2. we define the i and j **communicate** if:

$$i \to j, j \to i$$

and we write $i \leftrightarrow j$. Notice the relation $\leftrightarrow$ defines a equivalent relation, therefore we may speak of the **communicating class**:

$$E = \dot{\bigcup} C_i$$

For i,j in 2 different communicating classes we have either $i \nrightarrow j$ or $j \nrightarrow i$.

3. we say a communicating class is **closed**, if

$$i \to j \Rightarrow j \in C_i$$

Remark. If C_k is not closed, C_k will be left eventually, in fact, we can never return. For the long run dynamic, "C_k should not matter".

Definition 3.4 (absorbing state and irreducible Markov chain). For a E-valued Markov chain X, we have

- $j \in E$ is an absorbing state: $\{j\}$ is closed
- X is **irreducible**: E is a communicating class.

3.3 First entrance time

Asumme $A \subset E$, we denote:

$$1)\mathbb{P}_i(B) = \mathbb{P}(B|X_0 = i)$$
$$2)\textbf{first hitting time of } A\textbf{: } H^A := \inf\{n \in \mathbb{N}|X_n \in A\}$$
$$3)\textbf{absorption probability: } h_i^A := \mathbb{P}_i(H^A < \infty) = \mathbb{P}(H^A < \infty|X_0 = i)$$
$$4)\textbf{expected time to reach:} k_i^A := \mathbb{E}_i H^A.$$

If A is close, then $H^A < \infty$ implies X_n will never leave A, which motivates us to compute the absorption probability.

Theorem 3.4. *h_i^A is the minimal non-negative solution of the linear system:*

$$\begin{cases} h_i^A = 1 & i \in A \\ h_i^A = \sum_{j \in E} p_{ij} h_j^A & i \notin A \end{cases}$$

Proof.
(i): $i \in A$, trivially we have $\mathbb{P}_i(H^A < \infty) = 1$
(ii): $i \notin A$,

$$\begin{aligned} h_i^A &= \mathbb{P}_i(H^A < \infty) \\ &= \sum_{j \in E} \mathbb{P}_i(H^A < \infty|X_1 = j)\mathbb{P}_i(X_1 = j) \\ &= \sum_{j \in E} \mathbb{P}_i(H^A < \infty|X_1 = j)p_{ij} \\ (Markov) &= \sum_{j \in E} \mathbb{P}_j(H_A < \infty)p_{ij} \\ &= \sum_{j \in E} h_j^A p_{ij} \end{aligned}$$

(iii)**minimal solution:** Let $\hat{h}_j$ be another solution

$$\begin{aligned} \hat{h} &= \mathbb{P}(X_1 \in A) + \mathbb{P}(X_1 \notin A, X_2 \in A) \ldots \mathbb{P}(X_1 \notin A, X_{n-1} \notin A, X_n \in A) + \sum_{p_{ij_1}, p_{ij_2}, \ldots, p_{ij_n} \notin A} p_{ij_1} \ldots p_{j_{n-1}j_n} \hat{h}_n^A \\ &= \mathbb{P}_i(H_A \leq n) + \ldots \end{aligned}$$

Then

$$\hat{h}_i \geq \lim_{n \to} \mathbb{P}_i(H^A \leq n) = \mathbb{P}_i(H^A < \infty) = h_i^A$$

□

Question: how long does it take in average until X reaches A? Namely $k_i^A := \mathbb{E}_i[H^A]$?

Theorem 3.5. *The vector $k^A := (k_i^A)_{i\in E}$ is the minimal non-negative solution of the linear equation system:*

$$\begin{cases} k_i^A = 0 & i \in A \\ k_i^A = 1 + \sum_{j\in E} p_{ij} k_j^A & i \notin A \end{cases}$$

Proof.
i) $k_i^A = 0$ for $i \in A$ is trivil
ii)

$$\begin{aligned} \mathbb{E}_i H^A &= \sum_{j\in E} \mathbb{E}_i[H^A | X_1 = j] \mathbb{P}_i(X_1 = j) \\ &= \sum_{j\in E} \mathbb{E}_j[H^A \circ \theta_1] p_{ij} \\ &= \sum_{j\in E} (1 + k_j^A) p_{ij} \\ &= 1 + \sum_{j\in E} k_j^A p_{ij} \end{aligned}$$

iii) We show $\mathbb{E}_i H^A$ is the only solution of the equation system that satisfies $\mathbb{E}_i H^A = 0$ for $i \in A$. □

Example (Birth and death). In a birth and death chain we define the transition probability:

$$p_{i,i-1} = q_i, \quad p_{i,i+1} = p_i \qquad \forall i \in \mathbb{N}$$

Starting from i, we calculate h_i^A, $A = \{0\}$. We have

$$\begin{aligned} h_i^A &= p_i h_{i+1}^A + q_i h_{i-1}^A & \forall i \in \mathbb{N} \\ h_i^A &= 1 & i = 0 \end{aligned}$$

Then compute:

$$h_{i+1}^A - h_i^A = \prod_{j=1}^{i} \frac{q_i}{p_i}$$

notice it's a telescope sum, therefore:

$$h_i^A - 1 = \sum_{j=0}^{i-1} \prod_{k=1}^{j} \frac{q_i}{p_i} (h_1^A - 1)$$

the solution is uniquely determined by h_1^A. Now we observe the expected iteration to return to the origin:

$$k_0^A = 0, \quad k_1$$

3.4 Recurrence and transience

Question: when will a state be visited infinitely often. And how much time will a Markov chain spent in this state.

Definition 3.5 (transient and recurrent state). We call a state i

1. **transient:** if $\mathbb{P}_i(X_n = i \text{ for } n \text{ infinitely often}) = 1$
2. **recurrent:** if $\mathbb{P}_i(X_n = i \text{ for } n \text{ infinitely often}) = 0$

Remark. Notice that we alway have the δ_i as the initial distribution in the definition since many Markov chain is not irreducible. In this case the recurrent state can only be visited infinitely often if it starts there.

The k-th visiting time:

$$T_i^0 = 0, \quad T_i^1 = \inf_{n\geq 0}\{n > 0 | X_n = i\}, \quad T_i^k := \inf_{n\geq 0}\{n > T_{k-1} | X_n = i\}$$

The recursion time:

$$S_k = \begin{cases} T_i^k - T_i^{k-1} & T_i^k < \infty \\ 0 & T_i^k = \infty \end{cases}$$

we are interested in if $\mathbb{P}(T_i^k < \infty) = 1$. we need Markov property to study this stopping time. We now interested in $\{X_{T_i^{(k+1)}+n}\}_{n\in\mathbb{N}}$ on $\{T_i^{(k+1)} < \infty\}$

Theorem 3.6 (Generalised Markov property). $Y : \Omega \to \mathbb{R}$

$$\mathbb{E}_\mu[Y \circ \theta_n | \mathcal{F}_n] = h(X_n).\mathbb{P}_\mu a.s. \quad h(X) = \mathbb{E}_x Y$$

Proof.

1. First let be the simple function $Y : \mathbf{1}(X_0 = i_0, \ldots, X_n = i_n)$, $\forall A := \{(X_0, \ldots, X_n) \in B_n | B_n \subset S^{n+1}\} \subset \mathcal{F}_n$

$$\begin{aligned}
\mathbb{E}_\mu[Y \circ \theta^n \mathbf{1}_A] &= \sum_{b_0,\ldots,b_n \in B_n} \mathbb{E}_\mu[\mathbf{1}(X_n = i_0, \ldots, X_{n+m} = i_m)\mathbf{1}(X_0 = b_0, \ldots, X_n = b_n)] \\
&= \sum_{b_0,\ldots,b_n \in B_n} \mathbb{P}_\mu(X_0 = b_0, \ldots, X_n = b_n)\mathbb{P}_\mu(X_n = i_0, \ldots, X_{n+m} = i_m | X_n = b_n) \\
(time\text{-}homogeneous) &= \sum_{b_0,\ldots,b_n \in B_n} \mathbb{P}_\mu(X_0 = b_0, \ldots, X_n = b_n)\mathbb{P}_{b_n}(X_0 = i_0, \ldots, X_m = i_m) \\
&= \sum_{\ldots} \mathbb{E}_\mu[\mathbf{1}_{\ldots}\mathbb{E}_{X_n}[Y]] \\
&= \mathbb{E}_\mu[\mathbb{E}_{X_n} Y \mathbf{1}_A]
\end{aligned}$$

2. The cylinder sets of all the finite dimensional subsets forms a intersectional stable generator of Borel σ-algebra of the product space.

$$C_m := (X_0, \ldots, X_m)^{-1}(D_m), D_m \subset S^{m+1}$$

3. Apply measure theoretical induction.

□

Theorem 3.7 (Strong Markov Property). *Given a Markov chain and a **finite** stopping time τ, $X_\tau = i$, we have*

$$(X_{\tau+n})_{n\in\mathbb{N}} \textbf{ is a } \delta_{i,P} \textbf{ Markov chain}$$

*and is **independent** of $\mathcal{F}_\tau^X$, or*

$$\mathbb{E}_\mu[Y \circ \theta_\tau | \mathcal{F}_\tau] = \mathbb{E}_{X_\tau} Y \quad \mathbb{P}_\mu a.s. \quad on\ \{\tau < \infty\}$$

Proof. Let $A \in \mathcal{F}_\tau^X$, we need to show

$$\begin{aligned}\mathbb{P}(\{X_\tau = j_0, X_{\tau+1} = j_2, \dots X_{\tau+n} = j_n\} \cap A | X_\tau = i, \tau < \infty) &= \mathbb{P}(\{X_\tau = j_0 \dots\} | X_\tau = i, \tau < \infty)\mathbb{P}(A | X_\tau = i, \tau < \infty) \\ &= \delta_{i,j_0} p_{j_0 j_1} \dots p_{j_{n-1} j_n} \mathbb{P}(A | X_\tau = i, \tau < \infty)\end{aligned}$$

Indeed,

$$\begin{aligned}&\mathbb{P}(\{X_\tau = j_0, X_{\tau+1} = j_1, \dots, X_{\tau+n} = j_n\} \cap A | X_\tau = i, \tau < \infty) \\ &\overset{\tau \leq \infty}{=} \sum_{m \geq 0} \mathbb{P}(\{X_m = j_0, X_{m+1} = j_1, X_{m+n} = j_n\} \cap A \cap \{\tau = m\} | X_\tau = i, \tau < \infty) \\ (Bayes) &= \sum_{m \geq 0} \mathbb{P}(\{X_m = j_0, X_{\tau+1} = j_1, \dots, X_{m+n} = j_n\} \cap A \cap \{\tau = m\} \cap \{X_m = i\}) \frac{1}{\mathbb{P}(X_\tau = i, \tau < \infty)} \\ (past \perp future) &= \sum_{m \geq 0} \mathbb{P}(\{X_m = j_0, \dots, X_{m+n} = j_n\} | X_m = i)\mathbb{P}(A \cap \{\tau = m\} | X_m = i) \frac{\mathbb{P}(X_m = i)}{P(X_\tau = i, \tau < \infty)} \\ (Homogeneous) &= \mathbb{P}(X_0 = j_0, X_1 = j_1, \dots X_n = j_n | X_0 = i) \sum_{m \geq 0} \frac{\mathbb{P}(A \cap \{\tau = m\})}{\mathbb{P}(\tau < \infty, X_\tau = i)} \\ &= \delta_{i,i_0} p_{i j_1} \dots \mathbb{P}(A | \tau < \infty, X_i = i)\end{aligned}$$

□

In the next theorem we will see that the property of recurrence of transience can be determined by the first(or finite) returning time. We introduce the notation:

- S_i^k: **the excursion time between** $k-1$ **and** k**-th visits of the state** i
- $V_i = \sum_{n \geq 0} \mathbb{1}_{\{X_n = i\}}$: **#returns**

Theorem 3.8. *Let X be a (μ, P) Markov chain, then*

- *if $\mathbb{P}_i(T_i^1 < \infty) = 1$, then i is recurrent and $\sum_n p_{ii}^n = \infty$*
- *if $\mathbb{P}_i(T_i^1 < \infty) < 1$, then i is transient and $\sum_n p_{ii}^n < \infty$*

Proof. To show the recurrence is to show:

$$\mathbb{P}_i(V_i = \infty)$$

1) Observe the excursion time. On $\{T_i^{k-1} < \infty\}$ we have

$$X_{T_i^{k-1}} = i$$

By the strong Markov property

$$(X_{T_i^{k-1}+n})_{n \in \mathbb{N}}$$

is a Markov chain, thus (on $\{T_i^{k-1} < \infty\}$)

$$S_i^k = \inf\{n \geq 0 | X_{T_i^{k-1}+n} = i\}$$

so conditioned on the stopping time T_i^k we would still have the same (μ, P) Markov chain and the future state is independent of the filtration of the past.

$$\begin{aligned}\mathbb{P}_i(S_i^k < \infty | T_i^{k-1}) &= \mathbb{P}_i(S_i^1 < \infty) \\ &= \mathbb{P}_i(T_i^1 < \infty)\end{aligned}$$

2) if $X_0 = i$, we obtain the recursive formula of $\mathbb{P}_i(V_i > k)$:

$$\begin{aligned}\mathbb{P}_i(V_i > 0) &= \mathbb{P}_i(T_i^0 < \infty) = 1 \\ \mathbb{P}_i(V_i > k) &= \mathbb{P}_i(T_i^k < \infty) \\ &= \mathbb{P}_i(T_i^{k-1} < \infty, S_k < \infty) \\ &= \mathbb{P}_i(T_i^1 < \infty)\mathbb{P}_i(T_i^{k-1} < \infty) \\ &= \mathbb{P}_i(V_i > 1)\mathbb{P}_i(V_i > k-1)\end{aligned}$$

therefore:

$$\mathbb{P}_i(V_i > k) = \mathbb{P}_i(V_i > 1)^k$$

Then we have

$$\begin{aligned}\mathbb{E}V_i &= \sum_{k \geq 0} \mathbb{P}(V_i \geq k) \\ &= \sum_{k \geq 0} \mathbb{P}(T_i^1 < \infty)^k\end{aligned}$$

If i is recurrent, i.e. $\mathbb{E}_i V_i = \infty$, then $\mathbb{P}(T_i^1 < \infty) = 1$ must hold by the geometric series.
3)Furthermore:

$$\begin{aligned}\mathbb{E}V_i &\overset{def}{=} \sum_{n \geq 1} \mathbb{P}_i(X_n = i) \\ &= \sum_{n \geq 1} p_{ii}^n\end{aligned}$$

In the case of recurrence the sum must go to infinity. □

Remark. we have the $0-1$ event:

$$\mathbb{P}_i(V_i = \infty) \in \{0, 1\}$$

in another word

$$\mathbb{P}_i(V_i = \infty) > 0 \Rightarrow \mathbb{P}(V_i = \infty) = 1$$

Corollary 3.8.1. *Let C be a communication class of X, then either all states of C are recurrent or all state of C are transient.*

Proof. Let $i, j \in C$, then

$$p_{ij}^m > 0, \quad p_{j,i}^n > 0 \qquad \forall m, n \in \mathbb{N}$$

hence

$$p_{ii}^{n+k+m} \geq p_{ij}^n p_{jj}^k p_{ji}^m$$

Thus

$$\sum_{k\geq 0} p_{jj}^k \leq \frac{1}{p_{ij}^n p_{ji}^m} \sum_{k\geq 0} p_{ii}^k$$

if i is transient, then $\sum_{k\geq 0} p_{ii}^k < \infty$, we would have $\sum_{k\geq 0} p_{jj}^k < \infty$ □

Remark. If X is irreducible, then either all states are recurrent, or all states are transient.

Corollary 3.8.2.

1. *Every recurrent communication class is closed*
2. *Every **finite** closed communication class is recurrent.*

Proof.
1) Let C be a non-closed communication class,

$$\exists i \in C, j \notin C : p_{ij}^k > 0 \text{ for some } k \in \mathbb{N}$$

thus

$$\begin{aligned} P_i(V_i = \infty) &= \mathbb{P}_i(X_k \neq j, V_i = \infty) + \mathbb{P}_i(X_k = j, V_i = \infty) \\ &\leq \mathbb{P}_i(X_m \neq j) + 0 \\ &< 1 \end{aligned}$$

therefore $\mathbb{P}(V_i = \infty) = 0$. Contradiction.
2)since the communication class is finite, for all $i \in C$, there exists $j \in C$:

$$\begin{aligned} 0 < \mathbb{P}_i(V_j = \infty) &= \mathbb{P}_i(V_j = \infty | T_j^1 < \infty)\mathbb{P}_i(T_j^1 < \infty) \\ &= \mathbb{P}_j(V_j = \infty)\mathbb{P}_j(T_j^1 < \infty) \end{aligned}$$

Since the class is closed we have $\mathbb{P}_j(V_j = \infty) > 0$, consequently, $\mathbb{P}_j(V_j = \infty) = 1$ □

Example (irreducible but non-recurrent Markov chain).
transient: 3-d random walk

Example (Branching Process(Galton + Watson)). Given an organism producing offsprings with probabilities

$$\mathbb{P}(\text{i offsprings}) = \alpha_i \quad \text{before dying}$$

Each generation is independent.
X_0: size of the 0-th generation
X_n: size of the n-th generation
$(X_n)_{n\in\mathbb{N}}$ is a Markov chain on $\mathbb{N}_0$ with the transition matrix:

$$p_{i,0} = (\alpha_0)^i$$

since $p_{0,0} = 1$, $\{0\}$ is an absorbing state.
$\{0\}$ is recurrent, $\{1,2,\dots,\}$ is a class of transient states.(class of recurrent states are closed)
$\Rightarrow$ either the polulation dies out or goes to infinity.
let

$$\mu = \sum_{i\in\mathbb{N}_0} i\alpha_i$$

let K_i be the number of i-th organism of the $(n-1)$ generation, then we have

$$X_n = \sum_{i=1}^{X_{n-1}} K_i$$

By the Wald's identity:

$$\mathbb{E}X_n = \mu\mathbb{E}X_{n-1}\cdots = \mu^n$$

Let $\mu < 1$, then

$$\mathbb{P}(X_n \geq 1) \leq \sum_{i=1}^{\infty}\mathbb{P}(X_n \geq 1) = \mathbb{E}X_n = \mu^n \to 0$$

let $\mu > 1$, we have the equation:

$$\begin{aligned} p &= \mathbb{P}(extinction|X_0 = 1) \\ &= \sum_{i=0}^{\infty}\mathbb{P}(extinction|X_1 = i)\alpha_i \\ (\texttt{independent offsprings}) &= \sum_{i=0}^{\infty} p^i\alpha_i \end{aligned}$$

Example (Birth and Death). We revisit our example of birth and death. We want to analyse the recurrence of the Markov chain using martingale. To construct a martingale $\varphi(X_n)$, φ must satisfies

- **fair game:** $\varphi(k) = p_k\varphi(k-1) + r_k\varphi(k) + q_k\varphi(k+1)$
- **initial value:** $\varphi(0) = 0, \varphi(1) = 1$

rewrite the first equation we get a recursive formula:

$$\varphi(k+1) - \varphi(k) = \prod_{i=1}^{k}\frac{q^i}{p^i}$$

Combined with $\varphi(1) - \varphi(0) = 1$, $\varphi(0) = 0$, we get

$$\varphi(m) = \sum_{n=1}^{m-1}\prod_{i=1}^{n}\frac{q^i}{p_i}$$

Example. We summarise some results using this martingale. For $a < x < b$, define

$$T := T_a \wedge T_b$$

we can compute the probability

$$\mathbb{P}_x(T_a < T_b) = \frac{\varphi(b) - \varphi(x)}{\varphi(b) - \varphi(a)}, \quad \varphi(x) = \mathbb{E}_x\varphi(X_T)$$

Example (1-d symmetric random walk). $X_n \in \mathbb{Z}$, $\mathbb{P}(X_{n+1} = X_n + 1|X_n) = 1/2$, $\mathbb{P}(X_{n+1} = X_n - 1|X_n) = 1/2$, then

$$p_{0,0}^{2n} = \binom{2n}{n} p^n (1-p)^n$$

by Sterling:

$$p_{0,0}^{2n} = \frac{(2n)!}{n!n!} p^n q^n \sim p^n q^n \frac{4^n}{\sqrt{\pi n}}$$

$p = q = \frac{1}{2}$, then we have:

$$p_{0,0}^{2n} > \frac{1}{2\sqrt{\pi n}}, \quad \sum_{n \geq 1} p_{0,0}^{2n} = \infty$$

the 1-d symmetric Random walk is recurrent.

Example (2-d symmetric random walk).

Example (3-d symmetric random walk). Given $X_n = (x, y, z)$, then

$$X_{n+1} = \{(x+1, y, z), (x, y+1, z), (x, y, z+1)\}$$

$$\begin{aligned} p_{ij} &= \frac{1}{6} \quad |i-j| = 1 \\ &= 0 \quad else \end{aligned}$$

again we compute:

$$\begin{aligned} p_{0,0}^{2n} &= \sum_{i,j,k \geq 0, i+j+k=n} (\frac{1}{6})^{2n} \binom{2n}{i,i,k,k,j,j,} \\ &= \sum (\frac{1}{6})^{2n} \frac{(2n)!}{(i!)^2 (j!)^2 (k!)^2} \\ &= \binom{2n}{n} (\frac{1}{2})^{2n} \sum (\frac{1}{3})^{2n} \binom{n}{i,j,k}^2 \\ (n = 6m) &\leq C (\frac{1}{3})^n \binom{n}{m,m,m} \end{aligned}$$

then again by Sterling

$$p_{0,0}^{6m} << \infty$$

analog we get:

$$p_{0,0}^{6m} \geq (\frac{1}{6})^2 p_{0,0}^{6m-2}$$

Therefore the 3-d symmetric random walk is transient.

Example (Asymmetric 1-d random walk). $X_n \in \mathbb{Z}$, $\mathbb{P}(X_{n+1} = X_n + 1|X_n) = p$, $\mathbb{P}(X_{n+1} = X_n - 1|X_n) = 1 - p$, then

$$p_{0,0}^{2n} = \binom{2n}{n} p^n (1-p)^n$$

by Sterling:

$$p_{0,0}^{2n} = \frac{(2n)!}{n!n!} p^n q^n \sim p^n q^n \frac{4^n}{\sqrt{\pi n}}$$

if $p = q = \frac{1}{2}$, then we have:

$$p_{0,0}^{2n} > \frac{1}{2\sqrt{\pi n}}, \quad \sum_{n \geq 1} p_{0,0}^{2n} = \infty$$

the 1-d symmetric Random walk is recurrent.
If $p > \frac{1}{2}$, then we have $4^n p^n q^n \leq 2^n$, $\alpha < 1$, in this case

$$\sum_{n \geq 0} p_{00}^{2n} < \infty$$

The random walk is therefore transient.

If we start at 0, denote:

$$X_n = \sum_{i=1}^{n} Y_i$$

by the LLN $\mathbb{P}(Y_i = 1) = p, \mathbb{P}(Y_i = -1) = 1 - p$, $p > \frac{1}{2}$

$$\begin{aligned} \frac{1}{n} X_n &\to \mathbb{E} Y_1 > 0 \quad a.s. \\ &\Rightarrow X_n \to \infty \end{aligned}$$

we want to compute:

$$h_0^0 := \mathbb{P}(\text{ever returning to } 0 | X_0 = 0)$$

We abuse the notation and exclude the original hit of 0. By the theorem of first entrance we get

$$\begin{aligned} h_0^0 &= (1-p) h_{-1}^0 + p h_1^0 \\ &= (1-p) + p h_1^0 \end{aligned}$$

Now we compute h_1^0 :

$$h_1^0 = (1-p) h_0^0 + p h_2^0$$

By the **Markov Property:**

$$\begin{aligned} h_2^0 &= \mathbb{P}(H^0 < \infty | X_2 = 2) \\ &= \mathbb{P}(H^1 < \infty | X_2 = 2) \mathbb{P}(H^0 < \infty | X_1 = 1) \\ &= (h_1^0)^2 \end{aligned}$$

we have the solution:

$$h_1^0 = 1, or, \frac{1-p}{p}$$

By asymmetry of the random walk, 1 is not admissable, therefore we have:

$$h_0^0 = 2 - 2p$$

Example (M/G/1 quene).

3.5 Invariant distribution

Motivation: let (X_n) be i.i.d with $X_n \sim \mu$, by the SLLN we know:

$$\frac{1}{n}\sum X_i \to \mathbb{E}_\mu X_1 \quad a.s.$$

More generally, the empirical distribution satisfies

$$\frac{1}{n}\sum \delta_{Y_i} \to \mu \quad weakly$$

Question: can we expect same result for Markov chain, and if yes, what is μ. i.e. when do we have:

$$\frac{1}{n}\sum f(X_i) \to \int f(x)\lambda(dx)$$

For a Markov chain X and some λ.(section 3.6).
W.l.o.g for X irreducible, we expect some form of asymptotic loss of memory, e.g.

$$\mathbb{P}(X_n = j|X_0 = i) - \mathbb{P}(X_n = j|X_0 = i_0) \to 0$$

or

$$P^n_{ij} \to \pi_j$$

Definition 3.6 (Invariant distribution). Given a Markov chain $X = (P, \mu)$ with state space E, a vector is called **invariant measure** if:

$$\lambda = \lambda P, \quad \lambda_i = \sum \lambda_j p_{ij}$$

if in addition λ is a stochastic vector, then λ is an **invariant distribution**

Remark. We do not assume a priori that $\sum_{i\in E} \lambda_i < \infty$. We can always normalise λ such that $\sum \lambda_i = 1$.

Lemma 3.9. *If λ is invariant and X is a (λ, P) Markov chain, then $(X_{m+n})_{m\in\mathbb{N}}$ is a (λ, P) Markov chain. In another word*

$$\lambda = \lambda P = \lambda P^n$$

Furthermore, if $\lambda_i > 0$, then $\lambda_k > 0$ for all $k \in E$.

Proof.

$$\begin{aligned}\lambda_k &= \sum_{j\in E} \lambda_j p^n_{jk} \\ &\geq \lambda_i p^n_{ik} > 0\end{aligned}$$

□

Theorem 3.10. *If E is **finite** and suppose that for **some** $i \in E$:*

$$p^n_{ij} \overset{n\to\infty}{\to} \pi_j \quad \forall j \in E$$

Then π is an invariant distribution.

Proof. 1. π is a distribution

$$\begin{aligned}\sum_{j\in E}\pi_j &= \sum_{j\in E}\lim_n p_{ij}^n \\ (E finite) &= \lim_n \sum_{j\in E} p_{ij}^n \\ &= 1\end{aligned}$$

2. π is invariant

$$\begin{aligned}\pi_j = \lim_n p_{ij}^n &= \lim_n \sum_{k\in E} p_{ik}^{n-1} p_{kj} \\ &= \sum_{k\in E}\pi_k p_{kj} \\ &\Leftrightarrow \pi = \pi P\end{aligned}$$

□

Example (The condition of finite state is necessary).

Example (compute the invariant distribution).

- compute p_{ij}^n for all j
- solve the linear system $\pi = \pi P$ with $\sum \pi_i = 1$.

Remark. $\lambda \equiv 0$ is always invariant. For the 1-d symmetric random walk:

$$p_{ij}^n \to 0$$

but clearly $\lambda \equiv 0$ is not a distribution. The fact that E is finite in the theorem is important.

Existence and Uniqueness of invariant distribution

We assume X is irreducible(nn recurrent), first we consider the finite case, we see the finite Markov chain always admits an invariant distribution.

Lemma 3.11. *Let X Markov chain and E finite, then there exists an invariant* ***measure***

Proof. Since P is a stochastic matrix,

$$PI_n = I_n$$

hence, 1 is an eigenvalue of P, therefore also an eigenvalue of P^T, and the eigenvector is the row vector of I_n. □

To consider the general case where the Markov chain could have countable many state. For this we define

- **First revisit of k:** $T_k^1 = \inf\{n > 0, X_n = k\} \quad (\inf\emptyset = \infty)$
- **Number of visit of i until first revisit of k** $\gamma_i^k = \mathbb{E}_k \sum_{n=0}^{T_k^1 - 1} \mathbb{1}_{X_n = i}$

Lemma 3.12 (Existence of the invariance measure). *Let X be irreducible and recurrent, then the following holds*

1. $\gamma_k^k = 1$

2. $(\gamma_i^k)_{i\in E}$ *is an invariant vector. Moreover*

$$\gamma_i^k = \mathbb{E}_k \sum_{n=0}^{T_k^1} \mathbb{1}_{X_n=i} = \sum_{n=0}^{\infty} \mathbb{P}(X_n = i, T \geq n)$$

3. $0 < \gamma_i^k < \infty$ *for all* $i \in E$.

Proof.
1) trivial
2) Since X is recurrent,

$$\mathbb{P}_k(T_k^1 < \infty) = 1, \quad X_{T_k^1} = k$$

For $j \neq k$, we have:

$$\begin{aligned}
\gamma_j^k &= \mathbb{E}_k \sum_{n=0}^{T_k^1 - 1} \mathbb{1}_{X_n=j} \\
&= \mathbb{E}_k \sum_{n=0}^{\infty} \mathbb{1}_{X_n=j} \mathbb{1}_{n \leq T_k^1 - 1} \\
&= \sum_{n=0}^{\infty} \mathbb{P}_k(X_n = j, n \leq T_k^1 - 1) \\
(!) &= \sum_{n=1}^{\infty} \mathbb{P}_k(X_n = j, n \leq T_k^1) \\
&= \sum_{n=1}^{\infty} \sum_{i\in E} \mathbb{P}_k(X_n = j, n \leq T_k^1, X_{n-1} = i)
\end{aligned}$$

The **recurrence** of k gives us a finite stopping time to be conditioned on. Using the strong Markov property, the future is independent form the past given the current state.

$$\mathbb{P}_k(X_n = j, n \leq T_k^1, X_{n-1} = i) = p_{ij} \mathbb{P}_k(n \leq T_k^i, X_{n-1} = i)$$

Hence

$$\begin{aligned}
(Fubini) &= \sum_{i\in E} p_{ij} \sum_{n=1}^{\infty} \mathbb{P}_k(n \leq T_k^i, X_{n-1} = i) \\
(cycling) &= \sum_{i\in E} p_{ij} \sum_{m=0}^{\infty} \mathbb{P}_k(m \leq T_k^i, X_m = i) \\
&= \sum_{i\in E} p_{ij} \mathbb{E}_k \sum_{m=0}^{T_k^1 - 1} \mathbb{1}_{X_m=i} \\
&= \sum_{i\in E} p_{ij} \gamma_i^k
\end{aligned}$$

3) Since X is irreducible, we know

$$p_{ik}^n > 0, \quad p_{ki}^m > 0$$

Thus

$$\begin{aligned}
\gamma_i^k &= \sum_{j\in E} \gamma_j^k p_{ji}^n \\
&\geq \gamma_k^k p_{ki}^m = p_{ki}^m > 0
\end{aligned}$$

Moreover

$$1 = \gamma_k^k = \sum_j \gamma_j^k p_{jk}^n \geq \gamma_i^k p_{ik}^n$$
$$\Rightarrow \gamma_i^k \leq \frac{1}{p_{ik}^n} < \infty$$

□

Theorem 3.13 (minimal bound of the invariance measure). *Let X be **irreducible**, let λ be any invariant measure with $\lambda_k = 1$(always satisfied for some k after normalisation). then $\forall i \in E$:*

$$\lambda_i \geq \gamma_i^k$$

*if X is also **recurrent**, then*

$$\lambda = (\gamma^k)_{i \in E}$$

Proof. since λ is invariant and $\lambda_k = 1$:

$$\lambda_j = (\lambda P)_j = \sum_{i_0 \in E} \lambda_{i_0} p_{i_0 j} = \sum_{i_0 \neq k} \lambda_{i_0} p_{i_0 j} + p_{kj}$$

now we use

$$\lambda_{i_0} = \sum_{i_1 \neq k} \lambda_{i_1} p_{i_1 i_0} + p_{k i_0}$$

and get

$$\begin{aligned}\lambda_j &= p_{kj} + \sum_{i_0 \neq k} p_{k i_0} p_{i_0 j} + \sum_{i_0, i \neq k} \lambda_{i_1} p_{i_1 i_0} p_{i_0 j} \\ &= p_{kj} + \sum_{i_0 \neq k} p_{k i_0} p_{i_0 j} + \cdots + \sum_{i_0, i_1, \cdots, i_{n-1} \neq k} p_{k i_{n-1}} \ldots p_{i_1 i_0} p_{i_0 j} k + \varepsilon \quad \varepsilon > 0\end{aligned}$$

Notice the equation represents the probability of hitting j without hitting k in $1, 2, \ldots,$ steps. Altogether we have

$$\begin{aligned}\lambda_j &\geq \sum_{m=1}^{n} \mathbb{P}_k(X_m = j, T_k^1 \geq m) \\ &= \mathbb{E}_k \sum_{m=1}^{n \wedge T_k^1} \mathbb{1}_{X_m = j}\end{aligned}$$

letting $n \to \infty$ and shifting the index

$$\lambda_j \geq \mathbb{E}_k \sum_{m=0}^{T_k^1 - 1} \mathbb{1}_{X_n = j} = \gamma_j^k$$

If X is recurrent, then γ^k is an invariant measure, hence

$$\tilde{\lambda} = \lambda - \lambda^k isinvariant, \quad \tilde{\lambda_k} = 0$$

since X irreducible

$$p_{jk}^n > 0$$

and so

$$\tilde{\lambda_k} = 0 = \sum_j \tilde{\lambda}_j p_{jk}^n \geq \tilde{\lambda}_i p_{ik}^n$$

since γ^k is the lower bound of γ we have $\tilde{\lambda} = 0$. Up to normalisation, $\lambda_k = 1$, we have the uniqueness. □

Example. The communication classes, we look at the recurrent classes and get invariant distribution. And the convex combination of which is also invariant distribution.

Now we have get a invariant measure, which is the expected number of visits between 2 visits of a recurrent state. But it's not yet an invariant distribution. To have a invariant distribution we need the measure to be finite:

$$\sum_{i\in E} \gamma_i^k < \infty$$

We may have $\sum_i \gamma_i^k = \infty$, the question is when is it finite.

Definition 3.7 (positive and null recurrent). Let X be an irreducible Markov chain on E, let

$$T_i^1 := \inf\{n > 0, X_n = i\}$$

- A state $i \in E$ is **positive recurrent:** $\mathbb{E}_i T_i^1 = m_i < \infty$
- A state $i \in E$ is **null recurrent**: $\mathbb{E}_i T_i^1 = m_i = \infty$.

The following theorem states, in the class of recurrent and irreducible Markov chain, existence of an invariant distribution and positive recurrence is equivalent.

Theorem 3.14. *Let X be irreducible, recurrent Markov chain, the following are equivalent*

1. *every state is positive recurrent*
2. *some state $i \in E$ is positive recurrent*
3. *there exists an (unique) invariant distribution π.*

In any of these cases,

$$m_i = \frac{1}{\pi_i}, \quad i \in E, \pi_i > 0$$

Proof.
1)⇒2):trivil
2)⇒3): we know (γ^k) is an invariant measure with $\gamma_k^k = 1$, hence

$$\begin{aligned}
1 < \sum_{i\in E} \gamma_i^k &= \sum_{i\in E} \mathbb{E}_k \sum_{n=0}^{T_k^1-1} \mathbb{1}_{X_n=i} \\
(1) &= \mathbb{E}_k \left[\sum_{n=0}^{T_k^1-1} \sum_{i\in E} \mathbb{1}_{X_n=i} \right] \\
&= \mathbb{E}_k T_k^1 \\
&= m_k \\
(\text{k positive recurrent}) &< \infty
\end{aligned}$$

(1) follows from monotone convergence theorem. We already know γ_i^k defines an invariant vector.

$$\sum_{i\in E} \gamma_i^k = \mathbb{E}_k T_k^1 < \infty$$

Therefore $\pi_i = \frac{\gamma_i^k}{m_k}$ defines an invariant distribution.
3)⇒1) assuming there exists an invariant measure π with $\pi_k > 0$.

$$\lambda_i = \frac{\pi_i}{\pi_k}$$

is invariant and $\lambda_k = 1$, then for any $l \in E$:

$$m_l = \sum_{j \in E} \gamma_j^l$$
$$\text{(minimal bound)} \leq \sum \lambda_j = \frac{1}{\pi_k} \sum_{j \in E} \pi_j = \frac{1}{\pi_k} < \infty$$

thus $k \in E$ is positive recurrent.

□

Example (Null recurrent Markov chain: 1-d symmetric random walk). The symmetric 1-d random walk is recurrent, $\pi = (\pi_j)$ with $\pi_j = 1$ is **the** invariant measure, $\pi = \gamma^k$ for all k, hence

$$m_k = \sum_{i \in E} \gamma_i^k = \infty$$

hence no invariant distribution

Remark. To summarise, we get: Given X is irreducible, recurrent,

- **positive recurrent**: unique invariant distribution
- **null recurrent**: unique(up to scalar multiplication) invariant measure, no invariant distribution.

X is irreducible, transient(E countable), anything can happen.

Example (transient Markov Chain).
1)let $E = \mathbb{N}_0$, $P_{i,i+1} = 1$ for all $i \in E$. $\pi_i \equiv 0$ is then the only invariant measure
2)let $E = \mathbb{Z}$, $P_{i,i+1} = 1$ for all $i \in E$, the invariant measure is $\pi_i \equiv c$ for some $c \geq 0$. We have a invariant measure but no invariant distribution.
3)Asymmetric 1d random walk:

Example (reflective 1d random walk.). (in general Y MC not imply $|Y|$ a MC). If X a standard random walk, $|X|$ is a reflective random walk, if $q \neq p$, then $|X|$ is transient.

Example (M/G/1 queue).

The detailed balance condition

Theorem 3.15. *let $X = (X_n)_{0 \leq n \leq N}$ be an irreducible (π, P) Markov chain, let π be an invariant distribution, then*

$$Y = (Y_n)_{0 \leq n \leq N}, \quad Y_n = X_{N-n}$$

is an irreducible $(\pi, \hat{P})$ Markov chain with

$$\hat{p}_{ij} = \frac{\pi_j}{\pi_i} p_{ji}$$

and π is invariant distribution for Y. In particular,

$$\forall i \neq j : \pi_i = \pi_j \Rightarrow \hat{P} = P^T$$

Proof.
1)$\hat{P}$ is a stochastic matrix, compute the row sum

$$\sum_j \hat{p}_{ij} = \frac{1}{\pi_i} \sum \pi_j p_{ji} = 1$$

2)π is invariant:

$$\sum_{i \in E} \pi_i \hat{p}_{ij} = \sum_{i \in E} \pi_j p_{ji} = \pi_i$$

3)Markov chain □

Definition 3.8 (The detailed balanced condition).

- A stochastic matrix P and a measure λ are in **detailed balance** if

$$\lambda_i p_{ij} = \lambda_j p_{ji} \quad \forall i, j \in E$$

- An Markov chain (λ, P) Markov chain X is called **reversible** if $\forall N \in \mathbb{N} : (X_{N-n})_{0 \leq n \leq N}$ is a (λ, P) Markov chain.

Theorem 3.16. *Given an irreducible Markov chain (λ, P) , the followings are equivalent:*

1. *X is reversible.*
2. *X satisfies the detailed-balance sheet condition.*

Proof. □

Example.

3.6 Convergence theorem of Markov chain

Aperiodic Markov chain

Definition 3.9. Let X be a Markov chain with transition matrix P, then

- the **period** of a state i is defined as $d_i := gcd\{n \in \mathbb{N} : p_{ii}^n > 0\}$ with $gcd\{\emptyset\} = \infty$.
- A state i is called **aperiodic**, if $d_i = 1$
- The MC X is called **aperiodic**, if $\forall i \in E : d_i = 1$

Lemma 3.17. *Let X be a Markov chain on the state space E with transition matrix P. We have:*

1. $i \leftrightarrow j \Rightarrow d_i = d_j$
2. *For $i \in E$, we have: $d_i < \infty \Rightarrow \exists n_0 \forall n > n_0 : p_{ii}^{nd_i} > 0$*

Proof. 1) Since $i \leftrightarrow j$, there exists $m,n \in \mathbb{N}$, such that $p_{ij}^m > 0, p_{ji}^n > 0$. For all $k \in \mathbb{N}$ with $p_{jj}^k > 0$:

$$p_{ii}^{m+n+k} \geq p_{ij}^m p_{jj}^k p_{ji}^n > 0$$
$$p_{ii}^{m+n+2k} \geq p_{ij}^m p_{jj}^k p_{jj}^k p_{ji}^n > 0$$

therefore $d_i l = m+n+k, d_i l' = m+n+2k \Rightarrow d_i(l'-l) = k \Rightarrow d_j | d_i$. Analog we can prove $d_i | d_j$.
2)w.l.o.g assume $d_i = 1$.
The key observation is that $N_i := \{n : p_{ii}^n > 0\}$ is closed under addition(not subtraction since it's not irreducible). First we show if N_i contains 2 subsequent integer $N, N+1$, then $\forall m \geq N^2 : m \in I_x$, indeed,

$$m - N^2 = kN + r \quad 0 \leq r < N$$
$$\Rightarrow m = N^2 + kN + r$$
$$m = N(N+k-r) + (1+N)r \in N_i$$

Now we prove it contains 2 subsequent integers. □

Corollary 3.17.1. *Let X be an irreducible, **aperiodic** Markov chain, then for all $i, j \in E$, we have*

$$p_{ij}^n > 0 \quad \textit{for sufficiently large } n$$

Proof. For $i, j \in E$. Let $n(i), n(j)$ be the natural number given by the last lemma,then we have:

$$p_{ij}^{n(i)+n(j)+1} \geq p_{ii}^{n(i)} p_{ij} p_{jj}^{n(j)} > 0$$

All components are positive by periodicity and irreducibility. □

Theorem 3.18 (Convergence theorem). *Given X an **aperiodic, irreducible** Markov chain with a invariant distribution, then*

$$\lim_{n\to\infty} \mathbb{P}_i(X_n = j) = \pi_j$$

Proof.
1)**Construct the coupling:** Let Y_n be an independent copy of X_n. We show (X_n, Y_n) is an irreducible positive recurrent Markov chain. Irreducibility requires **aperiodicity**. We define the transition probability as:

$$p_{(i,j)(k,l)} = p_{ik} p_{jl}$$

By the **aperiodicity** there exists a large N such that $\forall n \geq N$: $p_{ik}^n > 0, p_{jk}^n > 0$. We see all states in (X,Y) commutes, hence the irreducibility.

On the other hand, by the existence of an invariant distribution, we know the irreducible Markov chain is positive recurrent.
2)**Formulate the variational distance:** we want:

$$\sum_{j\in E} |\mathbb{P}_i(X_n = j) - \pi(j)| \to 0$$

By recurrence we know the Markov chain hits diagonal in finite time. Define $T := \inf\{n \in \mathbb{N} : \text{the MC hits the diagonal}\}$. By Markov property we have:

$$\begin{aligned}
\mathbb{P}(X_n = j, T \leq n) &= \sum_{m=1}^{n} \sum_{k \in E} \mathbb{P}(X_n = j, X_m = k, T = m) \\
&= \sum \sum \mathbb{P}(X_m = k, T = m) \mathbb{P}(X_n = j | X_m = k) \\
&= \sum \sum \mathbb{P}(Y_m = k, T = m) \mathbb{P}(Y_n = j | Y_m = k) \\
&= \mathbb{P}(Y_n = j, T \leq n)
\end{aligned}$$

therefore:

$$\begin{aligned}
|\mathbb{P}(X_n = j) - \mathbb{P}(Y_n = j)| &\leq 2\mathbb{P}(X_n = j, T \geq n) \\
&\leq 2\mathbb{P}(T \geq n)
\end{aligned}$$

3)**Show the convergence:** Notice the result above is entirely base on the property of the transition matrix, by the independence of X and Y we can assign different initial distributions. Let $\mu_X = \delta_i$ and μ_Y be the invariance distribution we have:

$$\sum |p_{ij}^n - \pi(j)| \leq \sum |\mathbb{P}(X_n = j) - \mathbb{P}(Y_n = j)| \leq 2\mathbb{P}(T \geq n) \to 0$$

□

Remark. The technique is called **coupling**. Usually used to prove convergence in probability.

Birkoff's ergodic theorem

Goal:

$$\frac{1}{n} \sum_{k=0}^{n-1} f(x_k) \to \int f(i) d\pi_i$$

Lemma 3.19. *let (Y_n) i.i.d $Y_n \geq 0$ and $\mathbb{E}Y_0 = \mu \in [0, \infty]$, then we have:*

$$\frac{1}{n} \sum_{i=1}^{n} \to \mu \quad a.s.$$

In particular, if

$$N_n \to \infty \quad a.s.$$

then

$$\frac{1}{N_n} \sum_{i=1}^{N_n} \to \mu \quad a.s.$$

Theorem 3.20 (Birkoff's Ergodic theorem). *Let X be an irreducible (λ, P) Markov chain. And let*

$$V_i^n := \sum_{k=0}^{n-1} \mathbb{1}_{X_k = i}$$

then

$$\frac{1}{n} V_i^n \to \frac{1}{m_i} \quad a.s.$$

If X is also positive recurrent($m_i < \infty$), then the Markov chain admits an unique invariant distribution π, then we have:

$$\forall f : E \to \mathbb{R} \textbf{ \textit{bounded}} : \frac{1}{n}\sum_{k=0}^{n} f(x_k) \to \bar{f} = \sum_{i \in E} \pi_i f(i)$$

Proof. let X be transient, then $m_i = \infty$ for all $i \in E$ and $V_i^n < \infty$ a.s. Hence

$$\frac{1}{n}V_i^n \to 0 = \frac{1}{\infty}(?)$$

2)let X be recurrent, then

$$\forall i, j \in E : \mathbb{P}_j(T_j^1 < \infty) = 1$$

By strong Markov $(X_{T_i^1+n})_n$ is a Markov chain, moreover

$$\frac{V_i^n}{V_i^n + V_i^{T_i^1}} \to 1$$

3)Let X be positive recurrent, then we know the invariant distribution is given be $\frac{1}{m_i}$. In this case

$$\frac{1}{n}V_i^n$$

□

4 General Theory of stochastic process

4.1 Construction of stochastic process

Question: Given a consistent family of finite dimensional distribution(as constructed in MC), does a measure $\mathbb{P}$ in the big space exists, such that

Definition 4.1 (projective family). T be an index set, $(S,\mathcal{S})$ be a measurable set, for each finite subset $J \subset T$ the probability $\mathbb{P}_J$ is given. If $(\mathbb{P}_J)_{J\subset T\, finite}$ satisfies the **consistency condition**:

$$\forall J \subset J' \subset T, A \subset \mathcal{S}^{\otimes J} : \mathbb{P}_{J'}(\pi_{J',J}^{-1}(A)) = \mathbb{P}_J(A)$$

Lemma 4.1 (Consistency in finite product probability space). *Let $(X_t, t \in T)$ be a stochastic process with state space $(S,\mathcal{I})$ and denote by $\pi_{J,I} : S^J \to S^I$ for $I \subset J \subset T$ the coordinate projection. Then the finite-distributions $(P_J^X)_{J\in T}$ satisfies:*

$$\forall I \subset J \subset T \quad \forall A \in \mathcal{I}^{\otimes I} : P_J^X(\pi_{J,I}^{-1}(A)) = P_I^X(A)$$

Theorem 4.2 (Kolmogorov consistency theory). *Given a Polish measurable space $(S,\mathcal{B}_S)$, $T \neq \emptyset$ an index set and $(\mathbb{P}_J)$ be a projective family of S and T. Then there exists a unique probability measure $\mathbb{P}$ on the product space $(S^T, \mathcal{B}_S^{\otimes T})$ satisfying:*

$$\forall J \subset T, J < \infty, B \in \mathcal{B}_S^{\otimes J} : \mathbb{P}_J(B) = \mathbb{P}(\pi_{T,J}^{-1}(B))$$

Remark. Here the state space is Polish.

Proof. □

Example (Markov chain). Let $(E,\mathcal{E})$ be polish, μ a probability measure on the space, $P : E \times \mathcal{E} \to [0,1]$ transition kernal. We have the consistent family:

$$\{\mathbb{P}_J, J \subset H(\mathbb{N}_0)\} \quad H(\mathbb{N}_0) := \{J < \infty : J \subset H\}$$

Kolmogorov tells us the unique measure $\mathbb{P}$ exists, then we have

$$\pi_n(\omega) = \omega_n$$

is a Markov Chain.

Example (Brownian motion). The Brownian motion $(B_t)_{t\in[0,\infty)}$ on $(\Omega,\mathcal{F},\mathbb{P})$ is defined as

1. $\mathbb{P}(B_0 = 0) = 1$
2. stationary, normally distributed increments: $\forall 0 \le s \le t : B_t - B_s \sim N(0, t-s)$
3. independent increments: $\forall 0 \le t_1 \le t_2 \ldots t_n : B_{t_n} - B_{t_{n-1}} \ldots B_1 - B_0$ are independent.
4. $t \to B_t(\omega)$ is continuous a.s.

We construct a process $B = (B_t)_{t\in[0,\infty])}$ that satisfies 1),2),3).(Pre Brownian motion). Let $(E,\mathcal{E}) = (\mathbb{R}, B(\mathbb{R}))$, we construct a probability measure $\mathbb{P}$ on the functional space $(E^T, \mathcal{E}^T)$. For $T = [0,\infty)$,
For $\mu \in \mathbb{R}$, $\sigma^2 > 0$ let

$$C([0,\infty),\mathbb{R}) \notin \mathcal{E}^{[0,T)}$$

Corollary 4.2.1. *For any Polish space $(S, \mathcal{B}_S)$, and $T \neq \emptyset$ an index set. There exists to a prescribed projected family $(\mathbb{P}_J)$ a stochastic process $(X_t, t \in T)$ whose finite dimensional distribution is given by $\mathbb{P}_J$.*

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $X = (X_n)$ a MC, we have

$$\mathbb{P}$$

4.2 Continuity of stochastic process

So far we constructed the "pre-Brownian Motion" on

$$(\Omega, \mathcal{F}) = (E^{[0,\infty)}, \mathfrak{E}^{[0,\infty)})$$

Now we make the BM continuous on the space

$$(C([0,\infty), \mathbb{R}), B(C([0,\infty), \mathbb{R}))$$

instead of

$$(\mathbb{R}^{[0,\infty)}, B(\mathbb{R})^{[0,\infty)})$$

Definition 4.2 (Version and Indistinguishable). two process $(X_t)_{t\in T}), (Y_t)_{t\in T})$ on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called

1. **Indistinguishable**: $\mathbb{P}(\forall t \in T : X_t = Y_t) = 1$, i.e., there exists a set $N \in \mathcal{F}, \mathbb{P}(N) = 0$, such that

$$\{X_t \neq Y_t\} \subset N \quad \forall t \in T$$

   Notice the set may not be measurable.

2. **Version of each other** $\forall t \in T : \mathbb{P}(X_t = Y_t) = 1$

Example (Poisson Process). Let $(\tau)_{i\in\mathbb{N}}$ be independent, $\exp(\lambda)$ distributed.

$$\sigma_k = \sum_{i=1}^{k} \tau_i, k \in \mathbb{N}$$

and

$$N(t) = \sum_{k=1}^{\infty} \mathbb{1}_{[0,t]}(\sigma_k)$$

It models the number of events happens at time t. The Poisson process has right-continuous sample path with left limit(catalag). Let

$$N_-(t) = \sum_{i=1}^{\infty} \mathbb{1}_{[0,t)}(\sigma_k)$$

the left-continuous version of N, then

$$\mathbb{P}\Big(N(t) \neq N_-(t)\Big) = \mathbb{P}\Big(\bigcup_n \{\sigma_n = t\}\Big) = 0$$

Moreover, for any $t_1 < t_2 < \ldots t_n$

$$\mathbb{P}(\bigcup_{i=1} \{N(t_i) \neq N_-(t_i)\}) = 0, N \overset{d}{=} N$$

But have different sample path.

Definition 4.3. A stochastic process X taking value in (E,d) is called

- continuous in probability at t_0 if

$$X_t \to X_{t_0} \quad i.p. \quad t \to t_0$$

- continuous a.s. if

$$X_t \to$$

Remark. $\{\omega : t \to X_t(\omega)$ is continuous $\}$ may not be measurable

Example. The Poisson process satisfied

$$\mathbb{P}(N(t+s) - N(s) = n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$

In particular, N is continuous in probability

A probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called complete, if $\mathcal{F}$ contains all $\mathbb{P}$-null sets. If a probability space if complete, then the indistinguishability becomes

$$\mathbb{P}(X_t = Y_t \quad \forall t) = 1$$

Indeed

$$\bigcup_t \{X_t \neq Y_t\} \subset N, \mathbb{P}(N) = 0$$
$$\Rightarrow (\bigcup_t \{X_t \neq Y_t\})^C = \{X_t = Y_t \quad \forall t\} \in \mathcal{F}$$

and has measure 1. Conversely, if

$$\mathbb{P}(X_t = Y_t \quad \forall t) = 1$$

then

$$(\{\forall t : X_t = Y_t\})^C = \bigcup_t \{X_t \neq Y_t\} \in \mathcal{F}$$

Clearly, indistinguishable implies the 2 are versions. The converse is not true. However, if X, Y are version and have a.s. right-continuous sample paths, then

$$X_t(\omega) = \lim_{s \downarrow t, s \in \mathbb{Q}} X_s(\omega) \quad Y_t(\omega) =$$

Since X, Y are versions

$$\mathbb{P}(X_s = Y_s \quad \texttt{for all} \ s \in T \cap \mathbb{Q}) = 1$$

By right continuity, for any $t \in T$, $\omega \in \Omega_0$,

$$X_t = \lim_{s \downarrow t, s \in T \cap \mathbb{Q}}$$

Theorem 4.3 (Kolmogorov-Centsov-continuity theorem). *Let X_t be a stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$ such that*

$$\mathbb{E}|X_t - X_s|^\alpha \leq C|t-s|^\beta, t, s \in [0,T]$$

for $\alpha, \beta, c > 0$, then X has a continuous version $\tilde{X}$ such that

$$\mathbb{P}(\sup_{0 \leq t-s \leq}$$

Proof. w.l.o.g. $T = 1$. By Markov inequality

$$\mathbb{P}(|X_t - X_s| \geq \varepsilon) \leq \frac{\mathbb{E}|X_t - X_s|^\alpha}{\varepsilon^\alpha} \leq C$$

2. X is uniformly continuous on

$$D$$

We apply 1) to $t = \frac{k}{2^n}$, $s = \frac{k-1}{2^n}$, $\varepsilon = 2^{-\gamma n}$, we get

$$\mathbb{P}(|X_t - X_s| \geq 2^{\gamma n}) \leq C 2^{-\alpha \gamma n}$$

Hence

$$\mathbb{P}(\max_{0 \leq k \leq 2^n} |X_{k/2^n} - X_{(k-1)/2^n}$$

□

5 Ergodic Theorem

5.1 Stationary and Ergodic process

Definition 5.1 (stationary stochastic process).

$$(X_{t_1},\dots,X_{t_n}) \overset{d}{=} (X_{t_1+s},\dots,X_{t_n+s})$$

Definition 5.2 (measure-preserving map).

$$\forall A \in \mathcal{F} : \mathbb{P}(T^{-1}(A)) = \mathbb{P}(A)$$

Lemma 5.1.

1. *Every stationary process induces a measure preserving map*

2. *T measure preserving $\Rightarrow \forall Y$ S-valued: $X_n(\omega) := Y(T^n(\omega))$ is stationary*

Definition 5.3 (Invariant events Ergodic map and process).

1. Invariant event:

$$T^{-1}(A) = A \quad \mathbb{P}.a.s. \Leftrightarrow \mathbb{P}(A\Delta T^{-1}(A)) = 0$$

2. **Ergodic map**: the measure-preserving map with trivial $\mathcal{F}^T$ ($\mathcal{F}^T := \{A \in \mathcal{F} | \text{A is invariant}\}$).

3. **Ergodic process**: stationary process induced by ergodic map.

Lemma 5.2.

1. *$\forall A \in \mathcal{F}^T \exists B \subset A : T^{-1}(B) = B, \mathbb{P}(A\Delta B) = 0$. In particular*

$$\forall B : \mathbb{P}(B) \in \{0,1\} \Rightarrow T \text{ is ergodic}$$

Example (i.i.d random variables). Given a sequence of i.i.d random variables.

(X_n)i.i.d,$A \in \mathcal{F}^T$

$A = \theta^{-1}(A) = \cdots = \theta^{-n}(A) \in \sigma(\pi_k, k \geq n)$

$\overset{n\uparrow\infty}{\Rightarrow} A \in \mathcal{T}((\mathcal{A}_i)_{i\in I})$ (the σ-Algebra are independant)

$\overset{\text{0-1}}{\Rightarrow} \mathbb{P}(A) \in \{0,1\}$

$\overset{\text{lemma}}{\Rightarrow} \theta$is ergodic

Lemma 5.3. *We have:*

- *Y is $\mathcal{F}^T$ measurable $\Leftrightarrow Y$ is $\mathcal{F}^T$ a.s. invariant, in particular*
  T is ergodic $\Leftrightarrow$ Every invariant and bounded RV is constant

5.2 Recurrence and transience of Markov chains

Some notation:

- $(X_n)_{n\geq 0}$ the time-homogeneous Markov chain.
- $\Omega = S^{\mathbb{N}_0}$
- μ: initial distribution of the Markov chain (X_n).
- $\mathbb{P}_x := \mathbb{P}_{\delta_x}$ for a Markov Chain starting at x.

Proof. □

Remark. Y may depends on all future value of X_n.

aka, $(X_{\tau+n}$ is a (δ_{X_τ}, P) Markov chain. for any stopping time τ

Definition 5.4 (recurrent & transient state).
Set $\tau_y^0 := 0$:

1. time of kth return ro y : $\tau_y^k := \inf\{n > \tau_y^{k-1} | X_n = y\}$

Set $\tau_y := \tau_y^1$ $\rho_{xy} := \mathbb{P}_x(\tau_y < \infty)$

1. y recurrent state: $\rho_{yy} = 1$
2. y transient state: $\rho_{yy} < 1$

Theorem 5.4. *For $y \in S$, define $N(y) := \sum_{n=1}^{\infty} \mathbf{1}(X_n = y)$.*

1. $\mathbb{P}_x(\tau_y^k < \infty) = \rho_{xy}\rho_{yy}^{k-1}$
   y is recurrent $\Rightarrow \mathbb{P}_y(X_n = y$ infinitely ofen$) = 1$
2. *y is transient $\Rightarrow \forall x \in S : \mathbb{E}_x[N(y)] = \frac{\rho_{xy}}{1-\rho_{yy}}$;*
   y is recurrent $\Rightarrow \mathbb{E}_y[N(y)] = \infty$
3. *y is recurrent $\wedge\ \rho_{yx} > 0 \Rightarrow \rho_{xy} = \rho_{yx} = 1$ and x is recurrent(recurrent is infectious)*

Proof.

1. Define $Y := \mathbf{1}(\tau_y < 0)$, then

$$\begin{aligned}\mathbb{P}_x(\tau_y^k < \infty) &= \mathbb{E}_x[\mathbb{E}_x[Y \circ \theta_{\tau_y^{k-1}} | \mathcal{F}_{n-1}]\mathbf{1}(\tau_y^{k-1} < \infty)] \\ &= \mathbb{E}_x[\mathbb{E}_{X_{\tau_y^k}}[Y]\mathbf{1}(\tau_y^{k-1} < \infty)] \\ &= \mathbb{P}_x(\tau_y^{k-1} < \infty)\rho_{yy}\end{aligned}$$

   By induction we have:

$$\mathbb{P}_x(\tau_y^k < \infty) = \rho_{xy}\rho_{yy}^{k-1}$$

   Then we know:

$$y \text{ is recurrent} \Rightarrow \rho_{yy} = 1 \Rightarrow \forall k \in \mathbb{N} : \mathbb{P}_y(\tau_y^k < \infty) = \rho_{yy}^k = 1$$

2. Notice $N(y)$ is a random variable takes value at $\mathbb{N}_0$. Therefore:

$$\begin{aligned}\mathbb{E}_x[N(y)] &= \sum_{n=0}^{\infty} n\mathbb{P}_x(N(y)=n)\\ &= \sum_{n=0}^{\infty} \mathbb{P}_x(N(y)\geq n)\\ &= \sum_{n=0}^{\infty} \mathbb{P}_x(\tau_y^n<\infty)\\ &= \sum_{n=0}^{\infty} \rho_{xy}\rho_{yy}^{n-1}\end{aligned}$$

If y is recurrent the expectation goes to infinity. Otherwise $\frac{\rho_{xy}}{1-\rho_{yy}}$.

3.

$$\begin{aligned}0=\mathbb{P}_y(\tau_y=\infty) &\geq \mathbb{P}_y(\tau_y=\infty,\tau<\infty)\\ &= \mathbb{E}_y[\mathbb{E}_y[\mathbf{1}(\forall n\in\mathbb{N}: X_n\neq y)|\mathcal{F}_\tau]\mathbf{1}(\tau<\infty)]\\ &= \mathbb{E}_y[\mathbb{E}_{X_\tau}[\mathbf{1}(\forall n\in\mathbb{N}: X_n\neq y]\mathbf{1}(\tau_x<\infty)]\\ &= \mathbb{E}_y[\mathbb{E}_x[\mathbf{1}(\forall n\in\mathbb{N}: X_n\neq y]\mathbf{1}(\tau_x<\infty)]\\ &= (1-\rho_{xy})\rho_{yx}\end{aligned}$$

By assumption we know $\rho_{yx}>1$. Therefore $1-\rho_{xy}=0$, $\rho_{xy}=1$.
To prove that x is recurrent, notice $\tau_x\leq\tau_y+\tau_x\circ\theta_{\tau_y}$. Notice

□

Theorem 5.5. *The Markov chain (X_n) admit an initial invariant distribution $\pi \Rightarrow$ all states y are recurrent.*

Proof. Assume $\exists y$ transient, then

$$\mathbb{E}_\pi[N(y)] = \mathbb{E}_\pi[\sum_{n=0}^{\infty}\mathbf{1}(X_n=y)] = \sum_{x\in S}\pi(\{x\})\frac{\rho_{xy}}{1-\rho_{yy}} \overset{\text{y tansient}}{\leq} \frac{1}{1-\rho_{yy}}<\infty$$

And by the invariance of the initial distribution

$$\mathbb{E}_\pi[N(y)] = \mathbb{E}_\pi[\sum_{n=1}^{\infty}\mathbf{1}(X_n=y)] = \sum_{n=1}^{\infty}\pi(\{y\})\in\{0,\infty\}$$

Therefore $\pi(\{y\})=0$. □

Remark. The converse is not true.

Definition 5.5 (recurrent & irreducible Markov chain).

- recurrent Markov chain: all states are recurrent
- irreducible Markov chain: all states are pairwise connected
- connected states: $x\sim y \Leftrightarrow \rho_{xx}=1,\rho_{yy},\rho_{xy}>0$, which implies $\rho_{xy}=\rho_{yx}=1$

Theorem 5.6. *(X_n) recurrent Markov chain with invariant initial distribution π, $\mathcal{F}^\theta$ the invariant σ-field of the left shift θ, then*

$$\forall A \in \mathcal{F}^\theta \exists B \subset S : A = \{[X_0] \subset B\} \quad \mathbb{P}_\pi - a.s.$$

Proof. □

Corollary 5.6.1. *(X_n) recurrent Markov chain with invariant initial distribution π is ergodic $\Leftrightarrow$ π-almost all states all connected.*

Proof. □

5.3 Ergodic theorems

Lemma 5.7 (Maximal ergodic lemma). *$Y \in L^1$, T measure preserving, $S_n := \sum Y \circ T^k, S_0 = 0$, $M_n := \max\{S_0, \dots, S_n\}$. We have*

$$\mathbb{E}[\boldsymbol{I}_{M_n > 0}] \geq 0$$

Proof. □

Theorem 5.8 (Birkoff's ergodic theorem). *$X \in L^1$, T measure preserving map on $(\Omega, \mathcal{F}, \mathbb{P})$, then*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n-1} X \circ T^k = \mathbb{E}[X|\mathcal{F}^T] \quad \mathbb{P}.a.s.\& \text{ in } L^1$$

If T is ergodic,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n-1} X \circ T^k = \mathbb{E}X \quad \mathbb{P}.a.s.\& \text{ in } L^1$$

Proof. Define:

- The running average: $A_n : \frac{1}{n}\sum_{n=0}^{n-1} X \circ T^k$.
- limus supremum: $\overline{X} := \limsup A_n$
- limus infimum: $\underline{X} := \liminf A_n$

1. $\overline{X}, \underline{X} \in \mathcal{F}^T$:

$$\frac{n+1}{n} A_{n+1} = A_n \circ T + \frac{1}{n} X_n$$

Hence

$$\begin{aligned} \overline{X} &= \limsup A_n \\ &= \limsup \frac{n+1}{n} A_{n+1} \\ &= \limsup A_n \circ T \\ &= \overline{X} \circ T \end{aligned}$$

2. $\overline{X} = \underline{X}$ $\mathbb{P}$-a.s:

Use the notation in the last lemma. Define $Y := (X - b)\mathbf{1}_{\{\overline{X} < b, \underline{X} > a\}}$. We have

$$\begin{aligned} \lim \mathbf{1}_{\{M_n > 0\}} &= \lim \mathbf{1}_{\{(A_n - b)\mathbf{1}_{\{\overline{X} < b, \underline{X} > a\}} > 0\}} \\ &= \mathbf{1}_{\{\overline{X} < b, \underline{X} > a\}} \end{aligned}$$

3. (A_n) u.i:(lemma of u.i.)

$$\begin{aligned}\mathbb{E}|A_n| &= \frac{1}{n}\mathbb{E}|\sum X\circ T^k| \\ &\leq \frac{1}{n}\sum \mathbb{E}|X\circ T^k| \\ &= \mathbb{E}|X|\end{aligned}$$

and since $(X\circ T^k)$ are u.i. Taking the same A,ε,δ:

$$\begin{aligned}\mathbb{E}[|A_n||\mathbf{1}_A] &= \frac{1}{n}\mathbb{E}[|\sum X\circ T^k||\mathbf{1}_A] \\ &\leq \sup_k \mathbb{E}[|X\circ T^k||\mathbf{1}_A] \\ &< \varepsilon\end{aligned}$$

There for A_n is u.i.

4. $\overline{X} = \underline{X} = \mathbb{E}[X|\mathcal{F}^T]$:

$$\begin{aligned}\overline{X} &= \mathbb{E}[\overline{X}|\mathcal{F}^T] \\ &\overset{L^1}{\leftarrow} \mathbb{E}[A_n|\mathcal{F}^T] \\ &= \mathbb{E}[X|\mathcal{F}^T]\end{aligned}$$

□

Theorem 5.9. *$X \in L^p$, T measure preserving*

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n-1} X\circ T^k = \mathbb{E}[X|\mathcal{F}_T] \quad \mathbb{P}a.s.\& \text{ in } L^p$$

Corollary 5.9.1. *$(X_k; k\geq 0)$ ergodic process in L^1, then*

$$\frac{1}{n}\sum_{k=0}^{n-1} X_k = \mathbb{E}X_1 \quad \mathbb{P}.a.s \quad \text{and} \quad L^1$$

Corollary 5.9.2.

$$\textbf{\textit{T is ergodic}} \Leftrightarrow \forall A,B\in\mathcal{F} : \lim_n \frac{1}{n}\sum_{k=0}^{n-1}\mathbb{P}(A\cap(T^k\circ B)) = \mathbb{P}(A)\mathbb{P}(B)$$

5.4 Poisson process

Review 4 (Poisson, Gamma, Exponential , Beta distribution). Observe:

1. Poisson distribution
   - Assumption: In a fixed interval(could be any length), the occurrence of a certain event is proportional to the length of the interval, and independently distributed. And **the expected occurrence(namely also the proportion constant λ is fixed.**
   - Description: The probability of k times occurrence in the time interval.
2. Gamma distribution:
   - Assumption: Same as Poisson
   - Description: $Gamma_r(t)$ Probability of at least r occurrence in the time interval $(0,t)$. The expectation of a exponential distributed variable approximated the time till the next occurrence.
   - Density:
   $$\gamma_{\alpha,r} = \frac{\alpha^r}{\Gamma(r)} x^{r-1} e^{-\alpha x} \tag{II.1}$$
   The proof:
   $$\begin{aligned} \mathbb{P}_\gamma[(0,t)] &= 1 - \mathbb{P}_\alpha[\{1,\dots,\gamma-1\}] \\ &= 1 - \sum_{k=1}^{n} \frac{(\alpha t)^k}{k!} e^{-\alpha t} \\ (*) &= \int_0^t \frac{\alpha^r}{\Gamma(r)} x^{r-1} e^{-\alpha x} dx \\ \Gamma(r) &= \int_0^\infty y^{r-1} e^{-y} dy, r > 0 \end{aligned} \tag{II.2}$$
3. Beta-Distribution:

Definition 5.6 (counting process, jump time). Define a **monotone increasing** sequence of non-negative RV $(S_i)_{i=0}^n$, the process:

$$N_t := \sum_{k\geq 1} \mathbb{1}(S_k \leq t)$$

is called a counting process with jumping time(S_k).

Definition 5.7 (Poisson Process). A counting process N_t is called a **poisson process** if it satisfies:

1. $\mathbb{P}(N_(t+h) - N_t = 0) = \lambda h + o(h)$
2. $\mathbb{P}(N_(t+h) - N_t = 1) = 1 - \lambda h + o(h)$
3. (independant increament)
4. (stationary increament)

Remark. We can interpret S_k as the time of k-th occurrence of an event.

Theorem 5.10. *The followings are equivalent:*

1. *N is a poisson process*
2. *3, 4, are satisfied and $N_t \overset{d}{\sim} Poiss(\lambda t)$*
3. *The jumping times are i.i.d RV with distribution $Exp(\lambda)$*

6 Weak and functional convergence

Throughout $(S, \mathcal{B}_S)$ denote metric space with Borel $\sigma-$Algebra.

6.1 Fundamental properties

Theorem 6.1 (Continuous mapping). *$g : S \to T$ continuous T another metric space. then*

$$X_n \xrightarrow{d} X \Rightarrow g(X_n) \xrightarrow{d} g(X)$$

Theorem 6.2 (Portmanteau theorem). *For the probability measure $(\mathbb{P}_i)_{i\in\mathbb{N}}$ the followings are equivalent*

1. $\mathbb{P}_i \overset{weakly}{\to} \mathbb{P}$
2. $\forall f$ ***Lipschitz continuous:*** $\int f d\mathbb{P}_i \to \int f d\mathbb{P}$
3.

6.2 Tightness

Definition 6.1. A family of probability measures $(\mathbb{P}_i)_{i\in I}$ is called

- <u>(weakly)relatively compact</u>: Every sequence admits a weakly convergent subsequence
- <u>(uniformly)tight</u>:

$$\forall \varepsilon > 0 \exists K_\varepsilon \textbf{ compact} \forall i \in I : 1 - \mathbb{P}_i(K_\varepsilon) < \varepsilon$$

Theorem 6.3. *Any tight family of probability measure on a separable metric space is relatively compact.*

6.3 Weak convergence on $C([0,T])$, $C(\mathbb{R})$

We denote $C := C([0,T])$ and $\mathcal{B}_C$ the canonical Borel $\sigma - Algebra$

Theorem 6.4. *A sequence of probability measure $(\mathbb{P}_n)$ on $\mathcal{B}_C$ converges weakly to the probability measure $\mathbb{P}$ iff all the finite-dimensional distribution $\mathbb{P}_n^{\pi_1,\dots,\pi_m}$ converges to $\mathbb{P}^{\pi_1,\dots,\pi_m}$ and $(\mathbb{P}_n)$ are tight.*

Definition 6.2 (The modulus of continuity). For $f \in C([0,T])$, the modulus of continuity if defined as

$$\omega_\delta := \{\max |f(t) - f(s)| | s \in B_\delta(t), \quad s,t \in [0,T]\}$$

Theorem 6.5 (Arzila-Ascole). *A family of function in $C([0,T])$ is weakly compact if*

- *uniformly bounded:* $\sup_{f\in A} |f(0)| < \infty$
- *They are equi-continuous*

Theorem 6.6 (Kolmogorov, Centsov criterion). *Let $(X_n(t), 0 \le t \le T)$ be a sequence of continuous process. Then their laws are tight on $C([0,T)$ if*

1. $\lim_{R\to\infty} \sup_n \mathbb{P}(\{X_n(0) \ge R\}) \to 0$
2. $\exists \alpha, \beta > 0, K > 0, \forall n \in \mathbb{N}_0 \forall t, s \in [0,T] : \mathbb{E}[|X_n(t) - X_n(s)|^\alpha] \le K|s-t|^{1+\beta}$

7 Invariant principle and the empirical process

7.1 Invariant principle and the Brownian motion

Definition 7.1 (Brownian motion). A process $(B_t)_{t\geq 0}$ is called a Brownian motion if

- $B_0 = 0$ and $B_t \sim N(0,t)$
- stationary and independent increment:

$$(B_{t_n} - B_{t_{n-1}}, \ldots, B_{t_1} - B_{t_0}) \sim N(0, diag(t_n - t_{n-1}, \ldots, t_1 - t_0))$$

- B_t has a continuous sample

Theorem 7.1 (Invariance principle). *Suppose $(X_n)_{n\geq 0}$ are i.i.d. $X_n \in L^2$, $\mathbb{E}X = 0, Var[X] = 1$, $S_n = \sum X_k$, then the rescaled linear interpolated random walk*

$$Y_n(t) = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor} + \frac{\lfloor nt \rfloor - nt}{\sqrt{n}} X_{\lfloor nt \rfloor + 1}, \quad t \geq 0$$

satisfies

$$Y_n(t) \overset{n\to\infty}{\to} B$$

Proof. :

1. The finite dimensional distribution of Y_n converges to the Brownian motion

□

Remark. To prove the convergence in distribution to normal distribution: CLT
General technique:

1. continuous mapping

Chapter III

Stochastic Analysis

1 Gaussian random variable

Given X_n real random variables with $X_n \sim \mathcal{N}(m_n, \sigma_n)$. Suppose $X_n \xrightarrow{L^2} X$, then it holds

1. $X \sim \mathcal{N}(m, \sigma)$ with $\lim m_n \to m$ and $\lim_n \sigma_n \to \sigma$.
2. $X_n \xrightarrow{L^p} X$ for all $1 \leq p < \infty$.

2 Brownian motion

2.1 Construction

Definition 2.1 (The pre-brownian motion). $(B_t)_{t \geq 0}$ is a pre-brownian motion if

- $B_0 = 0$ almost surely
- $\forall 0 \leq s < t : B_t - B_s \sim \mathcal{N}(0, t-s)$
- $\forall 0 \leq s < t : B_t - B_s \perp (B_r)_{0 \leq r \leq s}$

Definition 2.2 (Haar wavelets).

$$\chi_{n,k}(t) := 2^{\frac{n}{2}} (\mathbf{1}_{[k2^{-n}], 2^{-n-1}(k+1)]}(t) - \mathbf{1}_{[2^{-n-1}(k+1), 2^{-n}(k+1)]}(t))$$

Remark. The Haar wavelet

$$(\chi_{n,k})_{n \geq -1, k \leq n}$$

defines an ONB of $L^2([0,1])$.

Definition 2.3 (The Schauder function). $\forall f \in C^1([0,1])$ with $f(0) = 0$, $f' \in L^2$ we have:

$$\begin{aligned} f(t) &= \int_0^1 f'(s) ds = \int_0^1 \mathbf{1}_{[0,t]}(s) \sum \langle f', \chi_{n,k} \rangle \chi_{n,k} ds \\ &= \sum \langle f', \chi_{n,k} \rangle \underbrace{\int_0^t \chi_{n,k} ds}_{:= \varphi_{n,k}(t)} \end{aligned}$$

The **Schauder function** is defined as

$$\varphi_{n,k}(t) := \int_0^t \chi_{n,k} ds$$

Observe the coefficients of the Schauder basis are the **pointwise evaluation:**

$$\begin{aligned}\langle f', \chi_{n.k}\rangle &= 2^{\frac{n}{2}}(2f(2^{-n-1}(2k+1) - f(2^{-n}k) - f(2^{-n}(k+1)) \\ &:= f_{n,k}\end{aligned}$$

And we have

$$f_n := \sum f_{n,k}\varphi_{n,k}$$

Convince yourself that $f_n = \sum f_{n,k}\varphi_{n,k}$ is a piecewise linear interpolation of the function f.

2.2 The path properties

Definition 2.4 (Sample paths).

Definition 2.5 (Modification and indistinguishable process). Given X and Y two probability processes. Then there are

1. **modification:** of each other if

$$\forall t \in I \exists N_t \texttt{ null set} : X_t = Y_t \texttt{ on } \Omega \backslash N_t$$

2. **indistinguishable:**

$$\exists N : X = Y \texttt{ on } \Omega \backslash N$$

Lemma 2.1 (Modification $\Rightarrow$ Indistinguishable). *Given X and Y modification of each other, if one of the following holds true, then X and Y are indistinguishable.*

1. *I is countable*

2. *X, Y are right continuous.*

Proof.
1): trivial
2): By right continuity:

$$\forall \omega \in N_t \texttt{ s.t. } \quad X_t(\omega) \neq Y_t(\omega) : X_r(\omega) \neq Y_r(\omega) \quad \forall r \in [t,s) \texttt{ for some } s \in \mathbb{Q}$$

We may merge the uncountable null set N_t into a collection of countable null sets. □

Definition 2.6 (Holder-Continuous functions). For $\alpha \in (0,1)$. The space of α-*Holder* continuous function is defined as

$$C^{\alpha}([0,T]), \mathbb{R}) := \{f : [0,T] \to \mathbb{R} \quad | \quad \|f\|_{\alpha} < \infty\} \qquad \|f\|_{\alpha} := \sup_{0 \leq s \leq t \leq T} \frac{|f(s) - f(t)|}{|t-s|^{\alpha}}$$

Lemma 2.2 (Ciesielski). *For a function f, given $\alpha \in (0,1)$ inheriting notations above*

- $\forall n \geq -1, 0 \leq k \leq n$:,

$$\sup_{n\geq 1} \max_{k\leq n} 2^{n(\alpha-\frac{1}{2})}|f_{n,k}| < C$$

then $\sum f_{n,k}\varphi_{n,k}$ *converges (abs)uniformly to a* $f \in C^{\alpha}([0,T])$.

- *Conversely*

$$\sup_{n\geq -1} \max_{k\leq n} 2^{n(\alpha-\frac{1}{2})}|f_{n,k}| \leq \|f\|_{\alpha}$$

Proof. 1. First we show the convergence. Given any point in the interval:

$$\begin{aligned}
\sum_{n,k}|f_{n,k}\varphi_{n,k}| &= \sum_{n}\max_{k}|f_{n,k}\varphi_{n,k}| \\
(\|\varphi_{n,k}\|_{\infty} = 2^{-\frac{n}{2}}) &\leq \sum_{n}\max_{k}|f_{n,k}|2^{-\frac{n}{2}} \\
&\leq \sum_{n} C2^{-n\alpha+\frac{n}{2}-\frac{n}{2}} \\
&\to f
\end{aligned}$$

by convergence of the geometric series we know $\sum f_{n,k}\varphi_{n,k}$ is absolutely and uniformly convergent to a f.

Then we show the α-continuity. Choose $2^{-n_0-1} \leq t < s \leq 2^{-n_0}$

$$\begin{aligned}
|f(s)-f(t)| &\leq \sum_{n\geq -1} |\sum_{k\leq 2^n} f_{n,k}\varphi_{n,k}(s) - f_{n,k}\varphi_{n,k}(t)| \\
&\leq \sum_{n\geq -1} 2\max|f_{n,k}|\|\varphi_{n,k}\|_{\infty} \\
&\leq \sum_{n\leq n_0} 2C2^{-n(\alpha-\frac{1}{2})}\|\varphi_{n,k}\|_{\infty} + \sum_{n>n_0} 2C2^{-n(\alpha-\frac{1}{2})}2^{-n_0} \\
&\leq 2C\sum_{n\leq n_0}
\end{aligned}$$

2. By the definition of $f_{n,k}$

$$\begin{aligned}
|f_{n,k}| &:= \langle f', \varphi_{n,k}\rangle \\
&= 2^{-\frac{n}{2}}(2f(\frac{2k+1}{2^{n+1}}) - f(\frac{k}{2^n}) - f(\frac{k+1}{2^n})) \\
&= 2^{-\frac{n}{2}}(|f(\frac{2k+1}{2^{n+1}}) - f(\frac{k}{2^n})| + |f(\frac{2k+1}{2^{n+1}}) - f(\frac{k+1}{2^n})|) \\
&\leq 2^{-\frac{n}{2}}\|f\|_{\alpha}2^{\alpha(-n-1)}2
\end{aligned}$$

up to a constant we showed the second result. □

Lemma 2.3. *Let* $\alpha \in \mathbb{R}, p \in [1,\infty]$, $f_{n,k}$ *are real numbers. We define*

Theorem 2.4 (Kolmogorov-Chentsov continuity theorem). *Given X a stochastic process and assume:*

$$\mathbb{E}|X_t - X_s|^q \leq |s-t|^{1+\varepsilon}$$

then X admits a modification $\tilde{X}$ *that is a.s.* α*-Holder-continuous with* $\alpha \in (0, \frac{\varepsilon}{q})$ *, i.e.,*

$$|\tilde{X}_t(\omega) - \tilde{X}_s(\omega)| \leq C(\omega)|t-s|^{\alpha}$$

Proof.
1). Use Markov inequality to bound the probability of discontinuity at a single point.

2): Use σ-subadditivity to prove probability at dyadic points are bounded. Infer by Borel-Cantelli:

$$K_\alpha(\omega) := \frac{|X_{i2^{-n}} - X_{(i+1)2^{-n}}|}{2^{-n\alpha}} < \infty$$

3): Use the lemma to infer $\forall s,t$ in the dyadic cube

$$|X_t - X_s| \leq C(\omega)|s-t|^\alpha$$

4) Extends process to $[0,1]$ for a.e. ω and define an arbitrary value on the null set. □

Corollary 2.4.1. *Every pre-Brownian motion admits a continuous modification.*

2.3 Excercise

Exercise 2.3.1. *Let B be a pre-Brownian motion and $f : \mathbb{R}^+ \to \mathbb{R}$ are defiend as $f = \sum \lambda_i \mathbf{1}_{[t_{i+1}-t_i]}$ and $\xi(f) := \sum \lambda_i (B_{i+1} - B_i)$*

c.2.11, finite not bounded. a.s. sense. the constant depends on omega

3 Filtrations and stopping time

Definition 3.1 ($\mathbb{F}$-stopping time). as in the discrete case, a stopping time w.r.t. the filtration $\mathbb{F}$ is a non-negative random variable on the probability space such that

$$\{\tau \leq t\} \in \mathcal{F}_t$$

Definition 3.2 (stopped σ-field(σ-field of the past before τ)). Define The stopped filtration for the stopping time τ is defined as

$$\mathcal{F}_\tau := \{A \in \mathcal{F}_\infty : A \cap \{\tau \leq t\} \in \mathcal{F}_t \quad \forall t \geq 0\}$$

Remark. Check that $\mathcal{F}_\tau$ is indeed a σ-field. Namely we should check 1. $\Omega \in \mathcal{F}_\tau$, 2. Inclusion of the complement, 3. Inclusion of the countable union.

Remark. $\mathcal{F}_\infty$.

Definition 3.3 (The right σ-field). We define for the filtration $\mathbb{F}$:

$$\mathcal{F}_\infty := \vee_{t\geq 0}\mathcal{F}_t \qquad \texttt{and} \qquad \mathcal{F}_{t^+} := \bigcap_{s\geq t} \mathcal{F}_s$$

Lemma 3.1.

Lemma 3.2. *Given a stopping time τ:*

1. *τ is $\mathcal{F}_\tau$-measurable*

2. ***(The trace σ-field)**: For the deterministic stopping time $\tau \equiv t$:*

$$\mathcal{F}_\tau = \mathcal{F}_t$$

Remark. Given a stopping time τ, $\tau + a$ is also a stopping time for $a \in [0, \infty]$.

Definition 3.4 (progressive σ-field). A stochastic process $(X_t)_{t\geq 0}$ is called **product measurable** if it's $\mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$ measurable. And it's **progressive measurable** if

$$\forall t : X_t \texttt{ is } \mathcal{F}_t \otimes \mathcal{B}(\mathbb{R}_+)\texttt{-measurable}$$

we define the **progressive σ-filed** w.r.t. the filtration $\mathbb{F}$:

$$Prog(\mathbb{F}) := \{A \in \mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+) : \mathbb{1}_A \texttt{ is progressively measurable}\}$$

Lemma 3.3.

Proof. X is progressively measurable
$\Rightarrow X^{-1}(B) \cap (\Omega \times [0,t]) \in \mathcal{F}_t \otimes \mathcal{B}([0,t])$
Define $A := X^{-1}(B)$, we see $\mathbb{1}_A$ is progressively measurable. □

Definition 3.5 (d-dim Brownian motion).

Proposition 3.4.

1. *Given X progressively measurable, then the terminal value of the stopped process X_τ is $\mathcal{F}_\tau$ measurable.*

2. *The stopped σ-field is the smallest σ field that generated by X_τ for a X progressively measurable*

Proof. ref. Protter □

Theorem 3.5 (Strong Markov property of Brownian motion). *Given X a $\mathbb{F}$-Brownian motion and τ a stopping time, then we have*

$$X_h^\tau = \mathbb{1}_{\tau<\infty}(X_{\tau+h} - X_\tau)$$

Proof. equivalent to show for arbitrary $p \in \mathbb{N}$:

$$\forall f \in C_b(\mathbb{R}^p) \forall A \in \mathcal{F}_\tau : \mathbb{E}[\mathbb{1}_A f(X_{t_1}^\tau, \ldots, X_{t_p}^\tau)] = \mathbb{P}(A)\mathbb{E}[f(X_{t_1}, \ldots, X_{t_p})]$$

taking $A = \Omega$ we see

$$(X_{t_1}^\tau, \ldots, X_{t_p}^\tau) \sim (X_{t_1}, \ldots, X_{t_p})$$

which shows that the lefthand side in independent of $\mathcal{F}_\tau$.

2) Approximate τ from right with dyadic points and transform to a problem of weak Markov property:

□

4 Martingale in continuous time

4.1 Path regularity

4.2 The stopping theorem

Theorem 4.1. *Let X be **right-continuous u.i.** martingale, S,T stopping time with $S \leq T$, then we have*

1. $X_S, X_T \in L^1$
2. $\mathbb{E}[X_T|\mathcal{F}_s] = X_S$

5 Continuous semartingale

5.1 process of finite variation

Definition 5.1 (finite variation). a continuous function $a : [0,T] \to \mathbb{R}$ with $a(0) = 0$ is said to have *finite variation* if

$$a(t) = \mu([0,t])$$

for some signed measure μ.

5.2 local martingale and the quadratic variation

we observe the (continuous) local martingale and its quadratic variation, namely:

$$M \in \mathcal{M}^c_{loc}, \quad \langle M \rangle := \langle M, M \rangle$$

Definition 5.2. An adapted continuous process M is called the local martingale if there exists a sequence of stopping time (τ_n) with $\tau_n \to \infty$ such that

$$M^{\tau_n} \texttt{ are martingale}$$

we write:

$$M \in \mathcal{M}^c_{loc}$$

Remark. we write $\mathcal{M}^c$ for the **uniformly integrable** continuous martingale.

1. many useful convergence, e.g., $\mathcal{M}^c_{loc}$ is stable under stopping.
2. more general definition: $(M - M_0)^{\tau_n}$ is uniformly integrable. (relaties to stochastic integral).

Proposition 5.1. *for $M \in \mathcal{M}^c_{loc}$ we have*

1. *$M \geq 0 \Rightarrow M$ is a supemartingale.*
2. *if $|M_t| \leq Z$ for all t with $Z \in L^1$, then $M \in \mathcal{M}^c_{loc}$*
3. *if $M_0 = 0$ or $M_0 \in L^1$, then*

$$\tau_n := \inf\{t \geq 0 | |M_t| \geq n\} \quad (\inf\{t \geq 0 | |M_t - M_0| \geq n\})$$

is a localising sequence for M.

Proof.
1): Fatou
2) By the assumption the process is uniformly integrable. Hence

$$M_s = \lim M_s^{\tau_n} = \mathbb{E}[\lim_n M_t^{\tau_n} | \mathcal{F}_s]$$

3) The natural localising stopping time to consider. Observe the equation of (indistinguishable) process

$$M^{\tau_n} = (M - M_0)^{\tau_n} + M_0^{\tau_n} \in \mathcal{M}^c$$

□

The intersection of continuous martingale and process of finite variation is almost trivial.

Lemma 5.2. *Let $M \in \mathcal{A} \cap \mathcal{M}^c_{loc}$, then*

$$M_t = M_0 = 0 \quad \forall t \geq 0$$

i.e., almost every trajectory of M is constant.

Theorem 5.3. *Let $M \in \mathcal{M}^c_{loc}$, then there exists an increasing process $\langle M, M \rangle := \langle M \rangle \in \mathcal{A}^+$, unique up to indistinguishability, such that*

$$M^2 - M_0^2 - \langle M, M \rangle \in \mathcal{M}^c_{loc}$$

Proof.
Uniqueness:

Existence:w.l.o.g $M_0 = 0$. Now we have $M_0 = 0$, M bounded and

$$M^2 - \langle M, M \rangle \texttt{ is a martingale}$$

we define

$$S_n := \inf\{t \geq 0 | |M|_t \geq n\}$$

Define

$$T_0^n := 0, T_{k+1}^n := \inf\{t \geq T_k^n : M_t \in 2^{-n}$$

□

Chapter IV

Stochastic Finanzmathe

1 Discrete Financial Modelling

1.1 1-period Financial Modelling

Arbitrage and martingale measure

Definition 1.1 (Arbitrage opportunity).

$$\exists \bar{\xi} \in \mathbb{R}^{d+1} : \bar{\xi}\bar{\pi} \leq 0, \quad \bar{\xi}\bar{S} \geq 0, a.s., \mathbb{P}(\bar{\xi}\bar{S} > 0) > 0$$

The next lemma shows we may w.l.o.g. observe arbitrage as portfolio of the risky assets.

Lemma 1.1. *The following arguments are equivalent*

- *there exists an arbitrage* $\bar{\xi} \in \mathbb{R}^{d+1}$.
- *there exists* $\xi \in \mathbb{R}^d : \xi \cdot Y \geq 0$ $\mathbb{P}$ *a.s.* $\mathbb{P}(\xi \cdot Y > 0) > 0$.

Definition 1.2 (Martingale measure and equivalent Martingale measure). A probability measure $\mathbb{P}*$ is martingale if:

$$\pi^i = \mathbb{E}^* \left[\frac{S^i}{1+r} \right] \quad \forall i = 1, \ldots, d$$

We denote the set of equivalent martingale measure by:

$$\mathcal{P} := \left\{ \mathbb{P}^* \approx \mathbb{P} | \mathbb{E}^* \left[\frac{S^i}{1+r} \right] \quad \forall i = 1, \ldots, d \right\}$$

Theorem 1.2 (The 1st Fundament Theorem of Asset Pricing). *The market model is free of arbitrage if and only if* $\mathcal{P} \neq \emptyset$. *In this case there exists a* $\mathbb{P}^*$ *with* $\frac{d\mathbb{P}^*}{d\mathbb{P}} \in L^\infty(\mathbb{P})$

We define all the possible payout the the market:

$$\mathcal{V} := \{\bar{\xi} \cdot \bar{S} : \bar{\xi} \in \mathbb{R}^{d+1}\}$$

the following theorem suggest if a payout is replicable and the market is free of arbitrage, then it admits only one price.

Theorem 1.3 (Law of 1 Price). *Any replicable payout in a arbitrage-free market has only one price, i.e. given* $v \in \mathcal{V}$ *with* $v = \xi_1 \cdot \bar{S} = \xi_2 \cdot \bar{S}$, $\xi_1, \xi_2 \in \mathbb{R}^{d+1}$, *then we have*

$$\xi_1 \cdot \bar{\pi} = \xi_2 \cdot \bar{\pi}$$

Proof. □

Contingent Claim

Definition 1.3 (Contingent Claim). A **claim** is a non-negative RV on the probability space. A **contingent claim** is a $\sigma(S_1, \dots)$ measurable claim.

Example (Put-Call parity). C - P = S - K

Theorem 1.4 (Arbitrage-free price of a contingent claim). *Given $\Pi(C)$ the set of all non-negative arbitrage-free price for the contingent claim C. If $\mathcal{P} \neq \emptyset$, then:*

$$\Pi(C) = \left\{ \mathbb{E}^* \left[\frac{C}{1+r} \right] : \mathbb{P}^* \in \mathcal{P} \text{ with } \mathbb{E}^* C < \infty \right\}$$

Remark. The non-negativity of the price of contingent claim is crucial for the non-arbitrage given a martingale measure.

Theorem 1.5 (Range of a contingent claim). *If $\mathcal{P} \neq \emptyset$, and C a contingent claim, then:*

$$\pi_{\inf}(C) := \inf \Pi(C) = \inf_{\tilde{P}} \tilde{\mathbb{E}}(\frac{C}{1+r}) = \max \left\{ m \in [0, \infty) : \exists \xi \in \mathbb{R}^n : m + \xi Y \leq \frac{C}{1+r} \mathbb{P} f.s. \right\}$$

$$\pi_{\sup}(C) := \sup \Pi(C) = \sup_{\tilde{P}} \tilde{\mathbb{E}}(\frac{C}{1+r}) = \min \left\{ m \in [0, \infty) : \exists \xi \in \mathbb{R}^n : m + \xi Y \geq \frac{C}{1+r} \mathbb{P} f.s. \right\}$$

Remark.
1. The fact that the price is a interval doesn't violate the law of 1 price since the contingent claim is in general not replicable.
2. The supremum is usually called the minimum super-hedging price.

Proof. The proof has the following steps:

1. $\sup_{P^*} \Pi(C) = \inf M$:
2. $\sup_{P^*} \Pi(C) = \sup_{\tilde{P}} \Pi(C)$: Convex combination of absolute continuous measure and equivalent measure belongs to equivalent measure. We break an absolute continuous measure into these 2 parts therefore it is bounded by the equivalent measure. Then take the ε to 0.
3. $\inf M = \min M$:

□

Definition 1.4 (Replicable contingent claim). A contingent claim is replicable, if there is a $\bar{\xi} \in \mathbb{R}^{d+1}$, such that $C = \bar{\xi} \bar{S}$ $\mathbb{P} - a.s.$

Remark. By law of one price, we know if that the price is uniquely determined by $\bar{\xi} \bar{S}$

Corollary 1.5.1. *Given C a contingent claim and $\mathcal{P} \neq \emptyset$.*

1. *C is replicable iff $|\Pi(C)| = 1$*
2. *C is not replicable, then $\Pi(C) = (\pi_{\inf}(C), \pi_{\sup}(C))$ is an open interval*

Complete market model.

Definition 1.5 (complete financial market). A market is complete if every contingent is claim replicable.

Theorem 1.6 (The 2nd fundamental theorem of Asset pricing). *A market is complete iff* $|\mathcal{P}| = 1$

Remark. If the market is complete, we have:

$$V = \{\bar{\xi}\bar{S} | \bar{\xi} \in \mathbb{R}^{d+1}\} = L^0(\Omega, \mathcal{F}, \mathbb{P})\}$$

Theorem 1.7. *Every complete market model admits a partition of* Ω *in mostly* $d+1$ *atoms.*

1-Period market model with randomised initial price

Theorem 1.8 (The 2nd Fundamental theorem of Asset Pricing, Version II). *The following statements are equivalent*

1. $K \cap L^0_+ = \{0\}$
2. $(K \cap L^0_+) \cap L^0_+ = \{0\}$
3. $\exists \mathbb{P}^* \in \mathcal{P} : \frac{d\mathbb{P}^*}{d\mathbb{P}} \in L^\infty$.
4. $\mathcal{P} \neq \emptyset$

Proof.
2) $\Rightarrow$ **3)**. □

1.2 Multi-Period Arbitrage theory

Arbitrage opportunity and Martingale measures

Theorem 1.9 (The Radon-Nikodym Theorem). *Given* $\mathbb{Q} \ll \mathbb{P}$ *and* $Z := \frac{d\mathbb{Q}}{d\mathbb{P}}$. *Then for all non-negative random variable X and sub-*σ*-Algebra* $\mathcal{F}_0 \subset \mathcal{F}$*:*

$$\mathbb{E}^{\mathbb{Q}}(X|\mathcal{F}_0) = \frac{\mathbb{E}^{\mathbb{P}}(XZ|\mathcal{F}_0)}{\mathbb{E}^{\mathbb{P}}(X|\mathcal{F}_0)}$$

Theorem 1.10 (Separating theorem of Hahn-Banach).

Definition 1.6 (self-financed strategies). A trading strategy is called self-financed, if

$$\bar{\xi}_t \bar{S}_t = \bar{\xi}_{t+1} \bar{S}_t \quad \forall t = 1, \ldots, T-1$$

Remark. For a trading strategy $\bar{\xi} = (\xi, \xi_0)$, having ξ fixed, we may select ξ_0 such that the trading strategy becomes self-financed, in that we choose the ξ_0 that satisfies the equation above.

The Value Process

Lemma 1.11 (Decomposition of the value process). *The following arguments are equivalent*

1. $\bar{\xi}$ *is a self-financed strategy*
2. $\bar{\xi}_t \cdot \bar{X}_t = \bar{\xi}_{t+1} \cdot \bar{X}_t, \forall t$

3. $V_t = V_0 + \sum_{k=1}^{t} \xi_k \cdot (X_k - X_{k-1})$

Definition 1.7 (Martingale measure). A martingale measure on $(\Omega, \mathcal{F})$ is a martingale measure, if the discounted price process X is a $\mathbb{Q}$-Martingale.

How do we find the equivalent market Martingale measure?

Theorem 1.12. *In a probability space the following arguments are equivalent*

1. *$\mathbb{P}$ is a martingale measure*
2. *For every self-financed strategy $\bar{\xi}$ with $\xi \in L^\infty(\mathbb{P})$, the induced value process is also a martingale*
3. *For a self-financed strategy, if the value process V_T^- is integrable(which mean we have bounded capital), then the value process is a martingale*
4. *For a self-financed strategy, if the value process is non-negative a.s., then $\mathbb{E}V_T = \mathbb{E}V_0$*

Theorem 1.13 (The first fundamental theorem of asset pricing). *The multi-period model is arbitrage-free iff $\mathcal{P} \neq \emptyset$.*

Definition 1.8 (replicable contingent claim).

Theorem 1.14. *The market is free of arbitrage and H is a discounted contingent claim, then:*

1. $\Pi(H) = \{\mathbb{E}^*(H) : \mathbb{P}^* \in \mathcal{P} and \mathbb{E}^* < \infty\} \neq \emptyset$
2. $\pi_{\inf}(H) = \inf_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*(H)$ *and* $\pi_{\sup}(H) = \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}^*(H)$.

Proof. □

Definition 1.9 (Arbitrage price).

Remark.

Cox-Ross-Rubinstein Modell(Binominal Model)

Theorem 1.15.
The CRR model is arbitrage-free $\Leftrightarrow a < r < b$. In this case $\mathcal{P} = \{\mathbb{P}^*\}$ with

$$p^* := \mathbb{P}^*(R_t = b) = \frac{r-a}{b-a}$$

And under $\mathbb{P}^*$, the returns in each period are independent.

Proof. □

Theorem 1.16 (Hedging Strategy).

$$\xi_t := \Delta_t(S_0, \ldots, S_{t-1}) = \frac{v_t(x_0, x_{t-1}, x_{t-1}(1+b)) - v_t(x_0, x_{t-1}, x_{t-1}(1+a))}{(1+r)^{-t} x_{t-1}(b-a)}$$

Choose ξ_t^0 such that the strategy $\bar{\xi}$ is self-financed with the initial capital $V_0 = \mathbb{E}^*(H)$

Proof. □

Exotic Derivative

Definition 1.10.

1. running maximum: $M_t := \max_{0 \le s \le t} Z_s$

Theorem 1.17 (Reflexion Principle). *Z_t standard random walk. $\mathbb{P}$ uniform distribution. $\forall k \in \mathbb{N}, l \in \mathbb{N}_0$,*

$$\mathbb{P}(M_T \ge k, Z_T = k - l) = \mathbb{P}(Z_t = k + l)$$

Proof. By using the obvious fact

$$k + l > k, \quad l \ge 0$$

to get rid of $M_T \ge k$. □

Theorem 1.18 (Reflexion Principle under $\mathbb{P}^*$).

$$\mathbb{P}^*[M_T \ge k, Z_T = k - l] = \left(\frac{1 - p^*}{p^*}\right)^l \mathbb{P}^*[Z_T = k + l]$$

Proof. Calculate the bounded density: $\frac{d\mathbb{P}^*}{d\mathbb{P}}$

$$\mathbb{P}^*(\omega) = (p^*)^{[T + Z_T(\omega)]/2}(1 - p^*)^{[T - Z_T(\omega)]/2} = \frac{d\mathbb{P}^*}{d\mathbb{P}} d\mathbb{P}(\omega) = \frac{d\mathbb{P}^*}{d\mathbb{P}} 2^{-T}$$

□

Theorem 1.19 (Up-and-in call option).

Proof.

1. Split the expectation
2. reflection principle
3. measure change

□

Black-Schole Convergence

Lemma 1.20 (Central limit theory for sum).

Lemma 1.21 (Slusky lemma).

Theorem 1.22 (Black-Schole Convergence theorem). *Assuming*

1. *the returns are i.i.d., (clear from 2.2.1)*
2. $\sum_{k=1}^{N} Var^*(R_N^k) \to T\sigma^2$
3. $-1 \le \alpha_N \le R_t^N \le \beta_N$ *with* $\lim \alpha_N = \lim \beta_N = 0$

We have:

$$S_N^{(N)} \xrightarrow{d} S_T := S_0 \exp\left((rT - \frac{1}{2}\sigma T) + \sigma W_T\right) \quad W_t \sim N(0, T)$$

Exercise 1.2.1. *down-and-in put option*

Exercise 1.2.2.

$$C_{BS}^{(N)}$$

Determine the price in the Nth CRR-Model at $t = 0$. Calculate the limit as $N \to \infty$.

American option

2 Time-continuous finance modelling

2.1 Ito-calculus

Definition 2.1 (Brownian motion). A stochastic process $(W_t)_{tgeq0}$,

1. $W_0 = 0$ $\mathbb{P}$-$a.s.$
2. independent increment:
3. normal distributed increment:
4. $\mathbb{P}$-a.s. continuous path

Remark.

$$\mathrm{Cov}(W_s, W_t) =$$

Theorem 2.1. *The following are martingales:*

1. (W_t)
2. $(W_t^2 - t)$
3. $\left(\exp(\sigma W_t - \frac{\sigma^2}{2}t)\right)$

Proof.

- The second

$$\begin{aligned}\mathbb{E}[W_t^2 - t|\mathcal{F}_s] &= \mathbb{E}[(W_t - W_s)^2 - W_s^2 - 2W_sW_t|\mathcal{F}_s] - t \\ &= \mathbb{E}(W_t - W_s)^2 + W_s^2 - t \\ &= W_s^2 - s\end{aligned}$$

- The third

$$\begin{aligned}\mathbb{E}[e^{\sigma W_t - \frac{\sigma^2}{2}t}|\mathcal{F}_s] &= e^{\sigma W_s - \frac{\sigma^2}{2}t}\mathbb{E}[e^{\sigma(W_t - W_s)}] \\ &= e^{\sigma W_s - \frac{\sigma^2}{2}t}\mathbb{E}[e^{x\sigma\sqrt{t-s}}] \quad (x \sim N(0,1)) \\ &= e^{\sigma W_s - \frac{\sigma^2}{2}s}\end{aligned}$$

□

Definition 2.2 (total variation).

$$TV_t(X) := \sup_{\Pi} \sum_{t_i \leq t} |X_{t_{i+1}\wedge t} - X_{t_1}]$$

a continuous and deterministic path $X : [0,\infty) \to \mathbb{R}$ is of

1. finite variation: $\forall t \in [0,\infty) : TV_t < \infty$.
2. bounded variation for $T \in [0,\infty)$: $TV_T < \infty$.

Lemma 2.2. *X is of finite variation $\Leftrightarrow \exists A^+, A^-$ monoton $: X = A^+ - A^-$*

Remark. Lebeague theory and define a integral $\int_0^t f(s)dX_s$

Definition 2.3 (Quadratic variation). $X_t : [0,\infty) \to \mathbb{R}$ continuous. For $\Pi_n \to 0, t \to \langle X\rangle_t$ continuous, we define the quadric variation of (X_t) as:

$$\langle X\rangle_t := \lim_n \sum_{t_i<t, t_i\in\Pi_n} (X_{t_{i+1}-t_i} - X_{t_1})^2$$

Remark. $\langle X\rangle_t$ depends on Π_n.

Lemma 2.3. *X is of bounded variation for $T \Rightarrow \langle X\rangle_T = 0$*

Corollary 2.3.1. *A is of bounded variation $\Rightarrow \langle X+A\rangle_T = \langle X\rangle_T$*

Lemma 2.4. *$\langle X\rangle_t(\Pi_n)$ is well defined, then $\forall g \in C(\mathbb{R}_+,\mathbb{R})$*

$$\sum_{t_i\in\Pi_n t_i<t} g(t_i)(X_{t_{i+1}} - X_{t_i})^2 \to \int_0^t g(s)d\langle X\rangle_s$$

Proof. $\langle X\rangle_t$ monotone and of finite variation(σ-finite measure)
$\Rightarrow \mu^n(0,t) := \sum_{t_i\le t, t_i\in\Pi_n}(X_{t_{i+1}\wedge t} - X_{t_i})^2$ defines a measure. And since the quadratic variation is well-defined it converges pointwise to the measure induced by $\langle X\rangle$.
$\Rightarrow$ It implies the weak convergence and by the definition of weak convergence we have the result. □

Lemma 2.5 (Excersice). *X of finite quadratic variation, then*

$$\forall g \in C^1 \forall t \in [0,T] : \langle g(X)\rangle_t := \sum_{t_i\le t, t_i\in\Pi_n} (g(X_{t_{i+1}\wedge t}) - g(X_{t_i}))^2 < \infty$$

and

$$\langle g(X)\rangle_t = \int_0^t g'(X_s)^2 d\langle X\rangle_s$$

Proof. By mean value theorem, for $\forall 0 < s < t < T \exists o \in (s,t)$:

$$\begin{aligned}
g(X_t) - g(X_s) &= g'(X_s)(X_t - X_s) + (g'(X_o) - g'(X_s))(X_t - X_s) \\
(g(X_t) - g(X_s))^2 &= g'(X_s)^2(X_t - X_s)^2 + [2g'(X_s)(g'(X_o) - g'(X_s)) + (g'(X_o) - g'(X_s))^2](X_t - X_s)^2 \\
&= g'(X_s)^2(X_t - X_s)^2 + [(g'(X_s) - g'(X_o))(g'(X_o) - g'(X_s))^2](X_t - X_s)^2 \\
&\Rightarrow g'(X_s)^2(X_t - X_s)^2 \quad \text{by the uniform continuity of g'(x) on the interval } [s,t]
\end{aligned}$$

□

Theorem 2.6. *$(W_t)_{0,T}$ Brownian motion, $\Pi_n \to 0$ partition of $[0,T]$ and $\sum|\Pi_n| <$ infty, then*

$$\lim_n \sum (W_{t_{i+1}} - W_{t_i})^2 \to t \quad \forall t \in [0,T] \quad \mathbb{P}\text{-}a.s.$$

Proof. □

Theorem 2.7 (Ito-Formal). *$X : [0,T] \to \mathbb{R}$ continuous and admits continuous quadratic variation. $A : [0,T] \to \mathbb{R}$ continuous and admits bounded total variation. Then $\forall F \in C^{1,2}(\mathbb{R}\times\mathbb{R},\mathbb{R})$*

$$\int_0^t F_x(A_s, X_d)dX_s := \lim \sum F(A_{t_i}, X_{t_i})(X_{t_{i+1}\wedge t_i} - X_{t_i})$$

is well defined and

$$F(A_t, X_t) = F(X_0, A_0) + \int_0^t F_a(A_s, X_s)dA + \int_0^t F_x(A_s, X_s)dX_s + \int_0^t F_{xx}(A_s, X_s)d\langle X\rangle_s$$

Corollary 2.7.1.

Lemma 2.8 (Associativity of Ito-formel). *$X : \mathbb{R} \to \mathbb{R} \in C(\mathbb{R})$ and $\forall t : \langle X \rangle_t < \infty$, $f \in C^2(\mathbb{R}), Y : \mathbb{R}_+ \to \mathbb{R}$ continuous, Assuming the integral exists:*

$$\int_0^t Y_s f'(X_s) dX_s = \lim_n \sum_{t_i < t, t_i \in \Pi_n} Y_{t_i} f'(X_{t_i})(X_{t_{i+1} - t_i} - X_{t_1})$$

and

Example.

Example.

Example (The geometric Brownian motion).

$$S_t = s_0 e^{\sigma W_t + (\mu - \frac{\sigma^2}{2})t}$$

2.2 Perfect dynamic hedge with out interest

Example (Black-Schole Model).

Example (Bachelier Model).

1. $X_t = X_0 + \mu t + \sigma W_t$
2. $\sigma(s,x) = \sigma$ constant

Assumption(Volatility profile):

- The quadratic variation of the price X_t could be expressed as

$$\langle X \rangle_t := \int_0^t \sigma(s, X_s) ds$$

-

The Heat equation

Lemma 2.9. *The boundary value problem:*

$$F_t(t,x) + \frac{1}{2}\sigma^2(x,t) F_{xx}(t,x) = 0 \tag{IV.1}$$

$$F(T,x) = f(x) \tag{IV.2}$$

If $F \in C^{1,2}((0,T) \times \mathbb{R}) \cap C([0,T] \times \mathbb{R})$ and the BVP admits a solution then the value process

$$H(\omega) := f(X_T(\omega)) = F(0, X_0) + \int_0^T F_x(s, X_s(\omega)) dX_s(\omega)$$

exist $\mathbb{P}$-a.s.

Theorem 2.10. *f continuous and satisfies*

$$|f(x)| \leq \alpha(1+e^{k|x|^{\beta}}) \quad \alpha,\beta < 0, k \in (0,2)$$

The the BVP

$$F_t(t,x) + \frac{1}{2}\sigma^F_{xx}(t,x) = 0, x \in [0,T) \tag{IV.3}$$

$$F(T,x) = \lim_{t \to T} f(x) \tag{IV.4}$$

has the solution

$$F(t,x) := \int_{\mathbb{R}} f(y)P(\sigma^2(T-t), x-y)dy$$

Chapter V

Interest Rate Modelling

1 Introduction

2 Stochastic basics

Theorem 2.1 (Bayes's rule for conditional expectation). *Let*

$$R = \frac{d\hat{\mathbb{P}}}{d\mathbb{P}}$$

be the Radon–Nikodym derivative associated with $(\Omega, \mathcal{F}, \mathbb{P})$ *and* $(\Omega, \mathcal{F}, \hat{\mathbb{P}})$*, and* X *a random variable. Then*

$$\mathbb{E}^{\hat{\mathbb{P}}}[X \mid \mathcal{F}_t] = \frac{\mathbb{E}^{\mathbb{P}}[RX \mid \mathcal{F}_t]}{\mathbb{E}^{\mathbb{P}}[R \mid \mathcal{F}_t]}.$$

Theorem 2.2 (Itô's Isometry). *Let* $\{W_t\}_{t\geq 0}$ *be a standard Brownian motion on a probability space* $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$*. Suppose* $X = \{X_s\}_{s\in[0,T]}$ *is an* adapted*, square-integrable process (predictable in the strict sense) such that*

$$\mathbb{E}\Big[\int_0^T X_s^2\, ds\Big] < \infty.$$

Then the Itô integral

$$I_T(X) \;=\; \int_0^T X_s\, dW_s$$

satisfies

$$\mathbb{E}\Big[\big(I_T(X)\big)^2\Big] \;=\; \mathbb{E}\Big[\int_0^T X_s^2\, ds\Big].$$

Sketch of Proof. We prove the result in two main steps:

Step 1: The case of simple (step) processes. Let X_t be a simple, predictable process of the form

$$X_t(\omega) \;=\; \sum_{k=0}^{n-1} \xi_k(\omega)\, \mathbf{1}_{[t_k, t_{k+1})}(t),$$

where $0 = t_0 < t_1 < \cdots < t_n = T$ and each ξ_k is $\mathcal{F}_{t_k}$-measurable. Then the Itô integral is

$$\int_0^T X_s\, dW_s \;=\; \sum_{k=0}^{n-1} \xi_k\big(W_{t_{k+1}} - W_{t_k}\big).$$

Since $W_{t_{k+1}} - W_{t_k}$ are independent increments with mean zero and variance $t_{k+1} - t_k$, we have

$$\mathbb{E}\Big[\Big(\sum_{k=0}^{n-1} \xi_k \big(W_{t_{k+1}} - W_{t_k}\big)\Big)^2\Big] \;=\; \sum_{k=0}^{n-1} \mathbb{E}\big[\xi_k^2\big]\, \mathbb{E}\big[(W_{t_{k+1}} - W_{t_k})^2\big].$$

The cross terms vanish because of independence and zero mean of each increment. Using $\mathbb{E}[(W_{t_{k+1}} - W_{t_k})^2] = t_{k+1} - t_k$,

$$\sum_{k=0}^{n-1} \mathbb{E}[\xi_k^2]\,(t_{k+1} - t_k) \;=\; \mathbb{E}\Big[\sum_{k=0}^{n-1} \xi_k^2\,(t_{k+1} - t_k)\Big] \;=\; \mathbb{E}\Big[\int_0^T X_s^2\, ds\Big].$$

Hence, for simple processes X_t,

$$\mathbb{E}\Big[\Big(\int_0^T X_s\, dW_s\Big)^2\Big] \;=\; \mathbb{E}\Big[\int_0^T X_s^2\, ds\Big].$$

Step 2: Approximation by simple processes. For a general (predictable) process X_t satisfying $\mathbb{E}[\int_0^T X_s^2\, ds] < \infty$, we can approximate X in L^2 by a sequence of simple processes $\{X^{(n)}\}$ such that

$$\int_0^T \mathbb{E}\big[(X_s - X_s^{(n)})^2\big]\, ds \;\xrightarrow[n\to\infty]{}\; 0.$$

Define

$$I_T(X^{(n)}) \;=\; \int_0^T X_s^{(n)}\, dW_s.$$

By the result for simple processes (Step 1),

$$\mathbb{E}\Big[\big(I_T(X^{(n)})\big)^2\Big] \;=\; \mathbb{E}\Big[\int_0^T \big(X_s^{(n)}\big)^2\, ds\Big].$$

One shows, using Itô isometry itself on the difference and dominated convergence arguments, that

$$\int_0^T X_s^{(n)}\, dW_s \;\xrightarrow{L^2}\; \int_0^T X_s\, dW_s$$

and

$$\int_0^T \big(X_s^{(n)}\big)^2\, ds \;\xrightarrow{L^1}\; \int_0^T X_s^2\, ds.$$

Thus, taking limits yields

$$\mathbb{E}\Big[\Big(\int_0^T X_s\, dW_s\Big)^2\Big] \;=\; \mathbb{E}\Big[\int_0^T X_s^2\, ds\Big],$$

as required. □

Theorem 2.3 (Ito's Lemma in One Dimension). *Let X_t be a one-dimensional Ito process satisfying the stochastic differential equation*

$$dX_t = \mu(t, X_t)\, dt \;+\; \sigma(t, X_t)\, dW_t,$$

where W_t is a standard Brownian motion. Suppose $f(t,x)$ is a function such that

$$f \in C^{1,2}([0,T] \times \mathbb{R}) \quad (\textit{once in } t, \textit{twice in } x).$$

Then $Y_t = f(t, X_t)$ satisfies

$$df(t, X_t) \;=\; \Big(\frac{\partial f}{\partial t} + \mu\, \frac{\partial f}{\partial x} + \tfrac{1}{2}\sigma^2\, \frac{\partial^2 f}{\partial x^2}\Big) dt \;+\; \sigma\, \frac{\partial f}{\partial x}\, dW_t.$$

Proof. **Step 1: Setup and Taylor expansion.**

Fix a small time increment $\Delta t > 0$. Let

$$\Delta X_t \;=\; X_{t+\Delta t} - X_t \;\approx\; \mu(t,X_t)\,\Delta t \;+\; \sigma(t,X_t)\,\Delta W_t,$$

where $\Delta W_t = W_{t+\Delta t} - W_t \sim \mathcal{N}(0,\Delta t)$.

Consider

$$\Delta f \;=\; f(t+\Delta t, X_t + \Delta X_t) \;-\; f(t,X_t).$$

By a second-order Taylor expansion around (t,X_t),

$$\Delta f \;\approx\; \frac{\partial f}{\partial t}(t,X_t)\,\Delta t \;+\; \frac{\partial f}{\partial x}(t,X_t)\,\Delta X_t \;+\; \tfrac{1}{2}\,\frac{\partial^2 f}{\partial x^2}(t,X_t)\,(\Delta X_t)^2.$$

Higher-order (small) terms are omitted as $\Delta t \to 0$.

Step 2: Substitute the SDE and use $(dW_t)^2 \approx dt$.

Since

$$\Delta X_t \;\approx\; \mu\,\Delta t + \sigma\,\Delta W_t,$$

we have

$$(\Delta X_t)^2 \;=\; (\mu\,\Delta t + \sigma\,\Delta W_t)^2 \;\approx\; \mu^2(\Delta t)^2 + 2\,\mu\,\sigma\,\Delta t\,\Delta W_t + \sigma^2\,(\Delta W_t)^2.$$

As $\Delta t \to 0$:

$$(\Delta t)^2 \;\to\; 0 \quad \text{(faster than } \Delta t\text{)}, \quad \Delta t\,\Delta W_t \text{ is } O((\Delta t)^{3/2}), \quad (\Delta W_t)^2 \;\approx\; \Delta t.$$

Hence the dominant term in $(\Delta X_t)^2$ is $\sigma^2\Delta t$.

Substitute back:

$$(\Delta X_t)^2 \;\approx\; \sigma^2(\Delta W_t)^2 \;\approx\; \sigma^2\,\Delta t.$$

Step 3: Collect terms in Δt and ΔW_t.

Therefore,

$$\Delta f \;\approx\; \frac{\partial f}{\partial t}\,\Delta t \;+\; \frac{\partial f}{\partial x}\,\big(\mu\,\Delta t + \sigma\,\Delta W_t\big) \;+\; \tfrac{1}{2}\,\frac{\partial^2 f}{\partial x^2}\,\sigma^2\Delta t.$$

Combine the terms:

$$\Delta f \;=\; \Big(\frac{\partial f}{\partial t} + \mu\,\frac{\partial f}{\partial x} + \tfrac{1}{2}\,\sigma^2\,\frac{\partial^2 f}{\partial x^2}\Big)\,\Delta t \;+\; \sigma\,\frac{\partial f}{\partial x}\,\Delta W_t.$$

Step 4: Pass to the limit as $\Delta t \to 0$.

In the limit $\Delta t \to 0$, we write in differential form:

$$df(t,X_t) \;=\; \Big(\frac{\partial f}{\partial t} + \mu\,\frac{\partial f}{\partial x} + \tfrac{1}{2}\,\sigma^2\,\frac{\partial^2 f}{\partial x^2}\Big)\,dt \;+\; \sigma\,\frac{\partial f}{\partial x}\,dW_t.$$

This is precisely the claimed *Ito's Lemma*, which generalizes the usual chain rule to the stochastic setting, accounting for the quadratic variation of Brownian motion. □

Remark. The extra term $\frac{1}{2}\,\sigma^2\,\frac{\partial^2 f}{\partial x^2}$ in the dt component appears because $(dW_t)^2 \approx dt$. This sets Ito's Lemma apart from the standard deterministic chain rule.

2.1 Financial Market Basics

Definition 2.1 (Financial Market). We assume p (dividend-free[1]) assets $X(t) = [X_1(t), \ldots, X_p(t)]^\top$, which are driven by Ito processes:

$$dX(t) = \mu(t,\omega)\,dt + \sigma(t,\omega)\,dW(t).$$

Definition 2.2 (Trading Strategy). A trading strategy represents a predictable adapted process (of asset weights)

$$\phi(t,\omega) = [\phi_1(t,\omega), \ldots, \phi_p(t,\omega)]^\top.$$

The value of the trading strategy (or corresponding portfolio) is

$$\pi(t) = \phi(t)^\top X(t).$$

Definition 2.3 (Numeraire). A Numeraire is a positive asset $N(t)$ of our market. An equivalent martingale measure(w.r.t. numeraire $N(t)$) is a measure $\mathbb{Q}$ such that the normalised asset price $\frac{X_i(t)}{N(t)}$ is a $\mathbb{Q}$ martingale. The Radon-Nikodym density from $\mathbb{Q}^N$ to $\mathbb{Q}^M$ is

$$\zeta(t) = \frac{M(t)}{N(t)} \cdot \frac{N(0)}{M(0)}.$$

Theorem 2.4 (The Fundamental Theorem of Asset Pricing).

3 Basic Fixed Income Modelling

In the next we introduce a central concept in the financial market.

Definition 3.1 (Short rate and (abstract)bank account). An asset with price $B(t)$ given by $B(0) = 1$ and

$$dB(t) = r(t)B(t)dt$$

therefore

$$B(t) = \exp\left(\int_0^t r(s)ds\right)$$

The short rate is normally considered as **risk-free rate**, set by the central bank. It is crucial to determine the equivalent Martingale measure. Accordingly, the most relevant assets are Zero Coupon Bonds, the short rate determine the most basis asset in financial market: money asset account, which serves as numeraire of financial assets).

Definition 3.2 ((ZCB)zero coupon bond). Define

$$P(t,T) \tag{V.1}$$

the price of a zero coupon bond at time t of maturity T with the payout at terminal time $1(P(T,T) = 1)$. It is a martingale with respect to w.r.t. $B(t)$ as Numeraire.

$$\mathbb{E}^{\mathbb{Q}}[\frac{P(t,T)}{B(t)}] = \mathbb{E}^{\mathbb{Q}}\left[\frac{P(T,T)}{B(T)}\right] = \mathbb{E}^{\mathbb{Q}}\left[\exp(-\int_0^T r(s)ds)\right].$$

Multiplying both sides by $B(t)$, it holds:

$$\mathbb{E}^{\mathbb{Q}}[P(t,T)] = \mathbb{E}^{\mathbb{Q}}\left[\exp(-\int_t^T r(s)ds)\right]$$

Definition 3.3 (Discounted Cashflow Methodology). Given V the cash flow stream, we have by FTAP

$$\frac{V(t)}{B(t)} = \sum_{i=1}^{N} \mathbb{E}^{\mathbb{Q}} \left[\frac{V_i}{B(T_i)} | \mathcal{F}_t \right]$$

We denote

$$\mathbb{E}^{T_i}[\cdot]$$

the expectation in T_i forward measure with zero coupon bond numeraire $P(t,T_i)$. Do the measure change

$$\frac{V(t)}{B(t)} = \sum_{i=1}^{N} \mathbb{E}^{T_i} \left[\frac{P(t,T_i)}{B(t)} \cdot \frac{V_i}{P(T_i,T_i)} \,\middle|\, \mathcal{F}_t \right].$$

We have therefore

$$V(t) = \sum_{i=1}^{N} P(t,T_i) \mathbb{E}^{T_i}[V_i | \mathcal{F}_t]$$

If the future cash flow is known, then

$$V(t) = \sum_{i=1}^{N} P(t,T_i) V_i$$

In general, the challenge lies in predicting the future cash flow.

4 Yield Curve and Linear Product

4.1 Static Yield Curve and Market Convention

Definition 4.1 (Yield Curve). A yield curve at an observation time t is the function of the **zero coupon bonds** $P(t,\cdot) : [t,\infty) \to \mathbb{R}^+$ for maturities $T \geq t$ to the IRR. The x-axis is the maturity, the y-axis is the calculated interest rate z. There are different methods to calculate z:

discrete compounded zero rate curve

$$P(t,T) = (1 + \frac{z_p(t,T)}{p})^{-p(T-t)}$$

simple compounded zero rate curve

$$P(t,T) = \frac{1}{1 + z_0(t,T) \cdot (T-t)}$$

continuously compounded zero rate curve

$$P(t,T) = \exp\{-z(t,T) \cdot (T-t)\}$$

Remark. Typically, we have $t = 0$.

Definition 4.2 (Continuous Forward Rate).

$$f(t,T) = -\frac{\partial ln(P(t,T))}{\partial T}$$

Remark.

1. The continuous rate is the spot rate at the maturity.
2. Holiday Calendar. Business day Convention.

Definition 4.3 (Day Count Convention). A day count convention is a function

$$\tau : \mathcal{D} \times \mathcal{D} \to \mathbb{R}$$

which measures a time period between dates in terms of years.

4.2 Multi-Curve Discounted Cash Flow Pricing

By simple calculation we determine the fair rate of the interbank lending rate:

Definition 4.4 (Spot Libor Rate). Fair rate for interbank lending deal with trade date T.

$$L(T;T_0,T_1) = \left[\frac{P(T,T_0)}{P(T,T_1)} - 1\right]\frac{1}{\tau}$$

Remark.

1. The Libor rate is daily updated.
2. The interval $[T_0,T_1]$ is typically 1m, 3m, 6m, 12m.

We observe the plain vanilla legs with DCF methodology:

$$V(t) = \sum_{i=1}^{N} P(t,T_i)\cdot \mathbb{E}^{T_i}\left[L(T_{i-1}^F,T_{i-1},T_i)\cdot \tau_i \mid \mathcal{F}_t\right]$$

The next theorem shows: Libor is a martingale in the terminal measure with the Numeraire of zero coupon bond of maturity T_1.

Theorem 4.1 (Martingale Property of Libor rate). *The Libor rate $L(T,T_0,T_1)$ is a martingale with respect in the T_1-forward measure. And we have:*

$$\mathbb{E}^{T_1}[L(T;T_0,T_1)|\mathcal{F}_t] = \left[\frac{P(t,T_0)}{P(t,T_1)} - 1\right]\frac{1}{\tau} = L(t;T_0,T_1)$$

Which allows us to price the Libor leg based on t:

$$V(t) = \sum_{i=1}^{N} P(t,T_i)$$

Proof. Fair Libor rate at fixing time T is

$$L(T;T_0,T_1) = \frac{P(T,T_0)}{P(T,T_1)} - \frac{1}{\tau}$$

The zero coupon bond $P(T,T_0)$ is an asset, and $P(T,T_1)$ is the numeraire in the T_1-forward measure. Thus, FTAP yields that the discounted asset price is a martingale, i.e.,

$$\mathbb{E}^{T_1}\left[\frac{P(T,T_0)}{P(T,T_1)}\middle|\mathcal{F}_t\right] = \frac{P(t,T_0)}{P(t,T_1)}.$$

Linearity of the expectation operator yields:

$$\mathbb{E}^{T_1}\left[L(T;T_0,T_1)|\mathcal{F}_t\right] = \mathbb{E}^{T_1}\left[\frac{P(T,T_0)}{P(T,T_1)} - 1\middle|\mathcal{F}_t\right]\frac{1}{\tau}.$$

$$= \left[\frac{P(t,T_0)}{P(t,T_1)} - 1\right]\frac{1}{\tau}.$$

$$= L(t;T_0,T_1).$$

□

4.3 The Tenor-basis Modelling

$$\frac{V(T)}{B(T)} = \mathbb{E}^{\mathbb{Q}}\left[-\mathbb{1}_{\{\xi_B > T_0\}} \cdot \frac{1}{B(T_0)} + \mathbb{1}_{\{\xi_B > T_1\}} \cdot \frac{1+K\cdot\tau}{B(T_1)} \,\middle|\, \mathcal{F}_T\right].$$

(all expectations conditional on $\mathcal{F}_T$)

$$V(T) = -P(T,T_0)Q(T,T_0)\mathbb{E}^{T_0}[1] + P(T,T_1)Q(T,T_1)\mathbb{E}^{T_1}[1+K\cdot\tau].$$

the hazard rate model

- **Survival Probability:**

$$S(t) = \mathbb{P}(\tau > t)$$

  The probability that the default event has not occurred by time t.

- **Hazard Rate ($\lambda(t)$):**

$$\lambda(t) = \lim_{\Delta t \to 0} \frac{\mathbb{P}(t \le \tau < t + \Delta t \mid \tau > t)}{\Delta t}$$

  The instantaneous rate of default at time t, conditional on survival up to t.

- **Relationship Between Hazard Rate and Survival Probability:**

$$S(t) = \exp\left(-\int_0^t \lambda(u)\,du\right)$$

  Survival probability decreases exponentially with the cumulative hazard.

- **Applications:**
  - **Credit Default Swap (CDS) Pricing:** Hazard rates are used to model expected cash flows from default protection.
  - **Bond Pricing with Default Risk:**

$$P(t) = \int_t^T \exp\left(-\int_t^u (r(s) + \lambda(s))\,ds\right) c(u)\,du$$

    where $r(t)$ is the risk-free rate, $\lambda(t)$ is the hazard rate, and $c(u)$ is the cash flow.
  - **Portfolio Credit Risk:** Used to model correlated defaults in portfolios.

- **Advantages:**
  - Flexibility to model time-varying hazard rates $\lambda(t)$.
  - Simplifies calculations using the exponential relationship with survival probabilities.
  - Intuitive interpretation of default intensity at any point in time.

- **Challenges and Limitations:**
  - Calibration of hazard rates from market or historical data can be difficult.
  - Assumes independent defaults, which may not hold during crises.
  - Correlations between multiple entities add complexity.

- **Extensions:**
  - **Stochastic Hazard Rates:** Models where $\lambda(t)$ is itself a stochastic process.
  - **Multi-State Hazard Models:** Allows transitions between different credit states (e.g., downgrade, default).
  - **Reduced-Form Credit Risk Models:** Focus on the likelihood of credit events rather than explicit asset dynamics.

Theorem 4.2. *Deterministic hazard rate assumption preserves the martingale property of the forward Libor rate.*

4.4 Vanilla Interest Models

Vanilla Interest Rate Option

Theorem 4.3 (Martingale representation theorem).

Theorem 4.4 (Bachelier Formula).

Theorem 4.5 (Black Formula).

5 Introduction To Risk Measures

Definition 5.1 (some properties). Given $\mathcal{X}$ the class of financial position, the function $\rho : \mathcal{X} \to \mathbb{R}$ is a risk measure if it satisfies

1. monotonicity: if $X \leq Y$, it holds $\rho(Y) \leq \rho(X)$.
2. translation invariant: $\rho(X+m) = \rho(X) - m$.
3. normalisation: $\rho(0) = 0$.

Remark. The cash invariance implies in particular

$$\rho(X + \rho(X)) = \rho(X) - \rho(X) = 0.$$

A risk measure can enjoy many properties, such as convexity (decentralised risk management), positive homogeneous.

Definition 5.2 (acceptance set). The acceptance set for the risk is defined as

$$\mathcal{A}_\rho(X) := \{X \in \mathcal{X} : \quad \rho(X) \leq 0)$$

Proposition 5.1. *positive homogeneity and convexity imply subadditivity.*

Positive homogeneity is strict. In the case where the risk grows nonlinearly w.r.t. position, we only focus on the convexity. In fact, subadditive functions are usually concave, which motivates the following definition:

Definition 5.3 (coherent monetary utility functional and coherent risk measure). A functional $\phi : \mathcal{X} \to \mathbb{R}$ is called a coherent monetary utility functional if it is monotone, concave, positive homogeneous and cash invariant. The functional

$$\rho(X) = -\phi(X),$$

is called a coherent risk measure.

The following theorem gives a robust representation of the risk measure ρ:

$$\rho(X) = \sup_{Q \in \mathcal{M}_{1,f}} \left(\mathbb{E}^Q(X) - \alpha(Q) \right)$$

Theorem 5.2. *A convex risk measure ρ on $\mathcal{X}$ is of the form*

$$\rho(X) = \max_{Q \in \mathcal{M}_{1,f}} \left(\mathbb{E}_Q[-X] - \alpha^{\min}(Q) \right), \tag{V.2}$$

where

$$\alpha^{\min}(Q) := \sup_{X \in \mathcal{A}_p} \mathbb{E}_Q[-X] \qquad Q \in \mathcal{M}_{1,f}. \tag{V.3}$$

$\alpha^{\min}$ dominate all the penalty term that admits the representation of ρ.

Proof.
I.

$$\rho(X) \geq \sup_{Q \in \mathcal{M}_{1,f}} \left(\mathbb{E}_Q[-X] - \alpha^{\min}(Q) \right) \quad \forall X \in \mathcal{X}$$

It holds in particular by the cash invariance

$$X' := X + \rho(X) \in \mathcal{A}_\rho.$$

Taking X' for the $\alpha^{\min}$ function, it holds

$$\forall Q \in \mathcal{M}_{1,f} : \quad RHS \leq \mathbb{E}_Q[-X] - \mathbb{E}_Q[X] - \rho(X) = \rho(X).$$

Hence the statement holds.
II. Next we show the inverse direction, i.e.,

$$\rho(X) \leq \mathbb{E}_{Q_X}[-X] - \alpha^{\min}(Q_X),$$

for some Q_X depends on X.

II.1, simplification w.o.l.g we prove for $X \in \mathcal{X}$ with $\rho(X) = 0$ and also we assume since ρ is cash invariant. (Why???) And we assume $\rho(0) = 0$.

Next, define

$$\mathcal{B} := \{Y \in \mathcal{X} : \rho(Y) < 0\}$$

Noted $\mathcal{B}$ is convex and open (why???). Since $\rho(0) = 0$, by Hahn-Banach separation theorem (of open convex subspace) there exist a functional l on $\mathcal{X}$ such that

$$l(X) \leq \inf_{Y \in \mathcal{M}} l(Y) := b.$$

Show the following claim:

$$\begin{aligned} &1).l(Y) \geq 0 \quad \forall Y \in \mathcal{B}, \\ &2).l(1) > 0. \end{aligned}$$

1). By the cash invariance and convexity and $\rho(0) = 0$, it holds

$$\rho(\lambda Y + (1-\lambda)\varepsilon) \leq \lambda \rho(Y) + (1-\lambda)\rho(\varepsilon)$$

Taking $\varepsilon \to 0$, by continuity it holds for $\lambda > 0$.

$$\rho(\lambda Y) \leq \lambda \rho(Y) < 0. \Rightarrow \lambda Y \in \mathcal{B}$$

By the cash invariance $1 + \lambda Y \in \mathcal{B}$, it holds

$$l(1 + \lambda Y) = l(1) + \lambda l(Y)$$

2) Since l it not trivial on $\mathcal{B}$, there exists $Y \in \mathcal{B}$ with $l(Y) > 0$, normalised Y such that $\|Y\|_{L^\infty} \leq 1$(???) and write $l(Y) = l(Y^+) + l(Y^-) \leq l(1)$. It holds

$$\begin{aligned} &l(1 - Y^+) \geq l(Y^-) > 0, \\ &l(Y^+) > 0, \end{aligned}$$

hence $l(1) = l(1 - Y^+) + l(Y^+) > 0$. By the representation theorem, there exists a content that represent the linear functional l, i.e., there exists a $Q \in \mathcal{M}_{1,f}$, such that

$$l(X) = \mathbb{E}_Q[X] \cdot l(1) \quad \forall X \in \mathcal{X},$$

hence

$$\mathbb{E}_Q[Y] = \frac{l(Y)}{l(1)} \quad \forall Y \in \mathcal{B}.$$

□

Example (entropic risk measure). Consider the penalty function defined on the space of measure

$$\alpha : \mathcal{M} \to (0, \infty]$$

$$\rho(X) := \sup_{Q \in \mathcal{M}_1} \left\{ \mathbb{E}_{\mathbb{Q}}[-X] - \frac{1}{\beta} KL(\mathbb{Q}|\mathbb{P}) \right\}.$$

Example (shortfall risk).

5.1 Convex risk measure on L^∞

5.2 V@R, AV@R and shortfall risk.

The Var Model is one of the most widely used model. It was invented by JP.Morgen, it indicated the quantile of loss, i.e., the loss of the position is at least X with probability at most λ. It improves the traditional variance-based risk measure in that it calculates the not only the volatility, but also the expected loss under the worst case. On the other hand, it is also criticised as pro-cycle since it's calibrated on the recent data.

Definition 5.4 (value at risk). Given the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and X a random variable on it, we observe the λ-quantile function

$$q_X(\lambda) := \{q \in \Omega : \quad \mathbb{P}[X \leq q] \geq \lambda \ \texttt{and} \ \mathbb{P}(X < q) \leq \lambda\}.$$

By the right-continuity of the cdf we define the value at risk as

$$V@R_\lambda(X) := -q_X^+(\lambda). \tag{V.4}$$

Proposition 5.3. *For any $X \in L^\infty$ and each $\lambda \in (0,1)$, it holds*

$$V@R_\lambda(X) = \min_\rho \{\rho(x) : \quad \rho \text{ convex, continuous from above, dominates } V@R\}.$$

Definition 5.5 (AV@R). The average value at risk is defined as

$$AV@R(X) := \frac{1}{\lambda}\int_0^\lambda V@R_\gamma(X)d\gamma$$

The Shortfall Risk

Given $u : \mathbb{R} \to \mathbb{R}$ the utility function, define the lost function

$$l(x) : \mathbb{R} \to \mathbb{R}, \quad x \mapsto -u(-x).$$

Define the acceptance set:

$$\mathcal{A} := \{X \in \mathcal{X} : \mathbb{E}[u(x)] \geq y_0\}.$$

We are able to define the convex monetary risk measure ρ (Why??? define as such)

$$\rho(X) := \inf_{m \in \mathbb{R}} \{m | \rho(X+m) \geq y_0\} \tag{V.5}$$

The following

Definition 5.6 (shortfall risk).

$$\alpha^{\min}(Q)$$

Chapter VI

Insurance Mathematics

1 Life Insurance

We start by defining the following notations

1. l_x: probability of living person at the age of x.
2. ${}_tq_x$: probability of dying of a x-years old within t years.
3. ${}_tp_x$: probability of living of a x-years old with t years.
4. v: technical interest rate

q_x^{aa}: died as active

Definition 1.1 (life expectancy). $\mathbb{E}[T(x)]$ the expected remaining life a x-years old person. T is the random variable denoting the remaining life.

Definition 1.2 (the hazard rate). The hazard rate μ_{x+t} describe the mortality of a given x-years old person died at $x+t$:

$$\mu_{x+t} := \lim_{\Delta t \to 0} \frac{P(t \leq T \leq T + \Delta t | T \geq t)}{P(T \geq t)}$$

Lemma 1.1 (Chapman-Kolmogorov in mortality). *It holds*

$${}_tp_x^r = \frac{l_{x+t}}{l_x}$$

Proof. □

Definition 1.3 (commutation value). The commutation value helps to calculate the present value of the future pension liability. It can be seen as weighted probability of survival.

$$D_x^s := v^x l_x^s.$$

Accordingly we defined the **discounted remaining life expectancy**

PV of the delayed pension and the

Notation von Barwert berechnen:

1. a_x^r: x-years old, old-age pension, life long
2. ${}_{z-x}a_x^{aA}$: x-years old, active to old-age pension at age z.
3. $a_{x,z-x\neg}^i$: x-years old, invalid to z.
4.

Lemma 1.2 (ewig annuity). *The present value of the lifelong old-age pension of the age x can be calculated as*

$$a_x^r = \frac{N_x^r}{D_x^r}$$

Proof. It holds

$$\begin{aligned} a_x^r &= \sum_{t\geq 0} {}_tp_{x+t}^r \cdot v^t \\ &= \sum_{t\geq 0} \frac{l_{x+t}}{l_x} \cdot v^t \\ &= \sum_{t\geq 0} \frac{l_{x+t} v^{x+t}}{l_x v_x} \\ &= \frac{N_x^r}{D_x^r} \end{aligned}$$

□

Lemma 1.3 (aufgeschobene annuity). *The invalid pension follows the same formula. However, it is the finite sum since the invalid annuity stops at the age of retirement, i.e.,*

$$a_x^i = \sum_{t=0}^{Z-t-1}$$

Lemma 1.4. *Other calculations*

1. ${}_{z-x}a_x^{iA} = (1-q_x^a - i_x)(1-q_{x+1}^a - i_{x+1})\dots(1-q_{z-1}^a - i_{z-1}) \cdot v^{z-x} \cdot a_z^r$

By pricing we established the PV of the insurance contracts at the moment. However, we need to assure the solvency is not jeopardised as the times marches.

Lemma 1.5 (projected Unit Credit Method). *Denote*

1. x_0*: age at the beginning of valuation*
2. *k: Entering data of the benefit*
3.

1.1 Valuation Methods

Projected Unit Credit Method

Definition 1.4 (Defined (Projected) Benefit Obligation by IFRS (GAAP)). Present Value of the pension plan at the reference data.

Proposition 1.6 (m/n degressiv method). *It is calculated as following*

1. *Calculate the*

Definition 1.5 (covering capital).

Definition 1.6 (embedded value).

$$EV = PVFP + ADNV$$

the adjusted net asset value includes capital gains, tex deduction.

Chapter VII

Population Genetics and Markov Process

1 Introduction

1.1 The Wright-Fisher random graph

Definition 1.1 (Wright-Fischer random graph). Define

1. $V := \mathbb{Z} \times N$ and $v(g,i)$ with $g \in \mathbb{Z}$, $i \in \{1,\dots,N)$
2. $\{U_v\}_{v\in V} \overset{d}{\sim} \mathbf{U}\{1,\dots,N\}$ are i.i.d random variables
3. $(g-1,U_v)$ is the parent of v.
4. The unoriented edge are denoted by $\langle U_v, v\rangle$. The set of the edges is
$$E := \{\langle (g_v - 1, U_v), v\rangle | v \in V\}$$
5. **The Wright-Fisher random graph** is represented by
$$G := \{V, E\}$$

Definition 1.2 (The natural two-type Wright-Fisher process). .

- type space of cardinality 2, such as $\{+,-\}$.
- μ a distribution on $\{+,-\}^N$
- The random map $t : V \to \{+,-\}$ with
  1. The initial distribution $(t(v)|g_v = g_0) \sim \mu$
  2. $\forall g(v) > g_0 : t(v) = t((g(v)-1, u_v))$
- The **natural 2-type Wright-Fisher process**
$$X^N(g_0,n) := \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{t(g_0+n,i) = "-"\}$$

Remark. .

- $X^N(g_0,n)$ is a Markov process because $(U_{v(g,i))})_{g\in\mathbb{Z}}$ are i.i.d. random vector
- For the same reason the Markov chain is time-homogeneous and we can write $X^N(g_0,n)$ as

$$X^N(n)$$

Definition 1.3 (Ancester and Common Ancester). .
$\tilde{v} \in V$ is an ancester of $v \in V$ if :

- $g(\tilde{v}) < g(v)$
- there exists a genelogical path from $\tilde{v}$ to v.

Define $V_g := \{v \in V | g(v) > g\}$. We define $\tilde{v} \overset{g}{\sim} v$ if

- $\tilde{v}, v \in V_g$
- $\tilde{v}$, and v have common ancester in generation g.

Theorem 1.1. *$\sim_g$ is an equivalence relation*

Remark. $\sim_g$ defines a partition on the set $[N]$.

Definition 1.4 (The genealogy process). Given a WFRG (V,E), we set

- μ a distribution on $([N], 2^{[N]})$, which is independent from U_v. (the sampling)
- $S \sim \mu$, $\Pi_0^N := \{\{S\}\}$, $k = |\Pi_0^N|$, $S = [k]$

The process $(\Pi^N(g_0,n))_{n\in\mathbb{N}_0}$ is the genealogy process of Π_0^N given by a partition of $[k]$ induced by $\sim_{g_0-n}$.

Remark.

Definition 1.5 (The block-counting process of geneology).

$$\forall n \in \mathbb{N}_0 : \quad A^N(g_0,n) = |\Pi^N(g_0,n)|$$

Definition 1.6 (Duality). The process $(X_i)_{i\in I}$ and $(A_i)_{i\in I}$ is called H-dual, if:

$$\forall i \in I, x \in E, a \in F : \mathbb{E}[H(X_i,a)|X_0 = x] = \mathbb{E}[H(x,A_i)|A_0 = a]$$

H is called the duality function.

Theorem 1.2.

$$\mathbb{E}[X^N(g_0,n)^a | X^N(g_0,0) = x] = \mathbb{E}[x^{A^N(g_0,n)} | A^N(g_0,0) = a]$$

induced by notation in Bayes we write:

$$\mathbb{E}_x[X_n^a] = \mathbb{E}_a[x^{A_n}]$$

1.2 The scaling limit

Definition 1.7 (the semi-group operator). Given a state space E and a continuous-time Markov process X_t, we define:

$$P_t : C^0(E) \to C^0(E), f \rightarrowtail \mathbb{E}_x[f(X_t)]$$

as the semi-operator.

Theorem 1.3 (Properties of the semi-group). *the semi-operator admits the properties of*

- ***associativity***: $P_t \circ P_s = P_{t+s}$.
- ***monoid:*** $P_0 = Id_{C_0(E)}$.

Proof. By Chapman-Kolmogrov we have:

$$\mathbb{E}_x[\theta_s \circ \theta_t \circ f(X)] = \mathbb{E}_x[\theta_{s+t} \circ f(X)]$$

and

$$\mathbb{E}_x[\theta_s \circ \theta_{-s} \circ f(X)] = \mathbb{E}_x f$$

□

Definition 1.8 (Generator of the semigroup). The generator of a semigroup P_t on the state x is defined as

$$A(f) : D(A) \subset C^0(E) \to \mathbb{R}, \quad f \rightarrowtail \lim_{t\to 0} \frac{P_t(f)|_x - P_0(f)|_x}{t}$$

or we write

$$A(f)|_x = \lim_{t\to 0} \frac{\mathbb{E}_x[f(X_t) - f(x)]}{t}$$

The domain where $A(f)$ is well-defined is denoted by $D(A)$, it includes the continuous function whose induced semi-group admits a generator.

Example (Jump process in finite state space). Consider Y_t a jump process in finite state space E. If the process jumps at exponential time, We have

$$\begin{aligned} A(f) &= \lim_{t\to 0} \frac{\mathbb{E}_y[f(Y_t) - f(y)]}{t} \\ &= \lim_{t\to 0} \frac{\sum_{w\in E} \mathbb{P}(N_t = 1)(f(w) - f(y))r_{yw}}{t} \\ (1) &= \sum_{w\in E} (f(w) - f(y))r_{wy} \\ (2) &= \int_E f(w) - f(y)\nu(w) \end{aligned}$$

(1) follows from that $N_t \sim Poisson(\lambda t)$. Therefore $\frac{\mathbb{P}(N_t=1)}{t} = \frac{\lambda t}{t} e^{-\lambda t} \overset{t\to 0}{\to} 1$.
(2) we define the measure $\nu(w) := \sum_{w\in E} r_{wy}\delta(y)$

Example (Deterministic ODE). Consider R a deterministic process. Furthermore we have the ODE:

$$\frac{dR}{dt} = \mu(R)$$

Calculate the generator:

$$\begin{aligned} A(f) &= \lim_{t\to 0} \frac{\mathbb{E}[f(R_t) - f(R_0)]}{t} \\ &= \lim_{t\to 0} \frac{f(R_t) - f(R_0)}{t} \\ &= f'(R_0)\mu(R_0) \end{aligned}$$

Example (Random Walk). Given a Poisson process N_t, we can write the generator:

$$A(f)|_N = N(f(n+1) - f(n))$$

A random walk admits the generator:

$$A(f)|_x = r(x)(f(x+1) - f(x)) + q(x)(f(x-1) - f(x))$$

If the random walk is **symmetric** indexed by the Poisson process, then the jumping time of the random walk is controlled by the Poisson process. The generator of the process $(S_{N_t})_t$ could be written as

$$A(f)|_{S_{N_t}} = \frac{N}{2}(f(x+1) - f(x)) + \frac{N}{2}(f(x-1) - f(x))$$

By rescaling the step size and applying the Taylor theorem, given $f \in C^3(\Omega)$, the process $\frac{S_{N_t}}{\sqrt{N}}$ admits the generator:

$$\begin{aligned} A(f) &= \frac{N}{2}\left(\frac{f''(x)}{N} + o(N^{-\frac{3}{2}})\right) \\ &= \frac{f''(x)}{2} + o(N^{-\frac{1}{2}}) \end{aligned}$$

Furthermore, the speed of the up-down movement could be controlled by the position of x, therefore we have:

$$A(f) = \frac{f''(x)\sigma^2(x)}{2} + o(N^{\frac{1}{2}}) \quad \xrightarrow{N\to\infty} \quad \frac{f''(x)\sigma^2(x)}{2}$$

The greater the $\sigma^2(x)$, the faster the Brownian motion moves.

Definition 1.9 (Convergence of Operator). A sequence of stochastic process $(S_t^N)_{t\in T}$ converges in operator if

$$\forall f \in D(A) : A(f)_{S_N} \xrightarrow{L^1} A(f)_S$$

Remark.

Theorem 1.4.

1.3 Wright-Fisher diffusion

Definition 1.10 (Diffusion).

Theorem 1.5 (Ito Formula).